

SAP–MM (MATERIAL MANAGEMENT)

Introduction to SAP (System application products in data processing)

SAP stand for system application products in data processing.

SAP is an ERP tools. (Enterprises resource planning).

SAP is a German software introduced in the year 1971 and implemented in the year 1972 by following 5 IBM engineers.

✚ Wallen Reuther

✚ HOPP

✚ Hector

✚ Plattner

✚ Tschira

in SAP EURO's currency is used.

What is SAP?

SAP is an ERP tool, application or leading software which is used to support each and every business process of the company such as purchasing, material services, selling and delivery goods to company's customers, finance, manufacturing activities (converting to RM to final products), logistics activities and warehouse management.

Why SAP?

SAP offers following advantages.

- ❖ It is Market leader in providing solution and services to every business requirement.
- ❖ The beauty of SAP is integration of modules.
- ❖ Data security.
 - SAP will capture each and every transaction tacking place in the company from starting to end.
- ❖ SAP will provide solution to all the department.

Why not SAP?

SAP suffers from following disadvantages.

- ❖ High cost.

Who will use SAP?

SAP can be used all type of industries such as

- ❖ Pharmacy.
- ❖ Chemical industry.
- ❖ Construction industry.
- ❖ Engineering work and procurement companies.
- ❖ Automobiles.
- ❖ Banking.
- ❖ Insurance.
- ❖ Retail companies etc.

SAP will bring all the received department on single platform to do the business.

Modules in SAP: -

In SAP each object is called module.

Each and every business process of the company will be treated as module in SAP. In SAP we have functional, technical and business modules.

The people who are working in functional modules are called functional consultants and for this no coding knowledge is required.

The people working in technical modules like ABAP are called technical consultants or developers and for this coding knowledge is required.

List of functional modules: -

- ❖ FI - Finance.
- ❖ CO - Controlling.
- ❖ MM - Material Management.
- ❖ QM - Quality Management.
- ❖ SD - Sales and Distribution.
- ❖ PP - Production Planning.
- ❖ PM - Plant Maintenance.
- ❖ PS - Project Systems.
- ❖ WM - Warehouse Management.
- ❖ HCM - Human Capital Management.

From technical side we have ABAP – Advance Business Application Programming and BASIS module, SAP security etc.

Roles and responsibilities of individual modules consultants: -

FI - Finance:

This module helps in company's all financial activities such as.

- AP - Account payables,
- AR – Account receivables, and Taxations.

CO - Controlling:

This module helps in capturing cash flow of the company.

MM - Material Management:

This module supports all purchasing activities of materials and services, inventory activities and invoice verification before making payment.

QM - Quality Management:

This module helps in quality (QC) check of company's materials.

SD - Sales and distribution:

This module helps in all sales activities and delivery materials to customers.

PP - Production planning:

This module helps in all manufacturing activities of company's materials.

PM – Plant Maintenance:

This module supports all maintenance activities of the company.

PS – Project System:

Mainly this module will be used in EPC projects - Engineering works Procurement constructions. This module captures projects budgets.

WM – Warehouse Management:

This module helps in storing company's materials physically.

HCM – Human Capital Management:

This module helps in all HR activities.

ABAP – Advance Business Application Programming:

This module consultants will write new programs to new data in SAP. This module consultants are called developers.

What happens in real time projects: -

Once we receive requirements then we consultants check solution in standard SAP and if we got solution then we configure (setting/design) as per client's requirements. Once

configuration is done then we consultant and clients will be testing and if it working fine then will handover to clients. Clients will use to do regular business. If incase there is no solution to client's requirement then we consultants will approach technical consultant

ABAP with a requirement document called FSD – functional specification document, based on FSD ABAP consultants will customize new program in SAP.

What is ERP?

ERP stand for Enterprises Resource Planning.

It an application or a tool which support each and every business process of the company.

In market we have many ERP tools such as,

- Oracle,
- JD Edwards,
- People soft,
- BAAN,
- SAP etc.

ERP can be used by small sector industries to large sector industries based on their business size and strength.

Why ERP?

ERP following advantages

- ❖ It saves time and expenses
- ❖ It provides real time information whenever we required.

Why not ERP?

ERP suffers from following disadvantages.

- ❖ Sometime it is very difficult to map client's requirement in ERP's.

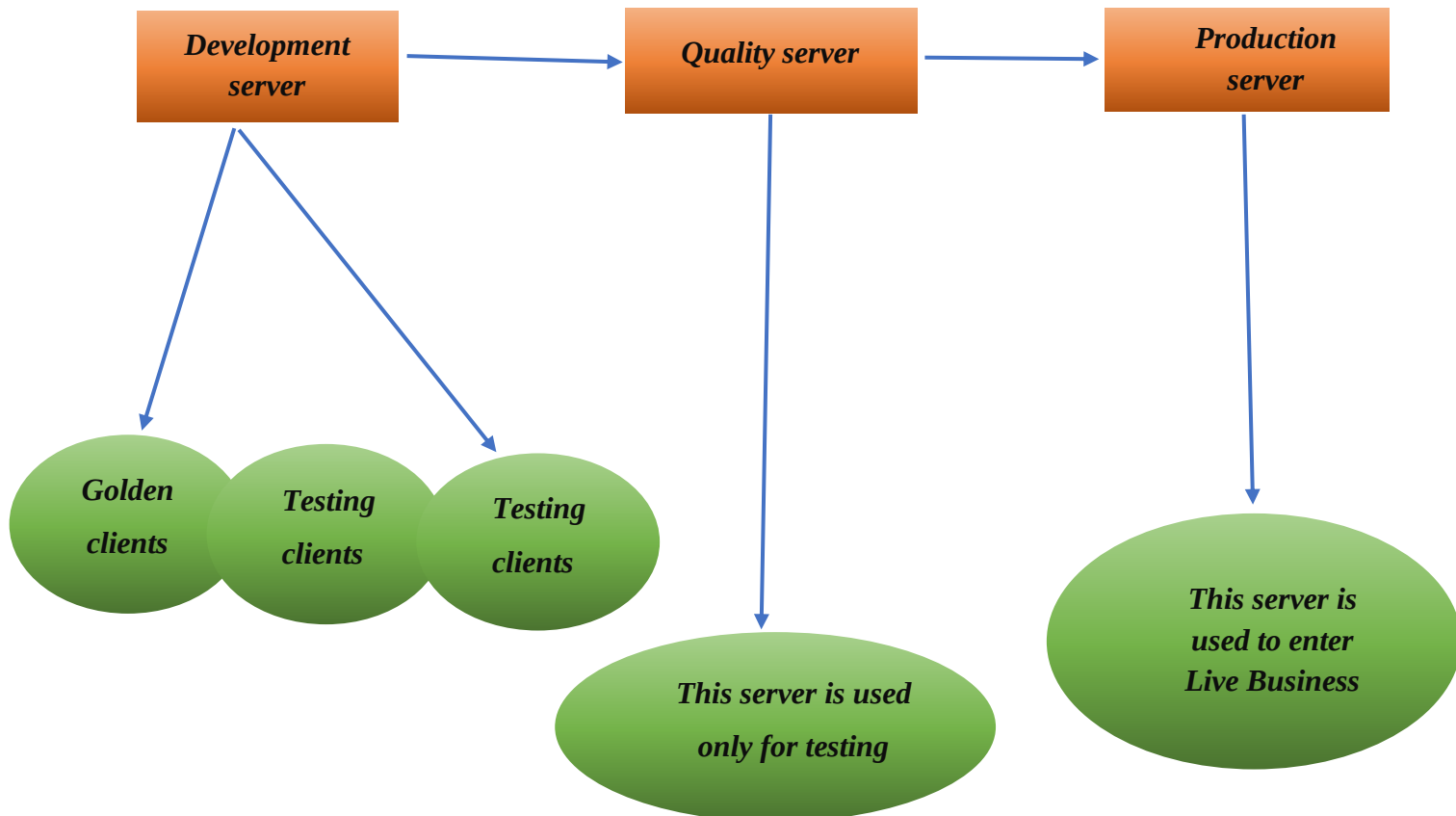
FSD – Functional Specification Document.

Under the heading we will learn how to operate SAP system as a beginner.

1. Landscape in SAP/server in SAP:

Server is computer or program or device that provides services to another computer program and to its users.

Landscapes/server in SAP



Squamate of server arrangement is called Landscape

1. Server in SAP

In server we have following servers.

- ❖ Development server
- ❖ Quality server
- ❖ Product server
- ❖ Sand box server – it used by consultants to learn to new things before providing solution to clients and it is not mandatory in all the projects/companies.
- ❖ Ideas/Dummy server - it is used for learning SAP. It is also called as rental server.

Basic consultants will design server purchased from SAP any login issues approach basic consultants.

1) Development server:

In development server we consultants we provide solution to client requirements. Every development server will have one golden client and 1 or more testing clients (based on no. of consultants). Golden clients and testing clients are just screen names available under development server.

In golden clients of development server will be used to maintain solution for client's requirement.

In testing clients of development server will used to do testing for the solution provided by consultant.

2) Quality server:

After providing solution in development server then those solutions will copy to quality server and in quality server client will do testing and sometimes consultants also will involve.

3) Production server:

In this server only clients will enter live business data.

Consultants don't have access to production server and clients don't have access to development server.

1. Used new page icon in login pad to maintain server details. Server details will be given by basic consultant based on project allocated to you. Once we save server details in logon

pad system will add a server line under connection option which is used to login. Double click on sever line to login.

2. Each and every screen of SAP will be identified with 3 digits numeric code called client. Client 800 is designed by SAP standard which is used only for practice.

3. In real time projects we will use different client codes from 002 to 999 except 001 and 800, these client code will be decided by project manager and designed by BASIS consultant. Some of the real time client codes are as below (just for idea/sample code).

Example purpose:

Realtime client examples

Servers	Clients	Project-1	Project-2	Project-3
Development server	Golden client	100	801	250
	Testing client	101	802	251
Quality server		200	850	350
Production server		300	900	450

4. New password:

This option will be used to change password of SAP ID but will changing password we have to remember following 2 terms.

- ❖ We can change password once in a day (24hours).
- ❖ System will not accept last 5 used passwords.

**** Note** – In real time projects server details and login credentials will give by basic consultants. Basic consultants will help to change password or to unlock SAP if it locks etc... if we enter wrong password for 3 times then SAP will get lock. In this case approach institute for unlock.

5. SAP has given in default languages:

- ❖ EN – English
- ❖ DE – German

6. Maintain user name and password to login.

7. When we login into SAP in any server system will open a default screen called “SAP EASY ACCESS”. In this easy access both consultants and clients work based on server.

SAP easy access screen of production server will be used by clients to enter day to day business activities and consultants will use the same in development server for testing purpose.

In quality server easy access used by both clients and consultants only for testing.

8. Consultants will provide solutions to client's requirement in development server – golden client – IMG screen.

IMG screen for implementation guide.

9. We have 2 ways to enter any data in SAP. They are

- Path/Navigation (long route to enter data).
- T. Code / transaction code (short cut to enter data).

10. Navigation to IMG screen:

After login into golden client of development server Give SPRO T. code in command box → press enter → click on SAP reference IMG and system will display IMG screen where all module consultants will work.

SPRO - SAP project Reference object.

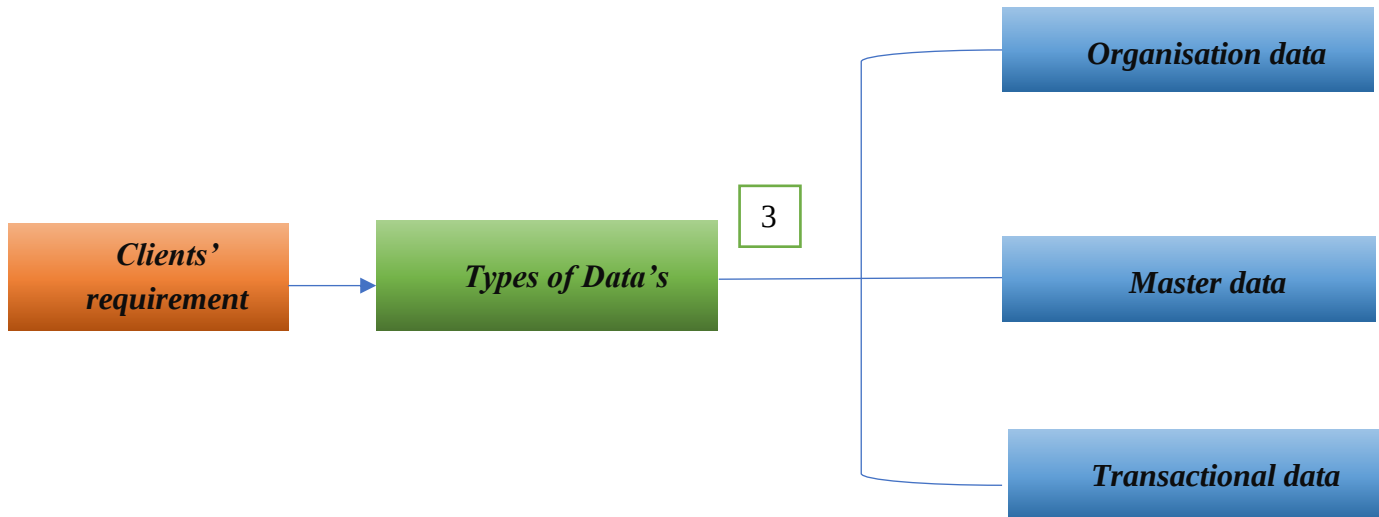
11. We can open 6 screens with one login ID.

12. How to Log off SAP?

- Go to systems in menu bar → then click on Log off option.
- Or directly we can close the screen.
- Or type T. code - /NEX then press enter to log off.

Types of Data in SAP

Whenever we consultants receive the requirement from clients then we will divide those requirements into following 3 types of data. They are:



Organization data:

Organization structure is always created by consultants in IMG screen of golden client – development server and it will be tested in easy access screen of testing client – development server.

Organization data is also called organization structure or enterprise structure.

Every organization will have a structure according to which entire business runs. Every organization structure will have organization level with their own roles and responsibilities.

SAP MM will have followed standard organization structure.

- Alphanumeric.
- Non case sensitives.

MM Organization Data

1. Company (6 CHAR)
2. Company Code (4 CHAR)
3. Purchasing organization (4 CHAR)
4. Plant (4 CHAR)
5. Storage Location (4 CHAR)

*This organization level will
be created by FI consultants
in Realtime projects.*

*This organization levels will
be created by MM
consultants in Realtime*

Define company:

Company will be created by finance consultants in real time projects. Company is an organization formed by individual person or group of people to do business in market.

Company accepts maximum length (6 CHAR)

Use below T. code or path to create company in SAP.

T. code- OX15

(OR)

Path:

SPRO→ IMG→ Enterprises structure→ definition→ financial accounting→ define company→ enter→ go to new entries→

- ❖ Given company unique code (6 CHAR)
- ❖ Company name,
- ❖ Street,
- ❖ Postal code,
- ❖ City,
- ❖ Country (IN),
- ❖ Language (EN),
- ❖ Currency (INR),
- ❖ Press enters.
- ❖ Click on save icon to save new company in SAP system.

Note:

Use same T. code OX15 or the above path to create, edit, check and delete company.

Define company code:

Each and every business of the company will be created as company code in SAP.

Company code is a legal entity with their own balance sheets and profit and loss A/c. company code is the sub division of company.

All financial transactions maintained and saved under company code only and company code will be created by FI consultants.

As we are non-FI consultants se don't know how to figure FI setting so we will use SAP standard company code 1000. In this 1000 company code SAP defined all FI setting for non-FI consultants to practice SAP.

We can use standard company code 1000 or we can create new company code by copying standard 1000. If we do copy then all the FI setting from 1000 company code will copy to our own company code.

Use below T. code or path to copy company code.

OX02– create/edit/delete company code (FI will USE in projects)

EC01 – copy company code.

Path:

SPRO→ IMG→ Enterprises structure→ definition→ financial accounting→ Edit, copy, delete, check company code→ double click on copy, delete, check company code line → click on copy org. object icon→ then give FROM company code 1000 and give TO company code (our own 4 CHAR code) → Enter→ click yes to copy G/L account → click yes to change currency, maintain currency (Ex. INR) → continue for 4 times→ click NO to copy number range→ continue→ click cancel(X) → click copy→ read the information and continue→ Go back→ Double click on edit company code line→ click on position→ Give your own company code→ Enter→ Double click on your company code→ maintain company name, city, country, language, key then press enter→ click on address icon, maintain company code address in detail (address will be given by clients in real time project but in practice we have to maintain o assumption) → press enter and save the company code.

Maintain purchasing organization:

Purchasing organization will be created by MM consultants even real time projects.

Clients purchasing department will be created as purchasing organization in SAP system.

Purchasing department are responsible for purchasing material and services required by the company.

T. code – OX08

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Material management→ Maintain purchase organization→ Enter→ Click on new entries→ Give your purchase organization (4 CHAR), maintain description→ Press enter→ Save.

Define Plant:

Plant will be created by MM consultants but it will be used by entire logistics. Plant is place where business happens or material manufacturing happens.

Whether it is a construction site, Brach, Metro station, manufacturing units, Depots will be created as plant in SAP system by using below T. code or path.

T. code – OX10

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Logistics General→ Define, copy, delete, check, Plant→ Define plant→ Press Enter→ Click on new entries→ Give your own plant (4 CHAR), maintain plant name→ Press enter→ Click on address icon and maintain plant address in details→ Press enters and save the plant.

Main storage location:

Storage location is the place where material is store physically inside the plant (within the plant). We can have multiple storage location under one plant to store different types of materials.

Use below t. code or path to define new storage location to the plant.

T, code – OX09

Path:

SPRO→ IMG→ Enterprises structure→ Definitions→ Material Management→ Maintain storage location→ Enter→ Click on new entries→ Give storage location (4 CHAR), maintain storage location→ Name of the storage location→ save.

Assignment of Organization Structure:

Under this heading will assign organization level to convert into organization structure.

Assign company code to company:

We can assign multiple company code under one company.

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Financial Accounting→ Assign company code to company→ Enter→ Click on position.

Assign plant to company code:

We can have multiple branches.

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Logistics General→ Assign plant to company code→ Click on New Entries.

Assign Purchasing organization to company code:

If we Assign purchasing organization to company code then it will become central purchase organization. It is responsible to buy material and services to all the branches.

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Material Management→ Assign purchasing organization to company code→ Enter→ Click on position.

Assign purchasing organization to plant (plant specific):

If we assign purchasing organization to plant then it will become plant specific purchase organization. It is responsible to buy material and services to only assigned branches.

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Material management→ Assign Purchasing organization to plant→ Enter→ click on new entries.

How to check organization structure in single screen:

- Go to T. code EC01.
- Click on structure.
- Click on navigation then continue.
- Click CTRL+F→ give your Own.
- Company code→ press Enter.
- Click on your company code.
- Again, double click on your company code and check the organization structure.

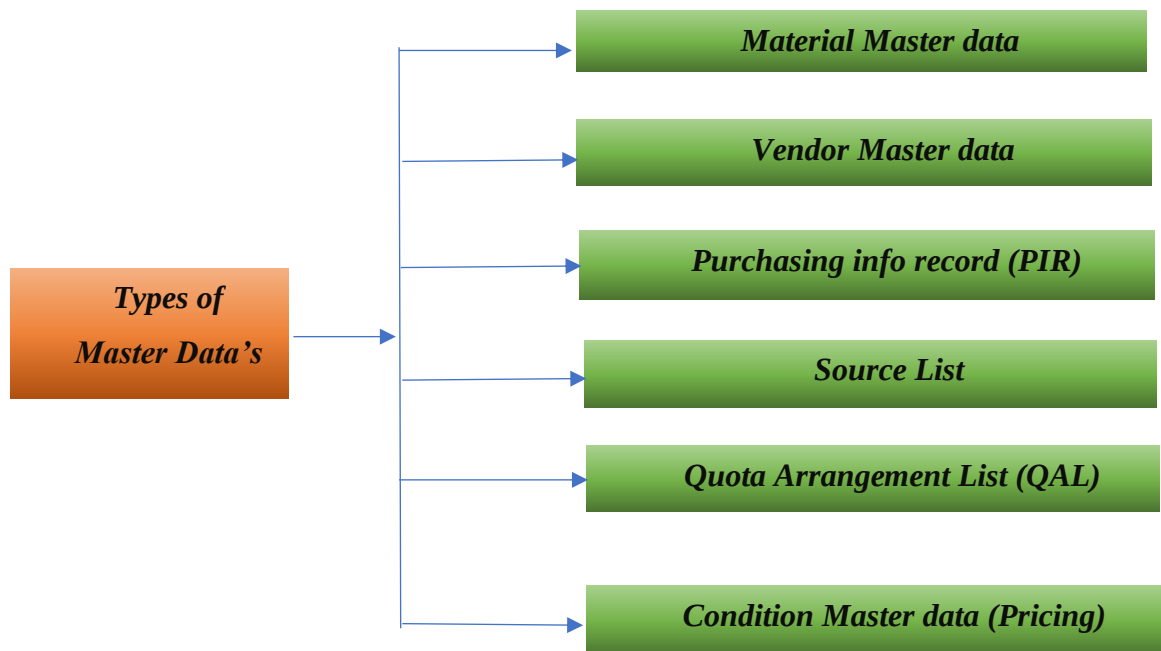
Master data:

Master data is a onetime activity created by clients in real time project in easy access screen production server and we consultants have to do backend setting of master data in IMG screen of development server – golden client.

The information of a material, vendor and customer maintained in SAP system is called master data.

Types of master Data in SAP MM

In SAP – MM we have following master data's



1. Material master data:

Material is one of the master data available in SAP MM.

Material is also called item, goods, products. The information of the material like material description, material UOM material weight etc. maintained in SAP system is called material master data.

Material is the first mater data available in MM.

Why we create material in SAP system:

If we want to buy, sell the check the material stock status throw SAP system then we have to create in SAP system.

Business requirement:

In real time project client deals with different types of material such as raw material, semifinished, finished material, spares, consumables, packing material, food material, operating material, trading material etc... we consultants have to provide solution different types of material in SAP system.

Consultant role in material master:

Under this heading we will learn how to provide solution to client requirement in material master, we will do these setting in IMG screen of golden clients – development server.

1. Define material type:

Material type differentiate the types of material in SAP system by generating material code to every material. SAP has designed below standard types.

- ROH – Raw Material
- HALB – Semi finished material
- FERT – Finished Goods
- VERP – Packing Material
- ERSA – Spares
- FOOD – Food Material
- PIPE – Pipeline material
- PIEN – Services (In ECC version)
- SERV – Services (In S4 HANA)
- HAWA – Trading Material
- NLAG – Nonstock Material

UNBW – No valuated material:

We can use above standard material types or we can customise new material types by copying standard.

Examples of material type codes customization in live projects:

- ZRM – Raw Material
- ZSFG – Semi finished good
- ZFG – Finished good

- ZSPR – Spares
- ZCON – Consumables
- PIPE – Pipeline material
- SERV – Service material
- FOOD – Food Material
- ZPAK – Packing Material
- ZNVL – Non – Valuated material
- ZTG – Trading goods
- ZOTH – Other materials
- ZSCP – Scrap Material
- ZOM – Operation Material.

Use below T. code or Path to define new material type

T. code – OMS2 – Material type.

Path:

SPRO→ IMG→ Logistics general → Material Master→ Basic Setting→ Material Type→ Define attributes of material type→ Enter→ Click on Position→ Give standard material type ROH for raw material→ Enter→ select ROH line→ click on copy as icon→ Give you own material type (should start with Z or Y with 4 CHAR), maintain material type description→ press enter, click continue and save the material type.

Note: Use same above steps to configure other material types.

Material type control material master types of control:

- It controls number range of material.
- It manages fields of materials.
- It controls views of material.

Define number range of material:

Number range will be used to identify materials in SAP system. Each material will have unique code which will be used to buy, sell or to check material stock in SAP system. This unique code will generate based on material type.

We consultant will create number ranges and it will be assigned to material type. Clients while creating materials they will select material type, based on material type selection system will generate material code.

In SAP standards for each material type we can maintain 1 internal and 1 external number range.

Internal means system will generate automatically within the range defined in SAP system and while creating material if we are entering material code manually within the range defined then it is called external number range.

Material number range length is 18 digits maximums.

Internal number range will accept only numeric and external will except alphanumeric.

T. code – MMNR

Path:

SPRO→IMG→Logistics - general → Material Master→ Basic Setting→ Material Type→ Define number ranges for each material type→ Press Enter.

Define field selection of material master:

Field selection will be used to manage field of material. Managing field is nothing but we can make field mandatory, optional, display or we can hide the fields.

In SAP standard each material type should have one field selection to manage fields. SAP has designed below standard field selections.

- ROH – Raw material field selections
- HALB – Semi finished material
- FERT – Finished material

And for remaining all other standard material type SAP is used ROH as standard field selection.

We can use above standard field selections or we can customise new field selections by copying standard. (Same like material type).

Use below T. code or path to define, manage and assign field selections to material type.

T. code – OMS9

Path:

SPRO→ IMG→ Logistic General→ Material master→ field selection→ Maintain field selection for data screens→ Click on position→ Field reference ROH raw material selection→ Enter→ Select ROH line→ Then click on copy icon→ Field reference your own field selection (4 CHAR) → Press enter→ Double clicks on your own field selection→ Click on field selection group in (contract icon) → Check the field name and close→ Manage the field and save it→ Go back→ Material type screen (OMS2) → Click on position→ Give your own field selection→ Press enter→ Double click on own field selection→ Assign field reference to change own field selection→ Press enter→ Then save it.

Views of material master:

If any client wants to use the material, then we have to inform system by selecting views. SAP has designed below standard views to create material.

Views are nothing but departments available in the company/clients.

Views of Material Master

MM	FI	CO	SD	PP	WM	QM
Basic Data 1	Accounting 1	Costing 1	Sales Org 1	MRP 1	WH 1	QM
Basic Data 2	Accounting 2	Costing 2	Sales Org 2	MRP 2	WH 2	
Purchasing			Sales general/plant data	MRP 3		
Purchase order test			Sales text	MRP 4		
Plant Data/Storage 1						
Plant Data/Storage 2						

Clients' role in Material Master:

Under this heading we will learn how to create material manually but in real time project clients will create the material and we consultants have to provide training along with user manual.

User T. code- MM01 – Create material

Creation on Material with MM views

Path:

SAP Easy Access→ SAP Manu→ Logistics→ Material Management→ Material Master→ Material→ Create (General) → Immediately→ Double clicks on Immediately→ Create Material (Initial Screen).

1. Material:

In this field we have maintain material manually if we have using external number range or else leave blank if we have using internal number range.

2. Industry Sector:

In this field we have to select client industry.

Ex: Chemical industry, construction industry etc...

Click on down ▼ mark industry.

If our client industry not available in the list then we consultant can add new industry sector by using below T. code or path.

T. code – OMS3.

Path:

SPRO→ IMG→ Logistics General→ Material Master→ Field selection→ Define industry sector and Industry – Sector – Specific Field selection→ Press Enter→ Click on new entries→ Give me Industry Sector (1 CHAR) → Industry Description our own industry→ Field reference A→ Then save it.

3. Material type:

Own material type clicks on.

4. Copy from Material:

If we want create new material by copying existing material code then in this field maintain existing material code.

Select view(s) → view select on 6 views (Basic Data 1, Basic Data 2, Purchasing, Purchase order test, Plant Data/Storage 1, Plant Data/Storage 2) → Click on default setting→ Click on Org. level→ Organizational level→ Give me your own plant and Storage location→ click on tick mark.

1. Basic data view:

Under this basic data we will maintain material general information which is applicable to entire company as display below.

➤ Material description.

- Unit of measurement.
- Material group.
- Old material number.
- Gross weight/net weight.
- Weight unit.
- Volume.
- Volume unit.
- Size/dimension.
- Packing material details.

1. **Material description:**

Material description is also called as short text in SAP. In this field we have to maintain material name and it accept 40 CHAR.

2. **Base unit:**

In this field we have to maintain unit of measurement in which material is measured.

Ex: kgs, Litres, EA, SQ. Ft. no's, Bag, MT ETC... we consultant can add new units of measurement to fulfil client requirement by using below t. code or path.

T. code – CUNI

Path:

SPRO→ IMG→ SAP NETWEAVER→ General setting→ Check units of measurement → Enter→ Click on unit of measurement then click on create icon→ Give units of measurement (3 CHAR), maintain commercial and technical same, maintain description→ ISO code double click→ Press Enter→ and save it.

3. **Material group:**

Material group is a sub division of material type. Material with same feature grouping together is called material group.

In real time project we consultant have to add new material groups. Use below or path to define new material groups.

T. code – OMSF

SPRO→ IMG→ Logistic- general→ Material master→ Setting for key fields→ Define material group→ Enter→ Click on new entries→ Give your own material group (9 CHAR), maintain description→ Press Enter→ Save.

4. Old material number:

In this field we can capture client's non-SAP material code just to identify materials in SAP. This is not mandatory field.

5. Division:

This field we will discuss in SD module.

6. X-plant material status and valid from:

If we want to block the material from purchase then in this field maintain 01 – block for procurement and maintain valid from date.

7. Gross weight/Net weight/weight unit:

In this field we will maintain weight of the material along with weight unit (Grams/KGS etc...)

8. Volume/volume unit:

Volume will be used for liquid materials.

9. Size/Dimensions:

In this field we will maintain material size like height/width etc...

10. Packing material data:

In this field we have to maintain packing material details to pack main material. Packing materials are like polytene, paper packing, wooden packing, steel packing, bag packing etc... we have to create packing material in same t. code y using packing material type. Press enter and go to purchasing view.

Purchasing view:

In this view we will maintain material purchasing information. This information belongs to individual plant.

1. Base unit:

This field will copy automatically from basic data 1.

2. Order unit:

In this field we have to maintain unit of measurement in which material is ordering to vender/market.

If order uniting base unit are different then system will ask conversion factory when we press enter.

Ex: 1 dozed = 12 each

1 ton = 1000 kgs etc...

If order unit and base unit are same then leave order unit field blank because it will copy base unit automatically.

3. Purchasing group:

Purchasing group is also called as buyer group. The group of people from purchasing department who are involved buying material from market is called purchasing groups.

In real time projects we consultants must add new purchasing groups to fulfil our client requirement.

T. code: OME4

Path:

SPRO→ IMG→ Material management→ Purchasing→ create purchasing group→ Enter→ Click on new entries→ Give your own purchasing group (3 CHAR), Maintain description, Telephone→ Press enters save.

4. Material group:

This field will copy automatically from basic data 1.

5. Auto PO:

In SAP we can create purchase order manually and automatically. If we want to create PO automatically then first step is to activate this field in material master.

6. Batch management:

If we want batch number for the material then activate this field.

7. Purchasing value key:

7.1 1st Reminder/2nd reminder/3rd reminder:

If we want to send reminder to vendor for delivering material on time then in these field maintain reminder days and with the help of ABAP consultants, we will design reminder message.

7.2 Underdel/ overdely/unlimited over-delivery tolerance:

If we want receive material quantity less then ordered quantity then maintain underdeliver tolerance in %. If we maintain underdeliver tolerance than system will allow us to receive material quantity less – then ordered quantity. If we want receive material quantity more than ordered quantity then maintain overdeliver tolerance in %. If we maintain over delivery tolerance than system will allow us to receive material quantity more then ordered quantity. If we want receive material quantity more than order quantity without any limit than activate unlimited over-delivery tolerance.

7.3 Acknowledgment required:

If we want acknowledgment copy from the vendor for given material order then activate this field.

➤ Reminders/tolerance and acknowledgment will be updated by maintaining purchasing value key. We consultants can add new purchasing value key by using below path.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Material Master→ Define purchasing value key→ Enter→ Click on new entries→ Give purchasing value key (any 4 CHAR code), maintain reminder days, tolerance, activate acknowledgment as per client requirement→ Press enter→ Save.

This purchasing value key we have select in every material if we required.

8. GR processing time:

In this field we have maintain no of days required time goods to storage location after GR (Goods receipt) and quality check.

9. Source list:

If we want to assign fixed vendor to a material then activate source list and we will maintain fixed vendor in 4th master data.

10. Manufacturing part number:

In this field we can maintain material model numbers.

11. Manufacturer:

In this field we can capture material manufacturer details and this is not mandatory field.

-----**Press enter**-----

Purchase order text view:

In this view we can maintain any material information in the form of text.

Ex: Handle with care etc...

Plant data/storage 1:

In this view we will maintain material storage information in storage location under plant. This information belongs to individual storage location available in the plant.

1. Base unit:

This field will copy automatically from basic data 1.

2. Unit of issue:

In this field we have to maintain unit of measurement in which material is issue from storage location to internal departments to external customer.

If unit of issue and base unit are different then system will ask conversion factor when we press enter.

Ex: 1 dozed = 12 each

1 ton = 1000 kgs etc...

If unit of issue and base unit are same then leave unit of issue blank because it will copy from base unit automatically.

3. Storage bin:

Storage bin the exact location in the warehouse where material is stored physically storage bin will be created in warehouse module.

4. Batch management:

This field will copy from purchasing views.

5. Shelf-life data:

Under this heading we will maintain material maximum storage period minimum storage period shelf-life expiry date (SLED) along with time unit.

Important note:

The date of production must be entered in the goods receipt.

-----Press enters-----

Plant data/storage 2:

Profit centre:

This field we will discuss in controlling module.

After maintaining all required data click on save icon to save the material and system will generate material code at left bottom corner of the screen. We will use this material code for further process like purchase, sales, stock checking etc...

Extension of material:

We will extend existing material code for following levels.

- View of extension
- Organisation levels extension

Extend material to accounting views:

Sometime in real time project client will miss the view while creating material in this case we will use extension of material to views concept. Use below process to extend the material to views.

- T. code – MM01
- Go to T. code MM01.

- Give extension material code in material field.
- Press enters for 2 times.
- Select accounting views.
- Press enters.
- Give plant.
- Press enter again and maintain below information in accounting view and save the material.

Accounting views:

1. Valuation class:

Valuation class will be used to link MM and FI modules. It is also helps us to valuates the material.

2. Price control:

We can maintain 2 types of prices in material master.

- S – Standard price.
- V – Moving average price.
- Standard price is a fixed price maintained for only finished goods.
- Moving average price is a fluctuating price maintain for all material which are buying from outsider (vender).

Extend material to new plant and storage location:

Under this heading we will learn how to extend material to new plant and storage location if we have a multiple plants and storage location using same material then follow below step to extend product to new org. levels.

- Go to T. code MM01.
- Give extension material code in material field.
- Press enters.
- Give same material code in copy from materials.
- Press enters.
- Select views.
- Give extension plant and storage location.
- Give copy plant and storage location.
- Press enters.
- System will copy data from copy plant to new plant in all the views if required edit the information then press and save the material.

The data maintained in SAP in any module will store in database (DB2, ORACLE, HANE etc..) in the form of tables and fields. We can check tables in blow T. code.

T. CODE OF MATERIAL MASTER

MM01→ Create/Extend material.
MM02→ Change/Edit material.
MM03→ Display material.
MM04→ Changes.
MM06→ Flag for Deletion/ Undeletion.
MM11→ Create schedule material.
MM12→ Change schedule material.
MM13→ Display schedule material.
MM14→ Changes schedule material.
MM16→ Flag for schedule material deletion.
MM17→ Mass maintenance.
MR21→ Change material price.
MMAM→ Change material type.
MMSC→ Extend material to storage location.
MM60→ Material List.

TABLES – SE16N

MARA→ Material general data.
MARC→ Material purchasing data (plant data)
MARD→ Material storage data.
MBEW→ Material accounting data.
MVKE→ Material sales data.
MAKT→ Material description.

-----Material Master Data completed-----

2. Vendor Master Data

Vendor is a supplier or seller or service provider. The information of the vendor like vendor name, address, purchasing info, accounting information maintained in SAP system is called vendor master data.

Vendor is an individual person or individual company or group of companies who sell material or who provides service.

We will have different types of vendors like regular vendor, fixed vendor or one time vendor.

Business requirement:

In real time project company deal with different type of vendors such domestics, foreign, service, transportation, etc... we consultants have to provide solution to differentiate types of vendors in SAP system.

Consultant role in vendor master:

Under this heading we will learn to provide solution to differentiate types of vendors in SAP system.

1. Define vendor account group:

Vendor account group differentiate the types of vendors by generating number (vendor codes) in SAP system. Each vendor will have a vendor code generates by SAP based on selection of vendor account group.

SAP has defined below standard vendor account group.

0001and LIEF- vendor

Wed consultant can use below standards or we can customise our own vendor account groups by copying standard.

Real time vendor account group examples:

- ZDMV – Domestic vendor.
- ZIV – Import vendor.
- ZSV – Service vendor.
- ZSCV – Subcontracting vendor.
- ZTRV – Transport vendor.
- ZOTV – One-time vendors etc...

Use below T. code or Path to define new vendor account groups.

T. code – OMSG

Path:

SPRO→ IMG→ Logistics general→ Business partner→ Vendor→ Control→ Define account groups and field selection (vendor)→ Enter→ Select 0001 line→ Click copy icon→ Give own vendor account group (4 CHAR) → General Data name description own account name→ Press enter→ Double clicks on your own group→ Data retent. Levels purchasing, vendor sub range relevant, plant level relevant tick→ Double click on general data→ Double click on address→ Manage vendor field→ save it.

Note: Vendor account group length is 4 CHAR.

***Vendor master will be controlled by vendor account group.

Types of control by vendor account group:

- It controls number range of vendor.
- It manages field of vendor.
- It controls partner functions.

1. Define number range of vendor:

Number range will be used to identify vendor in SAP system. Each number will have their own code and those code are generated by system based on vendor account group.

We consultants will create numbers and it is assigned to vendor account group. Client while creating vendor they will select vendor account group based on this system will generate code to vendor.

In SAP standard each vendor account group will have 1 internal or 1 external number range. Internal number range means system will generate automatically and external means clients have to enter manually while creating vendor but both should be assigned to vendor account group (mandatory).

Vendor number range length is 10 digit maximum.

T. code: OMSJ

Path:

SPRO→ IMG→ Logistics general→ Business partner→ Vendor→ Control→ Define number range for vendor master records→ Press enter→ Click on intervals→ Insert line→ Give serial number→ Then give from and to number range→ Press enter and save→ Go back → Click on number range option→ Click on position→ Give your vendor group→ Enter→ Assign number range→ Save it.

Define partner functions:

If multiple people or companies are involving in selling of a product then those multiple people or companies are called partner functions. In standard SAP we have following partner functions. We consultants have to configure to our client.

- OA – Ordering address.
- GS – Goods supplier.
- IP – Invoice presented by.
- FA – Forwarding agent.
- VN – vendor**.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Partner determination→ Partner role→ Define permissible partner roles per account group→ Click on new entries→ Give all your partner function→ And assign vendor account groups→ Press enter for every partner function→ Save it→ Go back→ Go back→ Select partner setting in vendor master record→ Define partner schemas→ Enter→ Click on new entries→ Give partners schemas (4 CHAR), maintain description (e.g.:- vendor partner) → Enter→ Select the partner schemas line→ Double click on partner functions in procedure→ Click on new entries→ Draw this box→ And save it→ Go back→ To vendor account group screen (OMSG) → Click on position→ Give your vendor account group→ Enter→ Double click on your vendor account group→ Assign your partner schemas in partner schema, purchase org field→ Save it.

Note:

We can assign same partner schemas in all the vendor account groups

EVENT	PURCHASING	FINANCE	CENTRAL (PUNCH. +FI)
CREATE VENDOR	MK01	FK01	XK01
CHANGE/EDIT	MK02	FK02	XK02
DISPLAY	MK03	FK03	XK03
CHANGES	MK04	FK04	XK04
BLOCK/UNBLOCK	MK05	FK05	XK05
FLAG FOR DELETION	MK06	FK05	XK06

Client's role in vendor master

Under this heading we will learn how to create vendor master in SAP but in Realtime projects clients purchasing dept/any end user will create vendor and we consultants will provide training along with user manual.

Use below T. Code or path to create vendor in ECC servers.

T. Code - XK01

Path:

SAP easy access screen→ SAP menu→ Logistics→ Materials management→ purchasing→ Master data→ vendor - Central→ XK01 - create→ Press enters.

Vendor initial Screen:

Vendor: In this field we have maintain vendor code manually if we are using external number range or else leave blank if we are using internal number range.

Maintain company code to add finance information

Maintain Purchasing organisation to add purchasing information.

Select vendor account group based on type of vendor we are creating.

-----Press Enter-----

Address Tab:

In this tab we will capture vendor name, address along with communication details. (This information we will get from vendor in vendor registration form (VRF)). - Enter.

Control Tab:

In this tab we will maintain vendor GST number in Tax number 3 field.

Payment Transactions Tab: In this tab we have to maintain vendor bank account details.

Contact Persons:

In this tab we will maintain contact person details from vendor side. We can capture multiple contact person details.

Accounting information Accounting:

In this tab we will maintain recon. Account. - 160000.

Payment Transactions accounting

Check Double Inv.:

If we activate this field then system check duplicate invoices.

Correspondence Accounting: -----NA-----

Purchasing Data Tab:

In this tab we will maintain purchasing information from vendor.

Order Currency:

In this field we have maintain currency in which vendor is accepting the order.

Terms of payment:

In this field we have to maintain payment terms agreed with vendor.

Examples: 50% Advance payment and 50% delivery payment

100% on delivery

100% advance payment ETC ...

We consultants can add new payment terms to fulfil clients' requirements.

Path:

SPRO → IMG→MM →Purchasing →vendor master →Define terms of payment →Enter
→Click on new entries →Give payment terms (4 CHAR), Maintain description, No. Of day,
activate account type →Press Enter →Click on save icon.

Incoterms: Incoterms stands for international commercial terms. This helps us to know the scope of delivery.

- EXW - Ex works
- FOB - free on board
- FCA - Free Carrier
- DDP - Delivered duty paid
- CIF - Cost insurance freight

Schema Group Vendor:

We will discuss this in 6th Master data.

Sales person/ Telephone:

In this field we have to capture sales person details from vendor side.

GR based Inv. Verification:

If we want to make the payment for only received goods then activate this field. If we activate this field then system will allow to make invoice payment only for goods receipt (GR) quantity.

Serv. Based Inv. Verification:

If we want to make the payment for only received services then activate this field. If we activate this field then system will allow to make invoice payment only for received services.

Acknowledgement Required:

If we activate this field then system will expect acknowledgement copy from vendor for given orders.

Automatic PO:

If we want to create PO automatically then we need to activate this field (same like material master).

Purchasing Group:

In this field we have select purchasing group.

Planned Deliv. time:

In this field we have to maintain no. of days taken by the vendor to deliver the product.

Partner Functions: In this tab system will copy partner functions from vendor account group.

After maintaining all required tabs and fields click on save icon to save the vendor and system will generate vendor code at left bottom of the screen.

Remaining T. Codes of Vendor:

- XK07 - Change vendor account group.
- XK99 - Mass maintenance.
- MKVZ - vendor's list.

***** This above vendor master concept is replaced by business partner (BP) in S4 HANA.**

Tables - SE16N

- LFA1 - Vendor general data.
- LFM1 - Vendor purchasing data.
- LFB1 - Vendor Company code data.
- LFBK - Vendor bank details.

-----**VENDOR MASTER DATA COMPLETED**-----
--

3. Purchasing info record (PIR)

Purchasing info record is also called as PIR or info record. It is used to capture purchasing information of a vendor and material combination.

Possibilities of creating PIR:

1. We can create info record manually by using T. code ME11 or we can update info record automatically by activating INFO UPDTATE check box in purchase order material data tab.
2. Info record can be created with material code or without material code but material description and material group are mandatory.

Consultant role in PIR:

1. Define number ranges:

As we are aware info record can be creating with and without material code. We consultants have to design solution to show difference between with and without material code info records.

Number range will be used to identify PIR in SAP system. Every material and vendor combination system will generate unique info record number.

When we are creating PIR with material code system will copy number from IL group if it is without material code then system will copy number range from IN group, for both IL and IN we can assign internal and external number range. (Some like material master)

T. code:

SPRO → IMG → Material management → Purchasing → Purchasing info record → Define number range → Enter → Click on maintain groups → Click on creating icon → Give group name (NR with material/NR without code,) Give FROM and TO number range for both internal and external (if required) → Press enter → Save → Go back → Click CLTR+F → Give number range (IL or IN) → Press enter → Click on IL or IN → Click on element/group → Search your number range group and double click on same group → Save.

Define field selection for info record:

As discussion vendor field selection will be used to manage field of info record.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchasing info record→ Define screen layout→ Enter→ Double clicks on ME11 line→ Double clicks on field selection group→ Manage field of info record→ Go back 2 time→ save it.

Client's role in PIR:

Under this heading we will learn how to create PIR manually and in real time project client and users will create.

Use T. code ME11 or below path to create PIR manually.

Path:

SAP easy access screen→ SAP menu→ Logistic→ material management→ purchasing→ Master data→ Info record→ ME11→ Create→ Enter.

PIR initial screen:

1. Maintain vendor.
2. Maintain material code or we can leave blank if we want to create PIR without material code.
3. Maintain purchasing organization.
4. Maintain plant.

5. Info record:

In this field we have to maintain info record number manually if we are using external number range or else leave blank internal number range.

6. Info category:

In SAP standard we can create info record in following categories.

- Standard.
- Subcontracting.
- Pipeline.
- Consignment.

As of now select standard category.

-----**PRESS ENTER**-----

PIR- General data tab:

1. **Short text** – If we are creating info record without material code then in this box, we have maintained material description manually.
2. **Material group** – In this field system will copy material group from material master and if we are creating info record without material then in this field, we have to maintain materials group manually.

NOTE: These above 2 field will copy automatically to info record screen if we are creating info record with material code.

1. 1st,2nd,3rd Rem./Exped:

In this field system will copy data from material master if required we can edit here which will copy to purchase order automatically.

2. Sales person and telephone:

In this field system will copy data from vendor master if required we can edit here which will copy to purchase order automatically.

3. Order unit and conversion:

In this field system will copy data from material master if required we can edit here which will copy to purchase order automatically.

4. Origin data:

Under this heading we will capture material origin certificate details given by vendor.

5. Available from and to:

In this field we will maintain info record valid dates.

6. Regular vendor:

If this info record vendor is a regular supplier, then activate this check box to inform system. If in case regular vendor is not available then we will reach prior vendor, maintain prior vendor. Press enters.

Purchase org. Data 1 Tab:

❖ Pl. Deliv. Time:

In this field system will copy data from vendor master if required we can edit here which will copy to purchase order automatically.

❖ Purch. Group:

This field will copy from material master

❖ Underdl., Overdl., unlimited delivery tolerance and acknowledgement required:

In this field system will copy data from material master if required we can edit here which will copy to purchase order automatically.

❖ **GR - BSD INV:**

This field will copy from vendor master.

❖ **Tax code:**

In this field we have to maintain tax code to calculate tax on purchase. If we miss to select tax code here then we can select in purchase order.

Tax codes will be created by FI consultants in T. code FTXP.

We can select VA or V0 tax codes for practise.

❖ **Condition:**

Maintain net price

❖ **Text Tab:**

In this tab system will copy text from material master purchase order text view. If required we can maintain text here also which will copy to purchase order.

After maintain all reward fields and tabs, click on save icon to save info record and seem will generate info record number at left bottom corner of the screen.

PIR T. codes:

- ME11 - Create info record.
- ME12 - Change info record.
- ME13 – Display info record.
- ME14 – Changes.
- ME15 - Flag for deletion.
- MEMASSIN - Mass maintenance.
- ME1M - List of info records by material.
- ME1L - List of info records by vendor.
- ME1W - List of info records by material group.

❖ **PIR Tables - SE16N.**

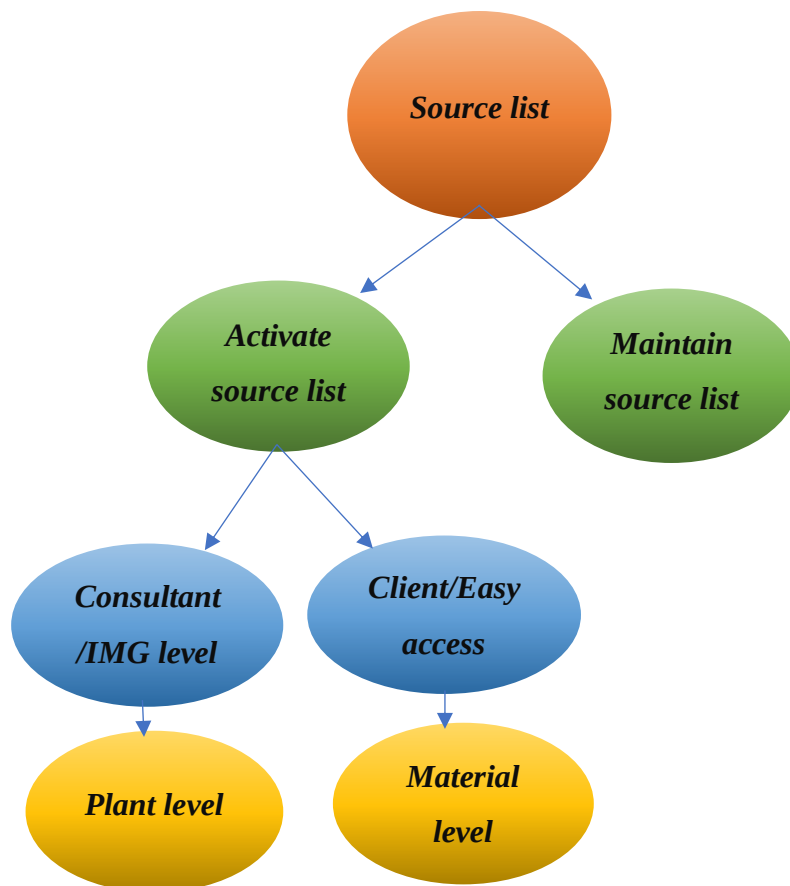
- ❖ EINA - Info record general data.
- ❖ EINE - Info record purchasing data.

-----PURCHASING INFO RECORD TOPIC COMPLETED-----

4. Source list

Source list is one of the master data available in SAP MM. it used to maintain fixed vendor to a material. Once source list is maintained then we have to buy material from source vendor only.

Source list process flow



Activate source list by consultant:

If consultants activate source, then it will be activated at plant level and it is applicable to all the material available in the plant.

T. code-OME5.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Source list→ Define source list requirement at plant level→ Enter→ Click on position→ Give your plant→ Activate source list (if required) →Save it.

Activate source list by client:

Client can activate source list only for required material in material master purchasing view.

T. code:

- MM01→ New creation material.
- MM02→ Existing material.

Client's role:

Maintain source list:

Under this heading we will assign fixed vendor to a source activate material.

T. code→ ME01.

Path:

SAP easy access screen→ SAP menu→ Logistic→ Material management→ Purchasing→ Master data→ Source list→ ME01-maintain→ Press enter→ Give source list activated material code→ Give plant→ Enter→ Maintain source valid date→ Who vendor source list→ Give own plant→ Enter→ And save it.

Source list t. code:

- ME01→ maintain/change source list.
- ME03→ Display source list.
- ME04→ Changes.
- MEOM→ List of source list by material code.

Tables-SE16N

EORD→ Source list.

5. Quota arrangement list (QAL)

This is one of the master data available in SAP MM. if our client is diving bulk quantities into ratios and ordering to multiple vendors then it is called quota arrangement list in SAP.

Quota arrangement t. code's:

- MEQ1→ maintain/change QAL.
- MEQ3→ Display QAL.
- MEQ4→ Changes.
- MEQM→ List of QAL.

Tables - SE16N

- EQUK→ Quota header table.
- EQUIP→ Quota item table.

What is Material management (SAP-MM)

SAP MM is the short form of SAP materials management module. SAP MM is one of the functional modules which supports all the aspects of material procurement i.e., purchasing, stock handling, i.e., inventory management (IM) and logistics invoice verification (LIV).

MM is the backbone of logistics.

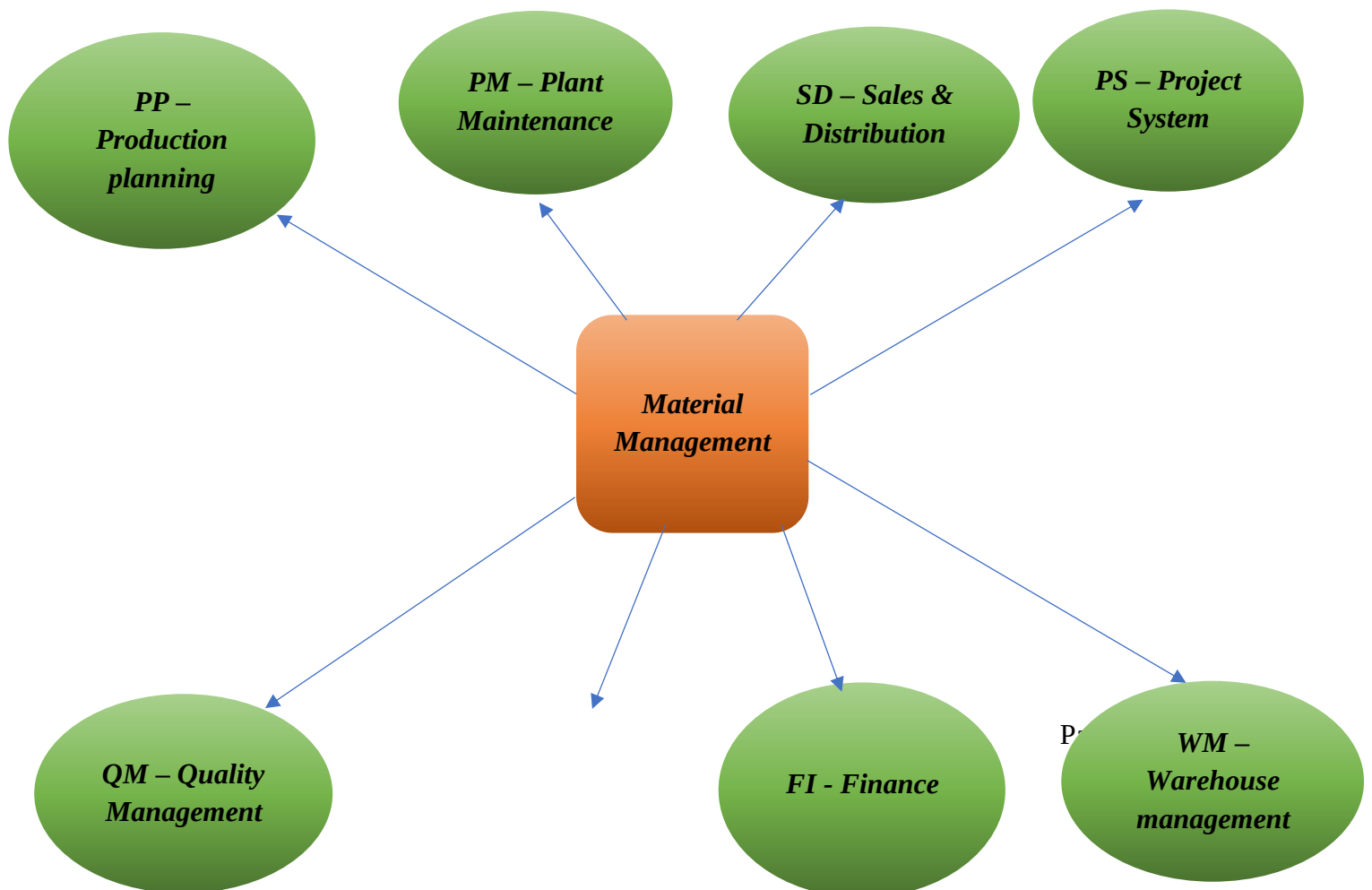
MM modules integrates with other modules such as SD, FI, CO, QM, PP, PM, PS, and WM.

MM as a process ensure no shortage of material or any gaps in supply chain of the company.

The business process or procurement process of MM is P2P cycle.

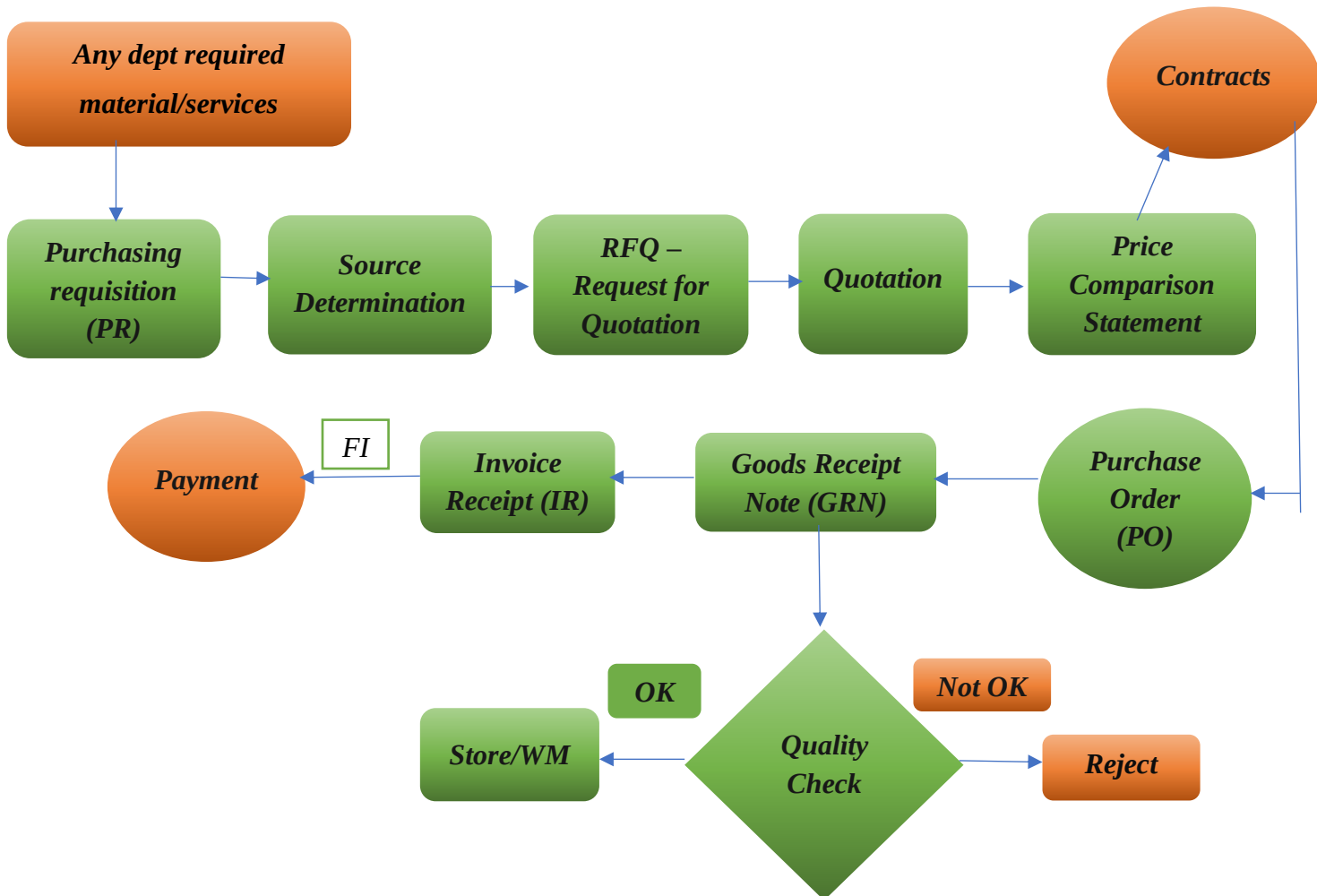
P2P stands for procurement 2 pay.

MM integrates with other Modules



CO -
Controlling

P2P – PROCUREMENT 2 PAY



P2P overview

Procurement process starts with purchase requisition and end with goods receipt and payments.

Whenever any department requires materials or services to fulfil their business needs then they will raise a request document called purchase requisition (PR). PR is an internal

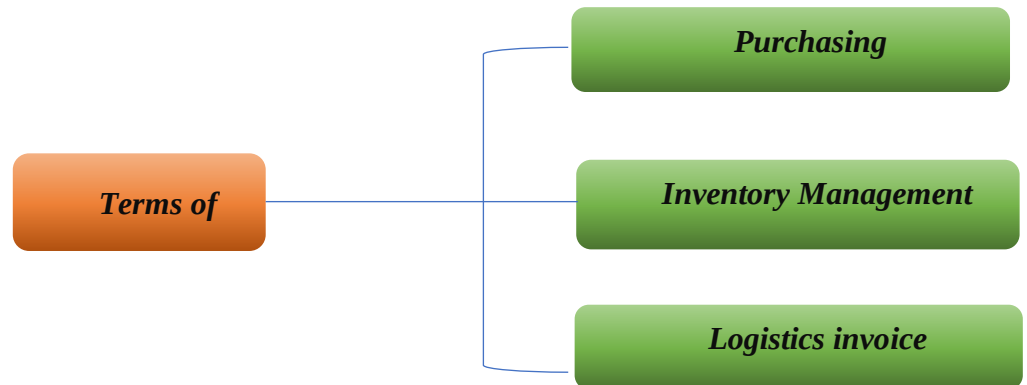
purchasing document which will be used to request materials to purchasing dept. once purchasing dept receives the request then will search vendor (supplier) in market called source determination. Once we get the source then we will raise price request to vendor called RFQ (Request for Quotation). RFQ will be created to vendor to know the price, terms and condition offered by vendor. Vendor will submit price, terms and condition in the form of a document called quotation. Once we receive quotation from multiple vendors then purchasing department will compare the prices (Quotations) to finalize the vendor. The process of comparing prices and finalizing vendor is called price comparison statement. Once vendor is finalized then the purchasing department will raise order called purchase order (PO).

P2P overview

Purchase order is an order document issued by purchasing department to the vendor for procurement of material and services. In the PO we will maintain material requirement, qty, delivery date, price and other terms and conditions. Once vendor receives the order then based on delivery date, he will deliver the goods to company and company will perform goods receipt (GR). Once company receives the material physically then goods receipt will be posted in SAP. After GR immediately our quality team will check the quality of material if everything looks ok then we will move to storage location or if it is not ok then will reject the stock. Vendor will submit invoice to company for payment called invoice receipt (IR). Company once receives the invoice then they will compare invoice with goods receipt document and purchase order (3-way match) if all are matching then they will submit the invoice to FI department for payment. Verifying invoice before making payment is called logistic invoice verification.

Terms of Material Management

MM deal with following terms



Components of MM along with T. codes

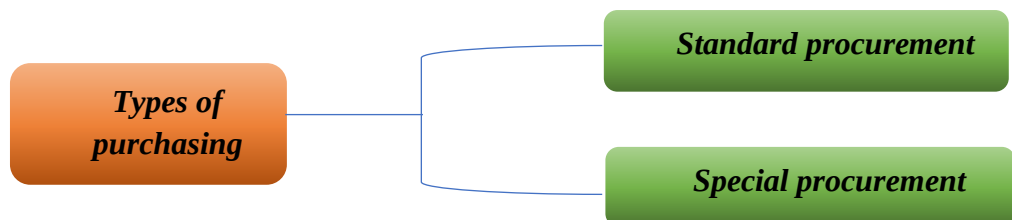
Components	T. codes
Purchase requisition (PR)	ME51N
RFQ	ME41
Quotation	ME47
Price Comparison Statement	ME49
Purchase Order	ME21N
Goods Receipt	MIGO
Invoice Receipt	MIRO

Purchasing

Purchasing is one of the components in SAP MM (MM – PUR). Purchasing concept will be used to procure materials and services from vendor/supplier to fulfil business requirements. The process of buying materials and services from vendor or dealers is called purchasing.

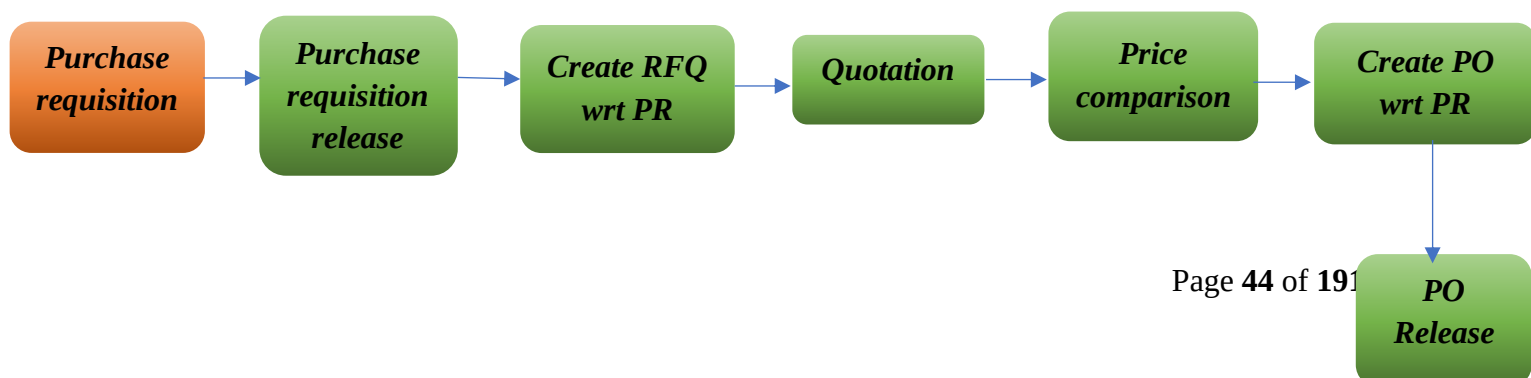
Types of purchasing:

We have below different types of purchasing in SAP MM:



Standard Procurement

Procurement process is used to procure materials and services and required by the company from external vendor. In this standard procurement company orders goods to vendor and vendor supplies the good to company (one to one relation). Standard procurement process flow is as below:

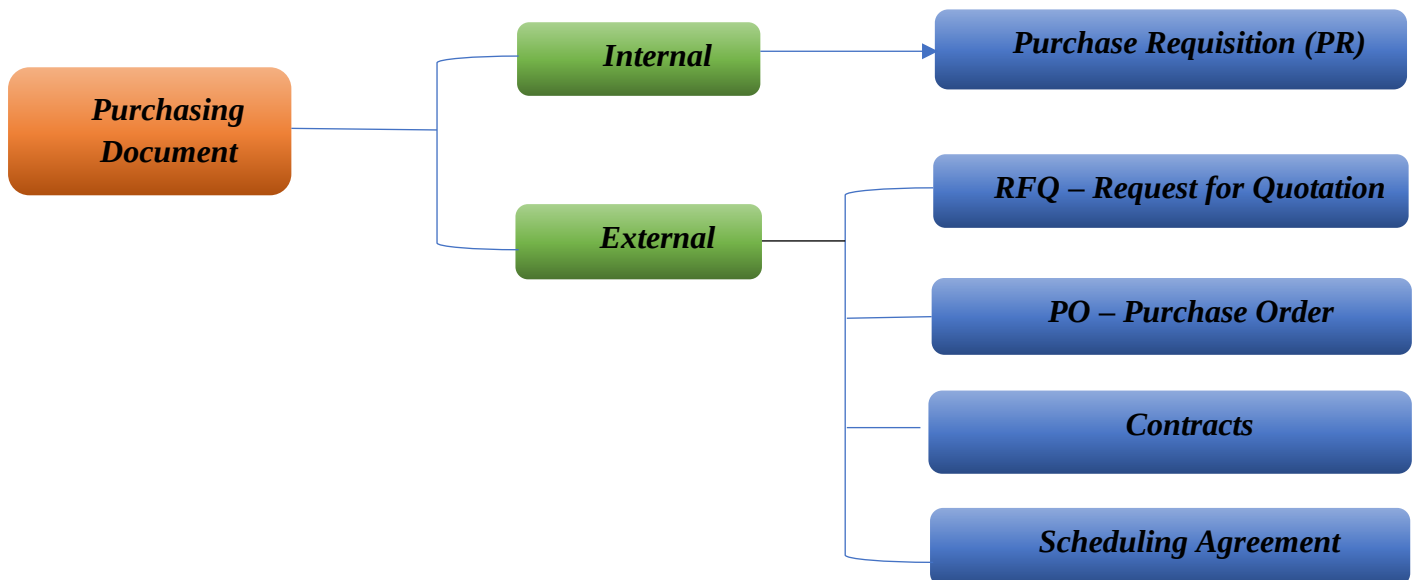




**RFO/Quotation/PC are not mandatory in SAP, project to project requirements will change.

*WRT – With reference to.

Types of purchasing document



PR – purchase Requisition

PR is the first purchasing document in SAP purchasing component.

It is an internal document.

If any department in the company required material or services then those departments will raise request to purchasing department in the form of a document called PR – purchase requisition to procure the same in market.

PR can be standard, Third party, Subcontracting, Stock transfer order (STO), Consignment and services.

PR – Business Requirements

Project Requirement:

In real time project client's internal departments will create purchase requisitions. We consultants have to provide solution to differentiate department wise PRs in SAP system. Example HR department, Sales, Purchasing, Stores, Design, Manufacturing, Maintenances Et...

PR – consultant role

Under this heading we will learn consultant's role in PR. They are.

- Define document type.
- Define number range to PR document type.
- Field selection of PR.
- Item categories.

Define document types:

Document types differentiates types of PR's SAP by generating PR number to each PR. SAP as defined standard document type.

NB – Standard purchase Requisition.

We consultants may use standard document type **NB** or we can customize types by coping standard. In real time project we will customize below document types and these are just examples. Project to project requirements may get change but the process is same.

PR – consultant role

We consultants can use standard document type **NB** or we can customize new document types by copying standard. In real time project we will use below document types and these are just examples:

- ❖ ZSPR – Sales PR's.
- ❖ ZHPR – HR PR's.
- ❖ ZMFP – Manufacture PR's.
- ❖ ZSTR – Store PR's.
- ❖ ZMNT – Maintenance PR's.
- ❖ ZPPR – Purchasing PR's Etc.

Path:

SPRO→ IMG→ Material Management→ Purchasing→ Purchase Requisition→ Define document type→ Enter→ Click on position→ Select NB line→ Click on copy icon→ Give your own document type (4 CHAR) → Maintain description→ Press enter→ Then click copy all→ Continue and save it.

PR – Consultant Role

Document type controls PR**

Types of controls by document type:

- ❖ It controls number range of PR.
- ❖ It manages fields of PR.
- ❖ It allows item category.

Define number range for PR:

Number range will be used to identify PRs in SAP system. In SAP standard we can create and assign internal and external number range for each document type. (Same like material type).

PR number range length is 10 digits.

T. code – OMH7.

Path:

SPRO→ IMG→ Material Management→ Purchasing→ Purchase Requisition→ Define number range→ Enter→ Click on change intervals→ Insert 2 lines→ Give serial number→

Maintain from and to number range→ For both internal and external→ Press enter→ Save it→ Go back to PR document type screen→ Enter→ Click on position→ Give your own document type→ Enter→ Assign serial number range→ press enter and save it.

Define field selection of PR:

As we are aware field selection will be used to manage fields of PR. In SAP standard each document type will have one field selection.

SAP has defined standard field selection **NBB** for PR. We can use standard field selection or we can customize our own based on document type. (If we customize document type then customize field selection also).

Document type+ number range (inter/external) +field selection+ item category = consultant role in purchasing.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchasing requisition→ Define screen layout at document level→ Enter→ Click on position→ Give standard field selection (NBB) → Select NBB line→ Click on copy icon→ Give your own field selection (4 CHAR) → Maintain description→ Then press enter→ Double click on your own field selection→ Double click on field selection group→ Manage field of PR→ Then go back 2 times→ And save it→ Then go back define document types screen→ Enter→ Click on position→ Give your own PR document type→ Enter→ Assign your own field selection→ save it.

Item category:

Item categories will be controlled by document type. Item category differentiates types of procurement in SAP system. SAP has designed below standard item categories for purchase and we consultant don't have any option of adding new item categories.

Types of item category:

- ❖ Blank – Standard.
- ❖ S – Third Party.
- ❖ L – Subcontracting.
- ❖ U – stock transfer order (STO).
- ❖ K – Consignment.
- ❖ D – Service.

Structure of purchasing documents:

All the purchasing documents in SAP divides into multiple structures. They are:

- ❖ Header data.
- ❖ Item overview.
- ❖ Item data.

The data maintained at header level is applicable to whole document.

In item overview we have to maintain list of materials to be procured.

The data maintained at item level is applicable only to particular lines items. Material to material item data will be different, for every material we need to maintain item data.

Client role in purchase requisition:

Under this heading we will learn how to create purchase requisition manually. But in real time project client internal department will create the PR we consultant train them.

Creation purchase requisition manually.

T. code – ME51N.

Path:

SAP access screen→ SAP Manu→ Logistics→ Material Management→ Purchasing→ Purchasing Requisition→ ME51N→ Create→ Enter→ Select Document type.

PR item overview:

Under this heading we will add all required materials, quantity along with below information.

1. Item:

In this field system will generate line-item serial no's automatically (ex: 10,20,30...)

2. A- Account assignment category:

If we are procuring materials for direct consumption (use) then we have to maintain account assignment category. Please find below are few standard account assignment categories.

- ✓ Assets.
- ✓ Sales order.
- ✓ Third party.
- ✓ E- customer individual requirement.
- ✓ F- Order.
- ✓ K- Cost center.
- ✓ N- Network.
- ✓ P- Project.
- ✓ Q- Project, makes to order.

As of now leave this field blank because we are creating stock PR.

3. I- Item category: Maintain item category (blank standard).

4. Maintain required material code.

5. Short text: In this field system will copy material description from material master.

6. Maintain PR Quantity.

7. UOM- Unit of measurement:

In this field system will copy UOM from material master basic data.

8. Delivery Date:

In this field we have to maintain material required date, if we leave this field blank then then system will take PR creation data as delivery date.

9. Material group:

In this field system will copy material group from material master.

10. Maintain plant and storage location.

11. purchasing group:

In this field system will copy purchase group from material master.

PR header data:

The data maintain in header level is applicable to entire purchasing requisition.

Tab 1- Text tab:

In this field we can maintain any PR information (or) instruction in the form of text.

PR item data:

Item data will be maintained for every material line available in PR item overview.

1. Tab 1: - Material data:

In this tab we can check material details.

2. Tab 2: - Quantities & Dates:

In quantity, quantity ordered, open quantity.

In this quantities field we will maintain required quantity or it will copy from item overview.

If we convert PR into PO then PO quantity will get update in quantity ordered field automatically and remaining quantity of PR after order will get update in open quantity field.

Closed:

If we want to close the purchase requisition line item from purchase then activate this filed manually.

3. Tab- 3: -Valuation Tab:

1. valuation price:

In this filed system will copy maintain price from material master.

2. Goods receipt: This field is in activate by default copied from item category. If this filed is in active then system will export goods receipt enter in SAP.

3. Invoice receipt:

If this field is in active then system will allow to enter invoice receipts.

4. Tab- 4: - Source of supply:

Assign source of supply:

If we click on this filed ten system will copy vendors in following sequence.

QAL→ Source list→ Contracts→ PIR.

5. Tab-5: - Status Tab:

In this tab we can check PR current status.

6. Tab- 6: - Contact person:

In this tab we can check PR created details.

In this tab system will copy text from material master. If required we can add new text here which will copy to purchase order automatically.

7. Tab-7: -Delivery address:

In this tab system will copy delivery address from plant address and if required we can exit field for delivering physical material by vendor.

After maintaining all required tabs and fields click a check icon. If system displays no message issued during check, then click on save icon to save the PR and system will display PR number in left bottom of screen.

T. code of tables of purchase requisition:

- ❖ ME51N→ Create Purchase Requisition.
- ❖ ME52N→ Change Purchase Requisition.
- ❖ ME53N→ Display Purchase Requisition.
- ❖ ME54N→ Release Purchase Requisition. (individual)
- ❖ ME55→ Release PR-collectively.
- ❖ ME5A→ List of PR's.
- ❖ ME5K→ List of account assignment PR's.
- ❖ ME5J→ Project PR's.
- ❖ MEMASSPR→ Mass maintenance.

Tables-SE16N:

- ❖ EBAN→ General Purchase Requisition.

- ❖ EBKN→ Account assignment PR's.

-----PURCHASE REQUISITION TAPIC COMPLETED-----

RFQ – Request for Quotation

- RFQ is one of the external purchasing documents.
- RFQ is a form of invitation sent to vendor thru email, fax or telephone to submit the price, term & condition for required material.
- RFQ contains details of a required materials, quantity, date of delivery, place of delivery and quotation submission last date.
- RFQ is used to know the price, term & condition offered by the vendor for a material.

RFQ- Business requirements:

In real time project clients requires quotations to different types of vendors such as domestics have provided solution to differentiate types of RFQ's in SAP system.

RFQ- Consultant role:

Define document types:

As discussed in PR document types differentiates types of RFQ's in SAP by generating RFQ number to each RFQ. SAP has defined standard document type.

AN-RFQ

We consultants can use standard document type **AN** or we can customize new document types by coping standard. In real time project we will use below document types and these are just examples.

- ❖ ZDRQ- Domestic RFQ.
- ❖ ZIR- Import RFQ.
- ❖ ZSR- service RFQ etc.

Path:

SPRO→ IMG→ Material management→ Purchasing→ RFQ/Quotation→ Define document types→ Enter→ Select AN line→ Click on copy as icon→ Give your own document types (4 CHAR) → Maintain description→ Press enter→ Click copy all→ Click on continue→ Save it.

RFQ- Consultant role:

Document type controls RFQ**

Types of controls by document type:

- It controls number range of RFQ.
- It manages field of RFQ.
- It allows item category.

Define RFQ number ranges:

As discussed in PR, each document type will have 1 internal and 1 external number. Number range will be used to identify RFQ's in SAP system.

T. code – OMH6.

Path:

SPRO→ IMG→ Material management→ Purchasing→ RFQ/Quotations→ Define number ranges→ Enter→ Click on change intervals→ Insert 2 line→ Give serial number→ Give from and to number range→ And both internal and external→ In bracket (10 CHAR) → Press enters and save it→ Go back RFQ document type screen→ Enter→ Click on position→ Give your own RFQ document type→ Assign number range serial number→ Press enter→ Save it.

Define filed selection for RFQ:

As we are aware field selection will be used to manage field of RFQ. SAP has defined standard filed selection ANA for RFQ standard document type. We can use standard or we can customize our own filed selections same like document types.

Path:

SPRO→ IMG→ Material management→ Purchasing→ RFQ/Quotation→ Define screen layout at document level→ Enter→ Click on position→ Give standard field selection ANA→ Select ANA line→ Click on copy as icon→ Give your own filed selection→ Maintain description→ Enter→ Double click on own field selection→ Double click on field selection group→ Manage the filed RFQ→ Go back 2 time→ Save it→ Go back to RFQ document type screen→ Click on position→ Give your document type→ Enter→ Assign your own field selection→ Save it.

Client role in RFQ:

Under this heading we will learn how to create RFQ's and in real time project clients purchasing department will create RFQ's and we MM consultants have to provide training.

Possibilities of creating RFQ's:

We can create RFQ's following ways:

- ❖ RFQ WRT PR
- ❖ RFQ WRT outline agreement (contracts).
- ❖ RFQ without any reference.

Creating of RFQ WRT PR:

If we create WRT PR then system will copy data from PR to RFQ screen automatically.

We can create multiple RFQ's to multiple vendor WRT one PR.

T. code- ME41.

Path:

SAP easy access→ SAP Menu→ Logistics→ Material Management→ Purchasing→ RFQ/Quotation→ RFQ→ ME41→ Create→ Enter.

Create RFQ – Initial screen:

1. RFQ type:

In this filed we have to select RFQ document type.

2. Language key: EN (English).

3. RFQ date:

In this filed we have to maintain RFQ request date.

4. Quotation Deadline:

In this filed we have to maintain last date of quotation submission.

5. RFQ:

In this filed we have to maintain RFQ number manually if we are using external number range or else leave this filed blank if we are using internal number range.

6. maintain purchasing org and purchasing group.

7. Click on reference to PR icon.

8. maintain PR number- press enter.

RFQ- Header data:

1. Collective no:

In this filed we can maintain any serial no. just to identify sequence of RFQ's created.

-----press enter-----

1. system will display PR line item.

2. Select PR line item then click on Adopt+ details icon.

3. maintain delivery date which should be greater than quotation deadline date- press enter.

4. Now we can see data copied from PR to RFQ screen.
5. Click on vendor address icon.
6. Give vendor code to whom we are requesting quotations.
7. Press enter and save the RFQ.
8. System will generate RFQ number at left bottom of the screen.

After creating RFQ's client will take the RFQ printout from SAP they will share with vendor via emails.

-----**REQUEST FOR QUOTATION TAPIC COMPLETED**-----
--

Quotation

After requesting quotations to vendor by using RFQ document. In reply to RFQ vendor will submit quotations thru emails, fax, courier, or directly vendor will submit in company's purchasing department.

Quotation contains price, term & conditions of the material offering by the vendor to company.

As below t. code or path to maintain quotation in SAP.

T. code- ME47

Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material Management→ Purchasing→ RFQ/Quotation→ Quotation→ ME47 maintain→ Enter→ Give RFQ no→ Select RFQ line item→ Click on item condition as icon→ Maintain gross price, discount, freight charges→ Go back→ And save it.

- ME47- Maintain/ change quotations.
- ME48- Display quotation.

Price comparison statement:

After maintain quotation of vendor we will use price comparison concept to compare the quotation.

Price comparison will do by purchasing dept in the company.

T. code- ME49

Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material management→ Purchasing→ RFQ/Quotation→ ME49 price comparison→ Enter→ Give your PO no→ And maintain RFQ no Quotation field→ Activate Include discount, include delivery cost→ And execute→ Check the price comparison statement→ And finalised the vendor.

Note: In SAP system vendor will be finalised based on price not based on quantity and discipline of the vendor.

T. code & Tables of RFQ

ME41→ Create RFQ.

ME42→ Change RFQ.

ME43→ Display RFQ.

ME45→ Release RFQ.

ME4M→ List of RFQ's by material.

ME4L→ List of RFQ's by vendor.

ME4C→ List of RFQ's by material group.

ME4S→ List of RFQ's by collective number.

ME4N→ List of RFQ's.

ME9A→ RFQ print.

Tables -SE16N

EKKO→ Purchasing header data.

EKPO→ Purchasing item data.

***Above table will store all the external purchasing documents data including RFQ. (PO, Contract, Scheduling agreement and RFQ).*

-----QUOTATION TAPIC COMPLETED-----

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PO- Purchasing Order

Purchase order is also called as order document. If we want to procure material and services then will place order to vendor in the form document called purchase order.

Possibilities of creating PO:

In standard SAP we can create po in following ways:

- ❖ PO WRT PR.
- ❖ PO WRT RFQ.
- ❖ PO WRT contracts.
- ❖ PO WRT PO.
- ❖ Manual creation of PO. (W/O any reference)

Types of purchasing order in SAP:

We have below different types of purchasing order in SAP standards.

- ❖ Standard PO
- ❖ Third party PO
- ❖ Subcontracting PO
- ❖ Consignment PO
- ❖ Stock transfer order (STO)
- ❖ Service order.

Business requirement:

In the real time project client creates different types of purchase order such as standard PO, domestics PO, import PO, third party PO, subcontracting po, service PO, consignment PO, STO etc. we consultants have to provide solution to show PO difference in SAP system.

Consultant role in purchase order:

1. Define document type:

As discussed in PR and RFQ document types will differentiate types of purchasing in SAP by generating different number to each purchase order.

SAP as define standard document type **NB** for PO; we consultants can use standard document type or we can customize our own by coping standard.

Real time project document type examples:

- ❖ ZDMP- Domestic purchase.
- ❖ ZIMP- Import purchase.
- ❖ ZSC- Subcontracting- ZSUB.
- ❖ ZTHP- Third party purchase.
- ❖ ZSER- service purchase.
- ❖ ZCON- Consignment PO.
- ❖ ZCAP- Capital purchase.
- ❖ ZSPR- Spares purchase.
- ❖ ZCPO- Consumables PO etc...

****Company to company PO types may be different.**

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchasing order→ Define document type→ Enter→ Click on position→ Give NB document→ Select NB line→ Click on copy icon→ Give your document type (4 CHAR) → Maintain description→ Press enter→ Click on copy all→ Then continue→ Save it.

***Document type control purchase order:**

Types of controls by PO document type:

- It controls number range of PO.
- It controls field selection of PO.
- It allows item category.

Define number range of PO:

As discussed in PR and RFQ each document type will have 1 internal and 1 external number range. Number range will be used to identify purchase orders in SAP.

T. code- OMH6

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchase order→ Define number ranges→ Enter→ Click on change intervals→ Insert 2 line→ Then give serial number→ Give from and to number ranges→ And both internal and external (10 CHAR) → Press

enter→ Save it→ Go back to define document types screen→ Enter→ Click on position→ Give your own document type→ Enter→ Assign your number range serial number→ Press enter→ save.

Define filed selection of PO:

As we are aware field selection will be used to manage fields of PO. SAP as defined standard field selection **NBF** for purchase order. We can use **NBF** or we can customize new field selections by copying standard.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchasing order→ Define screen layout at document level→ Enter→ Click on position→ Give standard field selection→ Select NBF line→ Click on copy as icon→ Give your filed selection (4 CHAR) → Maintain description→ Press enter→ Double click on your own filed selection→ Double click on filed selection group→ Manage the field of purchase order→ Go back 2 time→ Save it→ Go back to PO document type screen→ Enter→ Click on position→ Give your own document type→ Enter→ Assign your own field selection→ Save it.

Client role in purchase order:

Under this heading we will learn how to create purchase order but in real time project purchasing department end user will create the order and we consultants have to providing training.

Creation of purchase order (PO) WRT PR:

If we create order WRT PR then system will copy data from PR to PO screen and we can edit the data if required save the PO.

T. code- ME21N- create the purchase order:

Steps:

- Go to T. code- ME21N.
- Select document type based on type of purchase. (NB or new document type).
- Maintain vendor code to whom we are placing the order.
- Maintain purchasing org, Purchase group, and company code.

-----Press enters-----

PO item overview:

- Drag item overview screen to right and maintain PR number→ Press enters.
- System will copy data from PR to PO screen.
- Maintain the required filed like purchase price, delivery date and other required fields.

- Click on check icon and save the PO.

Notes:

We can create multiple purchase orders WRT one PR till PR quantity becomes 0.

We can create 1 PO WRT to multiple PR's if we are buying all PR products with same vendor.

Creation of purchase order W/O any reference:

Under this heading we will learn how to create purchase order W/O any reference document.

When we are not using any reference document then we have to maintain all the information in PO manually.

T. code- ME21N

Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material management→ Purchasing→ Purchase order→ Create ME21N→ Enter.

- Select document type based on type of purchase. (NB or new document type).
- Maintain vendor code to whom we are placing the order.
- Maintain purchasing org, Purchase group, and company code.

-----Press enters-----

PO item overview:

Under this item overview we will maintain list of required materials, qty, delivery, date along with below detailed information.

1. Item:

In this filed system will generate line-item serial no's automatically (ex: 10, 20, 30...).

2. A- Account assignment category:

If we are procuring materials for direct consumption (use) then we have to maintain account assignment category. Please find below are few standard account assignment categories.

- ✓ A- Assets.
- ✓ C- Sales order.
- ✓ 1- Third party.
- ✓ E- customer individual requirement.
- ✓ F- Order.
- ✓ K- Cost center.
- ✓ N- Network.
- ✓ P- Project.

✓ Q- Project, makes to order.

3. I- Item category:

Maintain item category (Blank – standard).

4. Maintain required material code.

5. Material description:

In this filed system will copy material description from material master.

6. Maintain PO quantity.

7. OUN- Order unit:

In this filed system will copy order unit from PIR or from material master.

8. Delivery date:

In this filed we have to maintain material required date. If we leave this filed blank then system will take PO creation date as delivery date.

9. Maintain net price.

10. Currency will copy from PO header data – invoice tab.

11. Material group:

In this filed system will copy material group from material master.

12. Maintain plant and storage location.

13. Info record:

In this filed system will copy info record number automatically if we maintain info record for the PO vendor and material. System will copy data from info record to PO screen if info record is available.

14. R – Return item:

If we want to return materials back to vendor by creating return purchase order then activate this field. If we activate this field then standard purchase order will become return purchase order.

If we are buying materials then leave this filed blank.

15. F – Free items:

If we are getting any materials free from vendor then activate this filed.

16. purchase requisition and requisition item:

In this field we will give PR number if we want to create PO WRT PR.

17. Outline arrangement and contracts item:

In this field we will maintain contacts number and contracts item if we want to create PO WRT contracts.

18. RFQ and RFQ item:

In this field we will maintain RFQ and item number if we want to create PO WRT RFQ.

19. Purchasing and item:

In this field we will maintain PO number if we want to create PO WRT PO.

PO Header data:

The data maintained at PO header level is applicable to entire purchase order.

1. Delivery/Invoice tab:

Payment term/ incoterms/ currency:

In this field data will copy from vendor master and if required we can edit here.

Exchange rate:

If orders currency and company code currency are different then system will copy currency exchange rate from OB08 T. code.

In this OB08 T. code FI consultants will maintain currency exchange rate.

Ex. 1 USD = 80 INR

2. Condition tab:

In this tab we can check total PO value.

3. Texts tab:

In this tab we can maintain any information in the form of text by selecting text types (left side of the tab).

4. Address tab:

In this tab system will copy address from vendor master.

5. Communication tab:

Salesperson/telephone:

In this field system will copy data from vendor master.

6. Partners tab:

In this tab we can maintain partners and their vendor codes if multiple people are involving in purchase.

7. Additional data tab:

In this tab we can maintain purchase order valid dates.

8. Org. data tab:

In this tab we have to maintain org levels like purchasing and company code.

9. Status tab:

In this tab we can check current status of PO like quantity ordered, delivered or invoiced quantity.

10. Payment processing tab:

DP category:

If we want to make down payment (advance payment) to vendor then in this field select option M – Mandatory down payment→ maintain down payment percentage, maintain due date.

PO item data:

As we are aware the data maintained in item tabs are applicable in individual line items. Every line-item data is mandatory.

1. Material data :

In this tab we can check material information.

2. Quantities/ Weight tab:

In this tab we can check ordering quantity copied from item overview.

3. Delivery schedule tab:

In this tab we can maintain multiple delivery dates for a single material.

4. Delivery:

Overdeliver. Tol. /Underdeliver. Tol. / Unlimited overdelivers tolerance/1st, 2nd, 3rd rem./

Expend:

Will copy from PIR or from material master.

Delivery completion:

If we don't want to receive material quantity from vendor then activate this field manually.

If we activate this field then system will treat delivery is completed.

5. Invoice tab:

Invoice receipt:

In we activate this field then system will expect to enter invoice document in SAP system.

Final invoice:

If we activate this field then system will not expect any further invoices from the vendor.

After activating this field, we can post only one invoice.

Against PO and that invoice will become final invoice.

GR- BSD INV:

This field will copy from PIR or else from vendor master.

Tax code:

In this field tax code will copy from PIR or we can maintain here to calculate taxes on purchase.

6. Conditions tab:

In this tab we can maintain material purchase prices.

7. Texts tab:

In this tab system will copy text from material master, PIR, purchase requisition etc... if required we can maintain text here in PO by selecting text types (left side of the tab).

8. Delivery address tab:

In this field system will copy delivery address from plant address. If required we can edit the address. In this address vendor will supply the materials.

9. Confirmation tab:

Acknowledgment required:

This field will copy from PIR or else from material master.

10. Condition control tab:**Print price:**

If we don't want to print material price on PO print out then deactivates this field.

After maintaining all required fields and tabs we will design output determination before saving the PO.

11. Retail tab.

PO output determination:

- Go to message icon.
- Maintain output type- NEU→ press enter.
- Select output line→ click on communication method icon.
- Maintain logical destination – LP01 (default printer).
- Activate print immediately.
- Maintain no of message 1 copy.
- Go back→ go back.
- Click on print preview icon and check the PO printout.
- Click on check icon and save the purchase order.

T. codes & Tables of purchase Order

- ❖ *ME21N- Create Purchase Order.*
- ❖ *ME22N- Change Purchase Order.*
- ❖ *ME23N- Display Purchase Order.*
- ❖ *ME28- Release Purchase Order collectively.*
- ❖ *ME29N- Release Purchase Order individually.*
- ❖ *ME9F- PO print.*
- ❖ *MEMASSPO- Mass maintenance.*
- ❖ *ME2L- List of Purchase Order by vendor.*
- ❖ *ME2M- List of Purchase Order by material.*
- ❖ *ME2K- List of account assignment PO's.*
- ❖ *ME2J- List of project Purchase Order.*
- ❖ *ME2C- List of Purchase Order by material group.*
- ❖ *ME2N- List of Purchase Order.*
- ❖ *ME1P- Purchase Order History (POH).*

Tables- SE16N

- ❖ *EKKO- Purchasing Header data.*
- ❖ *EKPO- Purchasing item data.*
- ❖ *EKBE- Purchasing history.*
- ❖ *EKKN- Account assignment purchasing.*
- ❖ *EKET- Purchasing schedule lines (Delivery dates).*
- ❖ *EKPA- Purchasing partner roles.*

-----PURCHASE ORDER TAPIC COMPLETED-----

Inventory management

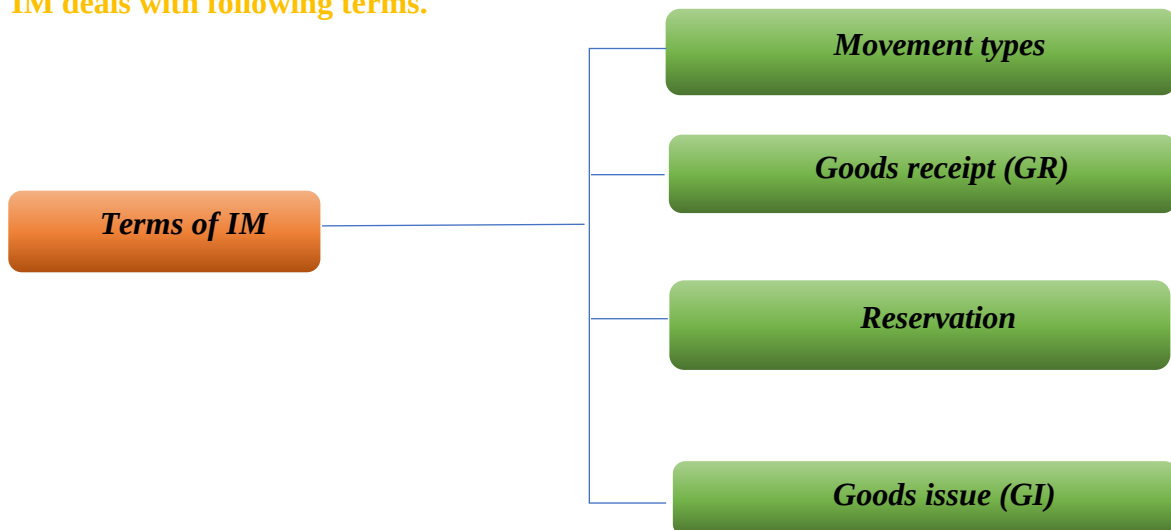
Inventory management is a sub module in SAP MM it's component of SAP MM-IM. Procurement process starts with purchase requisition and ends with goods receipt from vendor. After receiving goods from vendor, they need to be stored under companies' location (storage location) so that we can use whenever we required. Inventory management deals with placing and handling stock received from vendor.

Points to be noted in inventory:

- IM deal with managing stock either quantity or value basis.
- Goods movement creates a document which updates quantity and value in the inventory books called material document.
- Material document number range length 10 digits.

Terms of inventory management:

IM deals with following terms.



Movement types:

- ✓ Each and every goods movement will be captured with a specific digit code called movement types.
- ✓ Movement types help to identify goods movement direction.
- ✓ SAP defines below standard movement types which are used in real time projects also.

Movement types- OMJJ

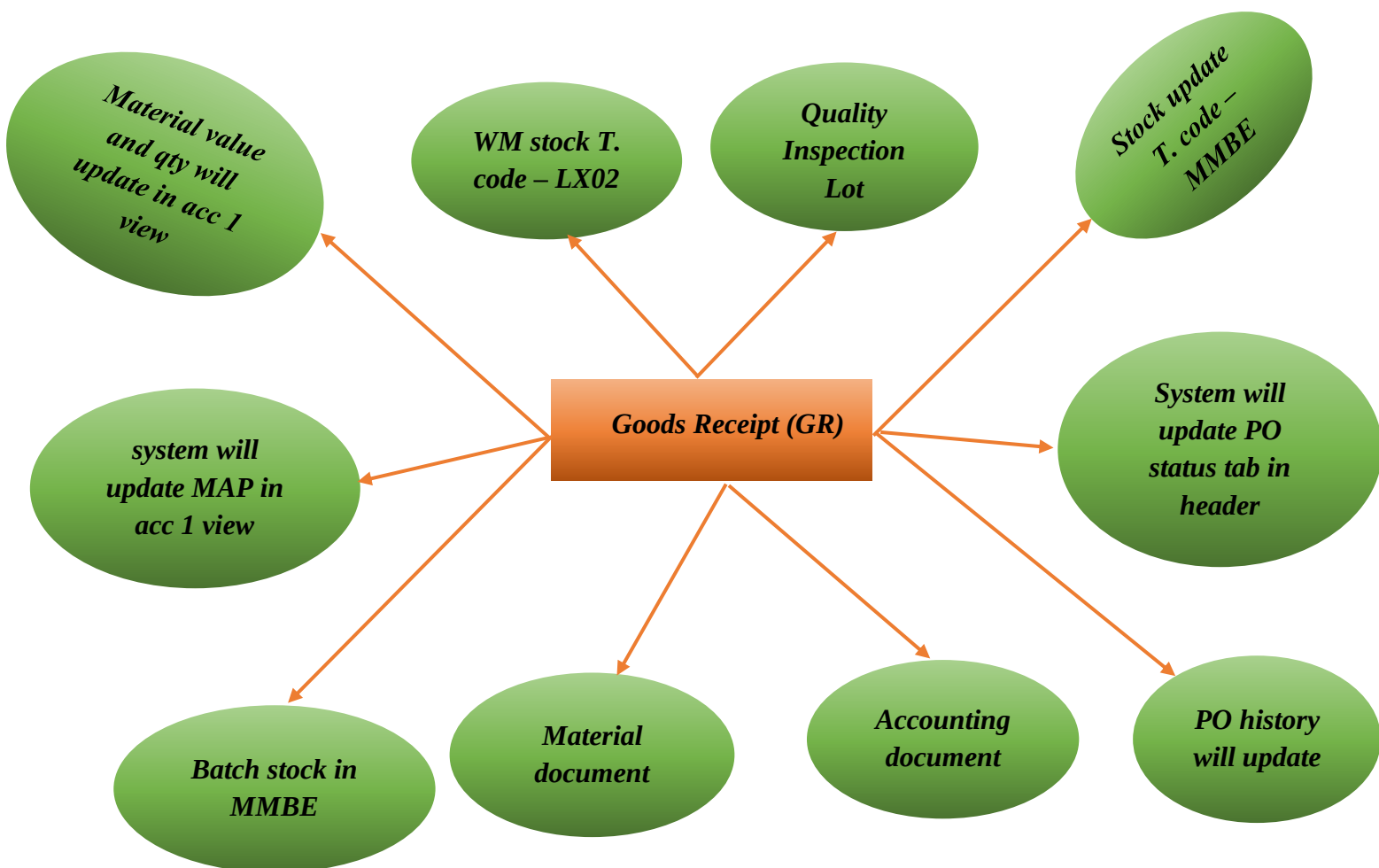
- ❖ 101- Goods receipt WRT PO.
- ❖ 102- Cancel GR.
- ❖ 103- GR WRT PO to blocked stock.
- ❖ 105- Release from GR blocked stock.
- ❖ 122- Return delivery to vendor.
- ❖ 124- Return delivery to vendor from GR blocked.
- ❖ 161- Goods return to vendor WRT return PO.
- ❖ 201- Goods issue to cost center.
- ❖ 202- GI cancelation.
- ❖ 221- GI to project.
- ❖ 251- Goods issue to sale (W/O sales order).
- ❖ 261- Goods issue for an order.
- ❖ 281- Goods issue to network.
- ❖ 291- Goods issue to any account assignment.
- ❖ 301- Plant to plant stock transfer 1 steps.
- ❖ 303- Plant 2 plant stock transfer 2 steps (stock removal).
- ❖ 305- Plant 2 plant stock transfer 2 steps (stock put away).
- ❖ 309- Transfer posting from material to material.
- ❖ 311- Storage 2 Storage stock transfer 1 steps.
- ❖ 313- Storage 2 Storage stock transfer 1 steps (stock removal).
- ❖ 315- Storage 2 Storage stock transfer 1 steps (stock put away).
- ❖ 321- Transfer of stock from inspection to unrestricted.
- ❖ 351- Goods issue to STO (without shipping).
- ❖ 411- Transfer of special stock own stock.
- ❖ 411 K- Transfer of consignment to own stock.
- ❖ 451- Return from customer (without shipping).
- ❖ 501- Goods receipt W/O Purchase order.
- ❖ 502- Cancel GR.
- ❖ 541- Stock issue to subcontracting.
- ❖ 543- Stock consumption of subcontracting.
- ❖ 545- Goods receipt of by- product from subcontracting.
- ❖ 551- Scrapping from unrestricted use.
- ❖ 553- Scrapping from inspection stock.
- ❖ 555- scrapping from blocked stock.
- ❖ 561- Initial stock upload.
- ❖ 581- Goods receipt of a byproduct for network.
- ❖ 601- Goods issue to delivery.
- ❖ 641- GI to STO (shipping)- intra 2 step.
- ❖ 643- GI to STO (shipping)- inter 2 steps.
- ❖ 645- GI to STO (shipping)- inter 1 step.
- ❖ 647- GI to STO (shipping)- intra 1 step.
- ❖ 651- Returns from customer (shipping).
- ❖ 657- Returns from customer to blocked stock (shipping).
- ❖ 701- Inventory difference (physically inventory).
- ❖ 702- Inventory difference (physically inventory).

Goods receipt (GR)

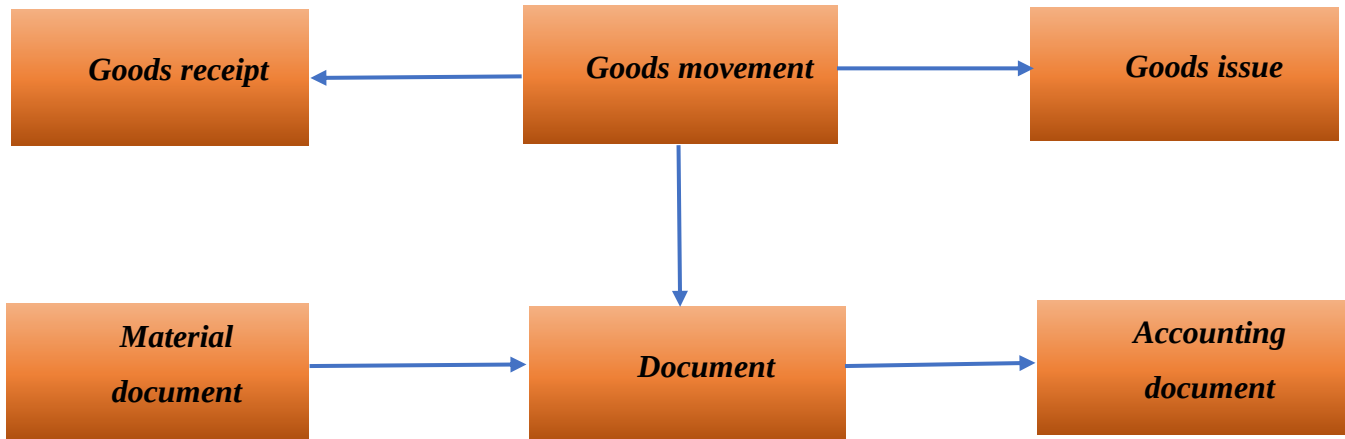
Goods receipt will be performed when the material is produced from production department or goods are procured from external vendor and placed in storage location. When we perform GR material quantity stock will increase.

Goods receipt is also called inwards movement of material.

Effects of Goods receipt



Goods movement document



Consultants' role in inventory management:

1. Define document types:

T. code – OBA7

- ❖ WE- Goods receipt.
- ❖ WA- Goods issue.
- ❖ IB- Physically inventory.
- ❖ RE- Invoice receipt.

In real time project we will use only standards no need to customize new document types for inventory and invoice.

2. Define number range:

As we know when we do any goods movement system will generate material and accounting document. So, we have to define number range for both.

Define material document number range:

Material document number range will be created by MM consultants and accounting document number range will be created by FI consultants in real time project.

Material document will be used to identify the goods movement and accounting document records the changes in the values in a company code arising from accounting transactions. Material and accounting document number range length is 10 digit and internal number range.

T. code- OMBT – Material number range:

Path:

SPRO→ IMG→ Material management→ Inventory management and physical inventory→ Number assignment→ Define number assignment for material and phys. Inv. Docs→ Enter→ Click on maintain group→ Click on create icon→ Give group name→ Give from and to number range and along with current year→ Press enters→ Save it→ Go back→ Click on CTRL+F→ Give document type (WE,WA,IB,RE) → Press enter→ Click on document type→ Click on element/group→ Search your number ranges group→ And double click on your number range→ And save it.

Define number range for accounting document:

As discussed, accounting document number range will be created by FI consultants in real time project. Number ranges will be defined yearly once below series.

T. code- FBN1.

Path:

SPRO→ IMG→ Material management→ Inventory management and physically inventory→ Number assignment→ Define number assignment for accounting document→ Enter→ Click on financial accounting number range→ Give your company code and change intervals→ Insert line→ Give serial number→ Maintain current year→ Give from and to number range→ Press enter→ And save it.

Inventory accounting number ranges:

	S.NO	YEAR	FROM	TO
GR	50	2022	5000000000	5099999999
GI/TP/other than GR	49	Current year	4900000000	4999999999
Invoice Receipt	51	Current year	5100000000	5199999999

Define field selection of goods movement:

T. code- OMJN

Path:

SPRO→ IMG→ Material management→ Inventory management and physically inventory→ Field selection for goods movement Initial/ Header screen→ Enter→ Manage the filed→ And save it.

Client roles Goods receipt (GI):

Under this heading we will learn how to perform goods receipt but in real time project store keeper will perform and we consultants will provide the training and user manuals.

Possibilities of goods receipt:

We can perform goods receipt following ways:

- ❖ GR WRT PO.
- ❖ GR W/O PO.
- ❖ GR WRT order (production module).
- ❖ GR WRT delivery (SD module – STO).

T. code:

- ❖ **MIGO- Goods movement.**
- ❖ **MIGO- GR- Goods receipt.**

Goods receipt WRT purchase order:

Once we place order to vendor then he will supply physical products to the company as per delivery date. Once we receive physical products then client's store keeper will perform goods receipt in SAP system. We can perform multiple times GR WRT one PO till PO quantity becomes zero (0).

- Go to T. code- MIGO.
- Select event A01- Goods receipts.
- Select reference document R01 – Purchase order.
- Give movement type- 101.
- Give purchase order number.

-----**Enter**-----

System will copy data from PO to GR screen.

Header data:

1. General tab:

Document data:

In this field we have to enter delivery challan data.

What is delivery challan (DC):

Delivery challan (DC) is a supporting document given by the vendors at the time of goods delivery. In this DC we can see the details of material, quantity etc...

Posting data:

In this field we have to maintain system entry data. As per SAP we can use current month dates and previous month dates as posting dates.

Delivery note:

In this field we have to capture delivery challan number.

Bill of loading:

It is also a kind of delivery document given by the foreign vendor customs.

Vendor:

In this field system will copy vendor from PO.

Header text:

In this field we can maintain any information in the form of text.

2. Vendor tab:

In this tab system will copy vendor from PO.

3. Doc. Info tab:

In this tab we can check accounting document in display mode not in creation mode.

Material tab:

In this tab we can check material details copied from PO.

Quantity Tab:

In this tab we have to maintain physically received quantity from vendor in field Qty in Unit of Entry.

Where Tab:

In this tab we have to maintain following details.

- ✓ Movement type - 101
- ✓ Plant will copy from PO.
- ✓ Storage location will copy from PO if we maintain or else, we can maintain manually here.

Goods recipient:

In this field we will capture store keeper name who received physical products.

4. Purchase Order Data:

In this tab we can check PO details.

5. Partner:

In this tab we can check goods supplier partner or vendor code.

6. Batch:

In this field system will generate Batch no. automatically based on batch settings. Batch

number will generate when we click on check icon.

Date of Manufacture:

In this field we have to maintain material manufacture date. This date we will get it on product packing.

After maintaining all required fields and tabs then activate ITEM OK check box then click on check icon and if document is OK then POST. Once document is posted system will generate material document and accounting document. (We can check accounting document only in display mode).

How to check accounting document after goods movement:

As we know once we post goods movement in SAP system at backend system will generate accounting document for FI.

Use below steps to check accounting document:

- Go to T. Code - MIGO
- Select event - Display
- Select reference document material document
- Maintain material document number (GR/GI etc) then press enter
- Go to doc.info tab
- Click on FI document options
- Double click on accounting document and check the accounting entries.

GR WRT PO accounting entries are as below:

- BSX - Inventory A/c - Dr (+)
- WRX - GR/IR clearing A/c - Cr (-)
- FR1 - Freight charges A/c -Cr. (-) ——— (only if we maintain freight in PO).

----- Goods receipt WRT purchase order topic completed-----

logistics invoice verification (LIV)

Logistics invoice verification is a part of P2P cycle. It is situated at the end of procurement that includes purchasing and inventory management.

Every company required materials and services to fulfil their business needs. Once materials are purchased from vendor. Then those material will store in company's location thru goods receipt. We need to make payment for received goods to the vendor. The amount to be paid along with the details of material, qty will be provide by vendor in the form of document called invoice/bill. Before making the payment, we will verify the invoice with PO and GR. (it is called as 3 phases in MM) the process of verifying invoice before making payment is called logistics invoice verification.

Points to be noted in invoice verification:

- Invoice verification marks the end of procurement after PO and GR.
- Invoice posting updates all finance documents.

Terms of LIV:

LIV deals with following terms:

- Posting invoice.
- Blocked invoice.

Consultant role:

1. Define number range for invoice posting.

T. code- OMRJ.

Path:

SPRO→ IMG→ Material management→ Logistics invoice verification→ Incoming invoice→ Number assignment→ Maintain number assignment for logistics document→ Maintain number range intervals for invoice documents→ Enter→ click on change intervals→ Insert a line→ Give your serial number, Maintain year→ Give from and to number range (10 digit) → Press enter→ And save it→ Go back 2 times→ Transaction-Assign number range→ Enter→ Assign your number range serial number in RD, RS transaction→ Press enter→ Save it.

Clients' role in invoice posting:

Post vendor invoice:

Posting of vendor invoice will be done after goods receipt. The invoice received from the

vendor will be verified with cost, quantity. Quality. If all are matching then clients will enter invoice in SAP system against PO.

T. code: MIRO

Path:

SAP easy access→ SAP Menu→ Logistics→ Material management→ Logistics invoice verification→ Document entry→ MIRO- Enter invoice→ Enter→ Give your company code→ Press enters.

Invoice header:

Basic data:

Invoice date:

In this field we have to maintain invoice date maintained on invoice copy.

Posting data:

In this field we have to maintain invoice entry date in SAP system.

Reference:

In this field we have capture invoice number.

Payment tab:

Base line date:

In this field we have to maintain payment due date.

Payment block:

In this field we have to select blocked for payment option if we want to block invoice or else select free for payment option (no block).

Details tab:

1. unpl del csts:**

What is delivery cost??

The amount spends on procurement of material other than material price is called delivery cost.

Ex: Freight charges, Loading and unloading charges, packing charges etc...

We have 2 types of delivery cost:

- PDC- Planned delivery cost.
- UDC- Unplanned delivery cost.

1. Planned delivery cost:

If the delivery cost is maintained at PO level in condition tab, then it is called planned delivery cost which we know at the time of order.

2. Unplanned delivery cost:

If we don't know the delivery cost at the time of order (PO) But we come know at the time of invoice then it is called UDC. Unplanned delivery cost, we will maintain UDC in details tab of invoice in unplanned delivery cost field. This will add to payment filed.

Tax tab:

In this tab we can check calculate tax amount.

Contacts tab:

In this tab we can check and user details who is responsible in posting invoice.

Go to item overview then click on PO reference tab and give PO number→ Enter→ System will copy data from PO to invoice screen based on goods receipt.

Go back to basic data tab:

- Select standard tax code VA or VO.
- Activate calculate tax check box to calculate tax amount.
- Copy balance and paste in amount filed.
- If balance amount becomes 0 with status indicator in green colour, then simulate and post the invoice.

Note:

Once invoice is posted in SAP then purchasing dept end users will hand over the original invoices to accounts department for payments.

LIV T. code:

- MIRO- Post vendor invoice.
- MIR4- Display invoice.
- MIR5- List of invoices.
- MIR6- Invoice overview.
- MIR7- Park invoice.
- MRBR- Release blocked invoice.
- MR8M- Cancel invoice.
- MR90- Invoice output.

LIV table- SE16N:

- RBKP- Invoice header.
- RSEG- Invoice item.

-----logistics invoice verification (LIV) topic completed-----

2. Subcontracting process

Special procurement

The name itself indicates special procurement type procuring material either internally or externally. In standard procurement vendor supplies goods to company after receiving purchase order but in special procurement case it is not mandatory. Sometimes goods will flow directly to customers, goods will be received from other plants within the company etc...

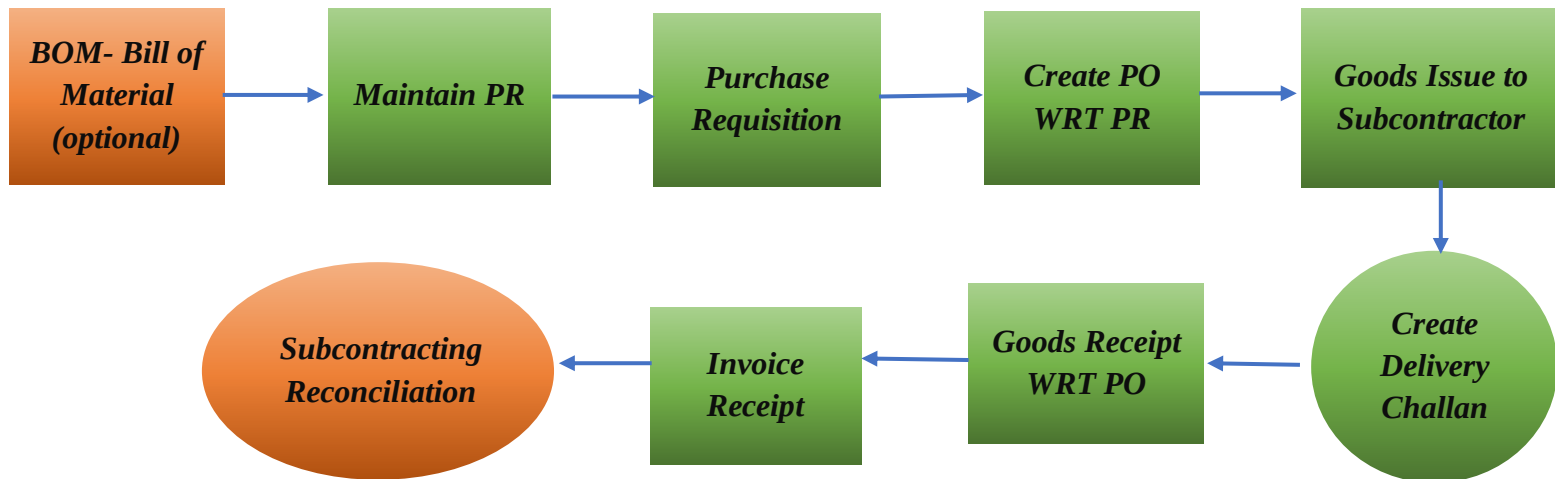
Types of special procurements:

- Third party process.
- Stock transfer order (STO).
- Subcontracting process – job work process.
- Consignment process.
- Pipeline process.

Subcontracting process:

Subcontracting is also called as job work is real time project. It is one of the special procurements in SAP MM. in subcontracting process companies will send their own components (any types of material) to outside vendor, outside vendor will convert those components into another product (final product) and sends back to company. Vendor will change payment for converting components into final products (another products).

subcontracting process flow



End user steps involved in subcontracting process:

- Create BOM- CS01 (optional).
- Create subcontracting info record- ME11.
- Create purchase- ME51N (optional).
- Create purchase order- ME21N.
- Goods issue to subcontractor – MB1B/MIGO.
- Create Delivery challan- J1IGSUBCON.
- Goods receipt (final project)- MIGO.
- Post vendor invoice – MIRO.
- Subcontracting reconciliation-J1IGRECON.

-----special procurement topic completed-----

Create BOM (Bill of Material)

BOM is used to maintain list of components required to manufacture finished products.

T. code:

Path:

Give final product code→ Maintain plant→ Maintain BOM usage 1→ Press enter→ Maintain item category L→ Maintain Qty, maintain valid date BOM→ Press enter→ And save it.

- CS01- Maintain BOM.
- CS02- Change BOM.
- CS03- Display BOM.

-----BOM topic completed-----

Creation of subcontracting info record

Subcontracting info record will be created for final product and vendor combination.

- Go to T. code- ME11.
- Give vendor.
- Give material.
- Give purchase order.
- Give plant.

Select info category→ Subcontracting enter→ System will copy data from vendor if required edit the information and save the info record.

----- Subcontracting info record topic completed -----

Creation of subcontracting purchase order

We can create subcontracting PO WRT to PR or without any reference.

Consultant role:

- Define document type by coping standard NB.
- Define assign number range to document type.
- Define, manage and assign field selection to document type by copying standard NBF.
- Item category- L.

Client role:

- Create PO W/O any reference:
- Path to create PO:
- Go to T. code-ME21N.
- Select subcontracting document type or standard NB.
- Maintain subcontracting vendor code.
- Maintain purchase org. purchase group and company code.

-----Press Enter-----

Go to item overview or expend item overview:

- Maintain item category- L.
- Maintain material code (final product).
- Maintain quantity, delivery date.
- Maintain price (subcontracting charges).
- Give plant and storage location→ press enter.

Subcontracting PR or PO error:

Error- Not possible to determine any components:

Solution:

This above error will trigger in all subcontracting PR's or PO's because if we miss to maintain components list.

Go to material data tab in PR or PO then click on explode BOM if we maintain BOM then from BOM components will copy otherwise click on components option then maintain components list (material code), quantity and supplying plant→ Press enter→ Then go back→ Click on check icon and save the subcontracting purchase order.

----- subcontracting purchase order topic completed-----

Goods issue to subcontractor (components)

Under this heading we will learn how to issue components to subcontractor with reference to subcontract purchase order.

Path:

- Go to T. code- MB1B- Transfer posting.
- Maintain document date.
- Maintain posting date.
- Maintain movement type 541- Goods issue to SC.
- Give plant.
- Click on purchase order icon.
- Give subcontracting PO no. Press enter.
- System will display components list from PO.
- Click on adopt + details icon.
- Maintain storage location and goods issue batch number (we will get batch number from MMBE T. code).
- Press enters save.

Note:

Once subcontracting goods issue to done then immediately system will add a line called **stock provide vendor** in MMBE T. code along with quantity provided. This shows how much our own stock lying in vendor (subcontracting) location.

----- Goods issue to subcontractor (components) topic completed-----

Create Delivery challan

Delivery challan is a supporting document use to send goods to outside of the company. In SAP standard we don't have DC printout.

We (SD consultant) will take the help of ABAP to design challan print out. Once challan is created will take the print and will transport physically components to subcontractor.

- Go to T. code J1IGSUBCON (this t. code is introduced only in India after GST).
- Give company code, supplying plant.
- Maintain material document (transfer posting document).
- Activate create challan radio button.
- Execute.
- Activate material document check box.
- Click on create icon and system will generate challan number. (These challan number configuration will done by consultant).
- We will take the challan print and will send physical goods to subcontractor.
- Once subcontractor receives the components then he will start converting those components into final product and sends back to company. Once company receives final product then they will do GR for final product and will inform about components consumption to SAP system.

----- Create Delivery challan topic completed -----

Subcontracting goods receipt

- Go to T. code MIGO.
- Select event goods receipt.
- Select reference document purchase order.
- Give subcontracting PO no. enter.
- System will copy data from PO to GR screen.
- Give document date, and posting date – enter.
- Expend final product (click on + icon).
- Go to quantity tab and maintain final product receipt quantity.

Where tab:

- Movement type – 101.
- Maintain storage location.
- Maintain goods recipient name.

Batch number:

- In this system will generate batch number automatically when we will click on check icon.

Activate item OK final product.

Components:

- Select components line.
- Go to quantity tab and maintain consumption receipt quantity.

Where tab:

- Movement type – 543.

Batch tab:

- In this tab we need to maintain goods issue batch number.

Activate item ok for components.

- Check and post the GR.

Note 1:

- We can check stock T. code MMBE.

Note 2:

Subcontracting accounting entry are:

- ❖ BSX – Inventory A/C – CR.

- ❖ GBB – VBO – Stock consumption A/C – DR.
- ❖ BSV – Change A/C – CR.
- ❖ FRL – SC charges A/C- DR.
- ❖ BSX – Inventory (final product) A/C – DR.
- ❖ WRX – GR/IR clearing A/C – CR.

----- Subcontracting goods receipt topic completed-----

Subcontracting Post vendor invoice

- Go to T. code MIRO.
- Give company code enter.
- Give invoice date, and posting date.
- Then invoice number reference field enter.
- Go to PO reference tab.
- Give subcontracting PO enter.
- System will copy data from PO to invoice tab.
- Go to basic data tab.
- Select tax code activate calculate check box.
- Copy balance amount.
- Posting amount field.
- If balance amount become 0.
- Then simulate and save the invoice.

----- Post vendor invoice topic completed-----

Subcontracting reconciliation

Under this heading end user will compare components issued (541 movement type) and components consumed (543 movement type).

- Go to T. code J1IGRECON.
- Give material document number (GR number).
- Give year.
- Then activate processing FIFO.
- Execute and check the reconciliation report.
- This step we be performed only if we create delivery challan by using T. code J1IGSUBCON.

----- Subcontracting reconciliation topic completed-----

Goods receipt W/O PO

Sometimes clients will request goods to vendor over phone-based priority. In this case will perform GR W/O PO after receiving physical stock from vendor.

- Go to T. code – MIGO.
- Select event goods receipt and reference document other – Enter.
- Give document date and posting date.
- Go to material tab and maintain material code – Enter.
- Go to quantity tab and maintain GR quantity.
- Go to where tab and maintain movement type – 501 plant and storage location.
- Go to partner tab and maintain vendor code.
- Go to batch tab and maintain date of manufacture, batch number will generate automatically when we click on check icon.
- Activate item ok then check and post.

Note:

- Check stock in T. code – MMBE.

GR W/O PO accounting entries are:

- ❖ BSX – Inventory A/c – DR.
- ❖ GBB – Off setting entry A/c – CR.

----- Goods receipt W/O PO topic completed-----

Opening stock entry

Opening stock is also called as initial stock. Closing stock of non-SAP will be the opening stock of SAP.

Path:

- Go to T. code – MIGO.
- Select event goods receipt and reference document other – Enter.
- Give document date and posting date. (Both can be same)
- Go to material tab and maintain material code – Enter.
- Go to quantity tab and maintain GR quantity, add stock value also in **amount** in LC field.
- Go to where tab and maintain movement type – 561 plant and storage location.
- Go to partner tab and maintain vendor code.
- Go to batch tab and maintain date of manufacture, batch number will generate automatically when we click on check icon.
- Activate item ok then check and post.

Note:

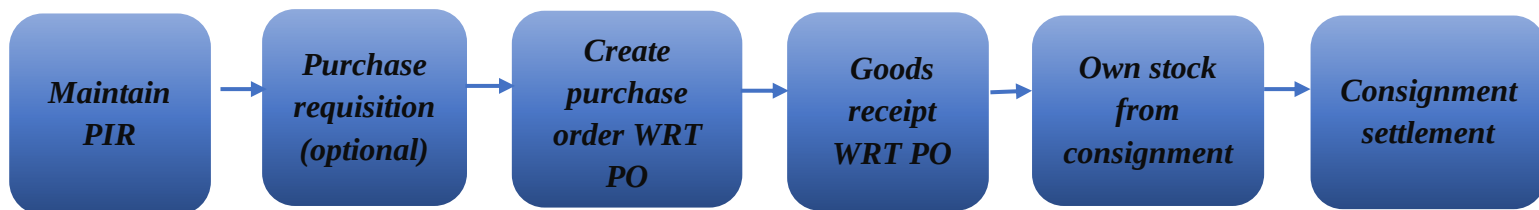
- Check stock in T. code – MMBE.
- Initial stock upload accounting entries are.
- ❖ BSX – Inventory A/c – DR.
- ❖ GBB – BSA-initial stock A/c – CR.

----- Opening stock entry topic completed-----

3. Consignment process

In consignment process vendor keeps his own material in company's location but ownership remains same with vendor until the company uses (withdraw) it. Once company uses the consignment stock then vendor will request for payment and company should make payments.

Consignment process flow



End user steps involved in consignment process:

- ❖ Create PIR – ME11.
- ❖ Create Purchase order with item category K – ME21N.
- ❖ Goods receipt – MIGO.
- ❖ Stock withdraws from consignment stock – MB1B/MIGO.
- ❖ Consignment settlement (payment) – MRKO.

Create consignment PIR:

- Go to T. code – ME11.
- Give your vendor.
- Give your material.
- Maintain purchase order.

- Maintain plant.
- Select info category - consignment – Enter.
- System will copy data from vendor in material master.
- If required edit that information.
- Maintain price, tax code (mandatory).
- Press enter and save it.

Note:

In consignment process price and taxes will copy from PIR screen to payment screen. But in general prices will copy purchase order. Not in consignment.

Consignment PO:

- We can create PO WRT PR or without any reference.
- Go to t. code – ME21N.
- Select consignment document type or standard NB.
- Give vendor code.
- Give purchasing org., Purchasing group and company code.

-----Enter-----

Expend item overview tab or go to item overview:

- Give item category K – consignment.
- Give material code, qty, delivery date.
- Maintain plant and storage location.

-----Enter-----

- Click on check icon and save the consignment PO.

Note:

In consignment PO net price field will be in disable mode because of item category K.

Condition tab and invoice tab will be in hide.

In consignment payment amount and taxes will copy from PIR instead of PO.

Create consignment goods receipt:

- Go to t. code- MIGO.
- Select event goods receipt and reference document PO.
- Give consignment PO- Enter.
- System copies from PO to GR screen.
- Give document date and posting date.
- Go to qty tab- Give receipt qty.

- Where tab- movement type- 101- K.
- Then plant will copy from PO.
- Storage location will copy from PO.
- If we maintain or maintain here.

Batch:

- ❖ In this system will generate batch no automatically.
- ❖ When we click on maintain manufacture date. (If required)
- ❖ Activate item OK click on check icon and post it.

Note:

- ✓ Check in consignment stock MMBE t. code.
- ✓ No accounting entries consignment GR.
- ✓ Double click vendor consignment details.

Transfer consignment stock to company stock (own stock):

Under this heading we will learn how to transfer consignment stock to own stock.

- Go to t. code- MB1B.
- Give document date, posting date.
- Maintain header text (if required).
- Maintain movement type – 411 and special stock – K.
- Give plant and storage location- Enter.
- Maintain consignment vendor.
- Maintain consignment material code.
- Maintain transfer qty.
- Maintain consignment stock batch no (if will get batch MMBE t. code).
- Press enter and save it.

Consignment accounting entries:

In consignment process accounting entries will generate at the time of accepting the stock ownership. (Movement type- 411 – K).

- ❖ BSX- inventory A/c – DR (+).
- ❖ KON – consignment payable – CR (-).
- ✓ Check stock in T. code- MMBE.

Consignment settlement:

In pipeline process and consignment process we will use T. code MRKO for invoice payment and in general we will used MIRO for invoice payment.

When we are executing MRKO t. code for consignment and pipeline settlement system will copy amount, tax code for PIR.

- Go to T. code- MRKO.
- Maintain company code.
- Maintain material document no- (411 no).
- Activate consignment check box.
- Activate settle radio bottom.
- Execute.
- Check document no.

-----**Consignment topic completed**-----

4. Pipeline process

Pipeline is one of the special procurements in SAP MM. The material which are flowing thru pipes, cables, wires, etc... are called pipeline materials. Ex. Electricity, water, gas, etc. if we want to make material payments thru SAP system then we will use pipeline process.

Pipeline process flow



End user steps involved in pipeline process:

- ❖ Create pipeline material by using material type PIPE – MM01.
- ❖ Create vendor or we can use existing vendor -XK01.
- ❖ Create PIR info category – PIPELINE – ME11.
- ❖ Record the consumption – MIGO.
- ❖ Pipeline settlement – MRKO.

1. Create pipeline material:

- Go to T. code – MM01.
- Select industry sector and material type pipeline material – PIPE.
- Select basic data, purchasing and accounting view.
- Press enters.
- Give your plant – Enter.
- Maintain material name and other field. In all the selecting view.
- Save the material.

2. create pipeline vendor (or) we can use any existing vendor.

3. create pipeline PIR.

- Go to T. code – ME11.
- Give vendor.
- Give pipeline material.
- Maintain PO, and maintain plant.
- Select pipeline info category – press enter.
- System will data copy from material and vendor.
- If required edit here.
- Maintain price and tax code which will copy to settlement screen.
- Press enter and save the info record.

4. record the consumption:

Under this heading we will record the pipeline material consumption (usage) quantity based on bill received from vendor.

- Go to T. code – MIGO.
- Select event goods issue and reference document other.
- Give document date and posting date.
- Go to material tab – give pipeline material code - enter.
- Heading quantity tab – maintain consumed quantity as per bill.
- Where tab – maintain movement type – 201 [P] then give your plant – enter.
- Partner tab – maintain vendor code.
- Accounting assignment tab – maintain cost center.
- Activate item OK and check and post it.

5. pipeline settlement:

In pipeline process and consignment process we will use MRKO t. code for invoice settlement. The amount and tax will be copied from PIR.

- Go to T. code – MRKO.
- Give company code.
- Maintain material document (goods issue document).
- Activate pipeline check box.
- Activate settle radio bottom and execute.

-----Pipeline process topic completed-----

Controlling Module

Controlling is one of the modules available in SAP. It provides the information to management or to decision maker to understand where company's money is spent.

CO module integrates with other modules like MM, SD, PP

FI etc ...

Components of Controlling module:

- ❖ Cost center accounting
- ❖ Cost element accounting
- ❖ Profit centre
- ❖ Internal order
- ❖ Profitability
- ❖ Product cost controlling
- ❖ Activity based costing

The controlling module helps to schedule, monitor, execute and report on cost.

Co modules takes care of cost management.

1. Maintain Controlling Area

Controlling area is an organisational unit that represents a closed system used for cost accounting purpose.

Controlling area will be assigned to multiple company codes to handle different currencies.

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Controlling→ Maintain controlling area→ Enter→ Double clicks on maintain controlling area→ Click on new entries→ Click on CO Area = C Code→ Give your company code→ Press enter→ Maintain assignment control→ 1 Controlling area same as company code→ Save it.

Assign controlling area and company Code:

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Controlling→ Assign company code to controlling area→ Press enter→ Click on position→ Give your company code→ Select controlling area line→ Double click on assignment of company code→ Click on new entries→ Assign your company code (we can assign multiple company code) → Press enter→ Save it.

----- Controlling Module topic completed-----

Cost center

Cost center is the exact location where the cost is occurred inside the company. The main purpose of cost center is used to track expenses of the company. Each and every work will be created as cost center in SAP.

EX: HR expenses, IT expenses, manufacture expenses, machinery expenses etc...

T. code: KS01.

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Controlling→ Create cost center→ Enter→ Double clicks on create cost center line→ Maintain your controlling area→ Enter→ Maintain cost center (10 CHAR) → Maintain valid date of cost center→ Enter→ Maintain cost center name, maintain person responsible, maintain cost center category 1, maintain Hierarchy area H1, maintain currency→ Save it.

T. codes:

- KS01- Create cost center.
- KS02- Change cost center.
- KS03- Display cost center.
- KS04- Delete cost center.

-----Cost center topic completed-----

Goods issue (GI)

Goods issue will take place when materials are consumed to produce finished goods or goods issue will be done to customers, to other plants of same company or internal department etc... when goods issue is done then system will decrease stock from SAP books and from physical storage location also.

Goods issue to cost center:

Path:

- Go to T. code MIGO.
- Select event goods issue and reference document – other → Enter.
- Give document date and posting date.
- Go to material tab and maintain material code – Enter.
- Go to quantity tab and maintain goods issue quantity.
- Go to where tab and maintain movement type – 201, plant and storage location.
- Go to batch tab and maintain goods issue batch number manually by copying from MMBE t. code.
- Go to account assignment tab and give cost center.
- Activate item OK then click and post.

Note:

- Check stock in T. code – MMBE.
- Goods issue to cost center accounting entries are.
- BSX – Inventory A/c – CR.
- GBB – VBR – Off setting entry A/c + DR (consumption A/c).

----- Goods issue (GI) topic completed -----

-

Reservation

Reservation in SAP will be used to block the material quantity for feature used.

Define reservation number range:

T. code: OMC2

Path:

SPRO→ IMG→ Material management→ Inventory management and physically inventory→ number assignment→ Define number assignment for reservation→ Enter→ Click on intervals→ Click on change intervals→ Insert line→ Give serial number→ Give from and to number range (internals 10 CHAR) → Save it→ Go back 2 times→ Click on no. range res/dep req→ Assign number range serial number→ Save it.

Client role:

T. code- MB21

Path:

SAP easy access screen→ SAP Menu→ Logistics→ Material management→ Inventory management→ Reservation→ MB21- create→ Enter→ Reservation initial screen:
Maintain base date.

Maintain movement type:

- 201- cost center.
- 221- Project (PS)
- 261- Order (PP)
- 281- Network (PS)
- Maintain cost center for 201.
- Maintain WBS element for 211.
- Maintain order no. for 261.
- Maintain network for 281.
- Maintain plant→ Enter→ Give material code, maintain QTY, maintain batch→ Enter→ Save it.

T. code:

- ❖ MB21- Create reservation.
- ❖ MB22- Change reservation.
- ❖ MB23- Display reservation.

- ❖ MB25- List of reservation.

Tables- SE16N:

- ❖ RESB- Reservation.
- ✓ We can check reservation stock in T. code MMBE along with MB25.

Goods issue against reservation:

- Go to T. code- MIGO.
- Select event goods issue and reference document – reservation→ Give reservation no→ Enter.
- System will copy data from reservation to goods issue screen.
- Go to quantity tab and maintain goods issue quantity.
- Activate item OK then check the post.

Note:

- Check reservation stock in T. code MMBE and MB25.
- Goods issue against reservation accounting entries are.
- BSX – Inventory A/c – CR.
- GBB – VBR – Off setting entry A/c – DR (consumption A/c).

-----Reservation topic completed-----

T. codes & Tables pf Inventory Management

- *MIGO– Goods Movement.*
- *A01– Goods Receipt.*
- *A02- Return delivery.*
- *A03- Cancel material document.*
- *A04- Display material document.*
- *A07- Goods issue.*
- *A08- transfer posting.*
- *MB1A- Goods issue.*
- *MB1B- Transfer posting.*
- *MB1C- Other than goods receipt.*
- *MB02- Change material document.*
- *MB03- Display material document.*
- *MBRL- Reverse material document.*
- *MBST- cancel material document.*
- *MB51- List of material document.*
- *MB52- Stock list with value.*
- *MMBE- Stock overview.*
- *MB5B- Stock on posting date.*
- *MBBS- List of special stock.*
- *MB5M- Shelf-life stock (Expiry stock).*
- *MMRV- Check and open previous posting period.*
- *MMPV- Open MM posting periods.*
- *OB52- FI posting periods.*
- *OMJJ- Movement types.*

Tables- SE16N:

- *MKPF- Material document header data.*
- *MSEG- Material document item data.*
- *BKPF- Accounting document header data.*
- *BSEG- Accounting document item data.*
- *MATDOC- Material document (S4 HANA tables)*
- *ACDOCA- Accounting document (S4 HANA).*

This T. code removed in S4 HANA:

- *MB1A- Goods issue.*
- *MB1B- Transfer posting.*
- *MB1C- Other than goods receipt.*
- *MB02- Change material document.*
- *MB03- Display material document.*
- *MBRL- Reverse material document.*
- *MBST- cancel material document.*

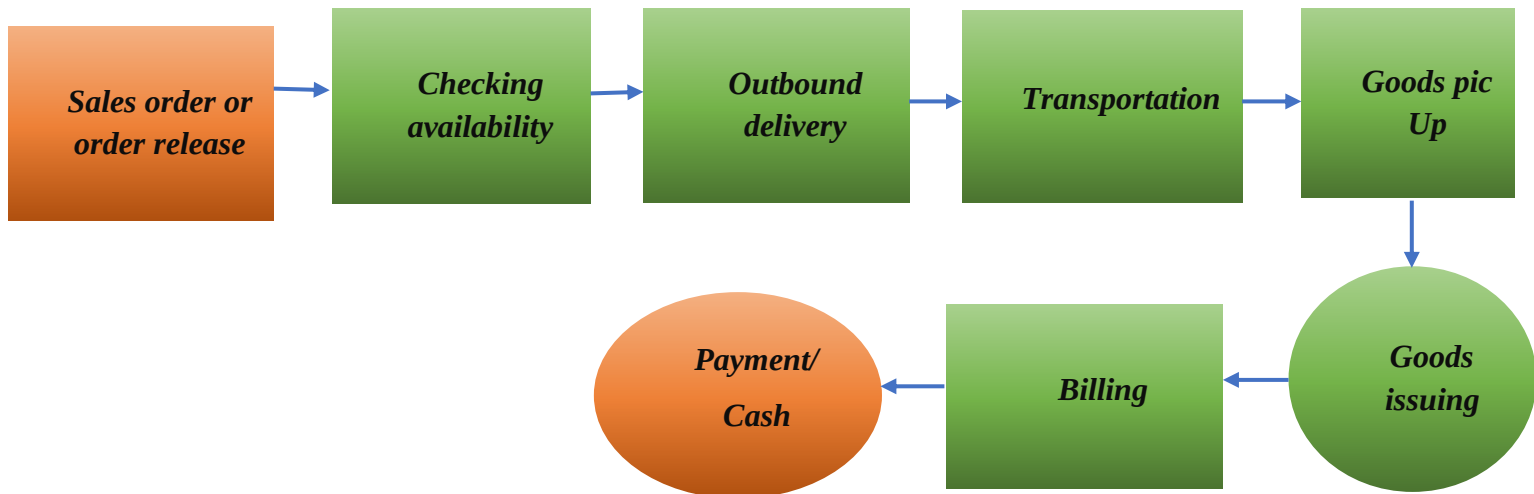
Sales and distribution

SD module one of the functional module available in SAP. SD module Supports all sales activity, delivery, shipping, and selling of goods and services.

This module integrates in other modules like MM, QM, FICO, WM etc...

This business cycle of SD modules is OTC – Order to cash.

OTC – Order to Cash



SD Organization Data

- Company (6 CHAR)
- Company code (4 CHAR)
- Sales organization (4 CHAR)
- Distribution channel (2 CHAR)
- Division (2 CHAR)
- Shipping point (4 CHAR)

These organization levels will be created by FI consultants in real time project.

These organization levels will be created by SD consultants in real time project

Define sales organization:

Sales organizations used to sell the goods and services to the customer.

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Sales and Distribution→ Define, copy, delete, check sales organization→ Enter→ Define sales organization→ Select 001 sales organization line→ Then click on copy as icon→ Give your sales organization (4 CHAR) → Maintain description→ Maintain currency→ Press enter→ Maintain sales organization address→ Press enter→ And save the sales organization.

Definition distribution channel:

It is used to send goods to the customer. In real time project SD consultants will define distribution channel based on client's requirement.

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Sales and distribution→ define copy, delete, check distribution channel→ Enter→ Double clicks on define distribution channel→ click on new entries→ Give distribution channel (2 CHAR) → Maintain distribution→ Press enter→ Save it.

Define Division:

Division is same like material group division is used for SD module and material is used for MM module.

The material with same future is material with same product line group together is called division.

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Logistics general→ Define copy, delete, check division→ Enter→ Double clicks on define division → Click on new entries→ Give division (2 CHAR) → Maintain distribution→ Press enter→ Save it.

Define maintain shipping point:

Shipping point is a place from where goods delivery will happen to customer.

Path:

SPRO→ IMG→ Enterprises structure→ Definition→ Logistics execution→ Define copy, delete, check shipping point→ Enter→ Double clicks on Define shipping point→ Select 0001 shipping point line→ Then click on copy as icon→ Give your own shipping point (4 CHAR) → Maintain distribution→ Press enter→ Maintain shipping point address (It most

of the cases shipping point address will be the plant address) → Press enter→ Save the shipping point.

Assignment of SD organization structure:

1. Assign sales organization to company code:

We can assign multiple sales organization under one company code.

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Sales and distribution→ Assign sales organization to company code→ Enter→ Click on position→ Give your sales organization and assign your company code→ Save it.

2. Assign distribution channel to sales organization:

We can assign multiple distribution channel under one sales organization.

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Sales and distribution→ Assign distribution channel to sales organization→ Enter→ Click on new entries→ Give your sales organization and assign distribution channel→ Press enter→ Save it.

3. Assign division to sales organization:

We can assign multiple division under one sales organization.

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Sales and distribution→ Assign division to sales and distribution→ Enter→ click on new entries→ give your sales and distribution and assign division→ Enter→ Save it.

4. Set up sales area:

Sales area = Sales organization+ Distribution channel+ Division.

Path:

SPRO→ IMG→ enterprises structure→ Assignment→ Sales and distribution→ Set up sales area→ Enter→ Click on new entries→ Give your Sales and distribution, Distribution channel, and Division→ Press enter→ Save it.

5. Assign sales organization – Distribution channel – plant:

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Sales and Distribution→ Assign sales organization – distribution channel - plant→ Enter→ Click on new entries→ Give your sales organization, and distribution channel, assign plant→ Press enter→ Save it.

6. Assign business area to plant/ valuation area and division:

Path:

SPRO→ IMG→ Enterprises structure→ Assignment→ Logistics general→ Assign business area to plant/ valuation area and division→ Enter→ Click on plant/ valuation area - Division→ Click on new entries→ Give your plant, Division, assign business area (0001)- Press enter→ Save it.

Assign shipping point to plant:**Path:**

SPRO→ IMG→ Enterprises structure→ Assignment→ Logistics execution→ Assign shipping point to plant→ Enter→ Click on CTRL +F→ Give your plant→ Enter→ Click on your plant→ Click on assign account→ Again click on CTRL+F→ Give your shipping point→ Enter→ Click on your shipping point→ Activate shipping point check box→ Press enter→ Save it.

Sales and distribution data's:

In sales and distribution, we have following master data.

- Material master.
- Customer master.
- Credit master.
- Pricing master.

1. Extension material to sales view:

- Go to t. code – MM01.
- Give extension material code.
- Press enters (2 time).
- Select sales view - press enter.
- Give plant, sales organization, distribution channel – enter.

Sales: sales org. 1 view tab:

Under this heading we will maintain sales information of the material.

1. Base unit of measure:

This field maintain basic data 1.

2. Maintain division.

3. Sales unit:

In this field we have to maintain unit of measurement in which material are sold to customer.

4. Maintain tax data 1 – enter.

Sales: sales org. 2 view tab:

Maintain item category group – NORM.

-----**press enter**-----

Sales: general/ plant view tab:

In this field we have to maintain transportation group – 0001, loading group – 0001.

-----**press enter**-----

Sales text:

In this view we can maintain any sales information in the form of text.

-----**save the material**-----

Note:

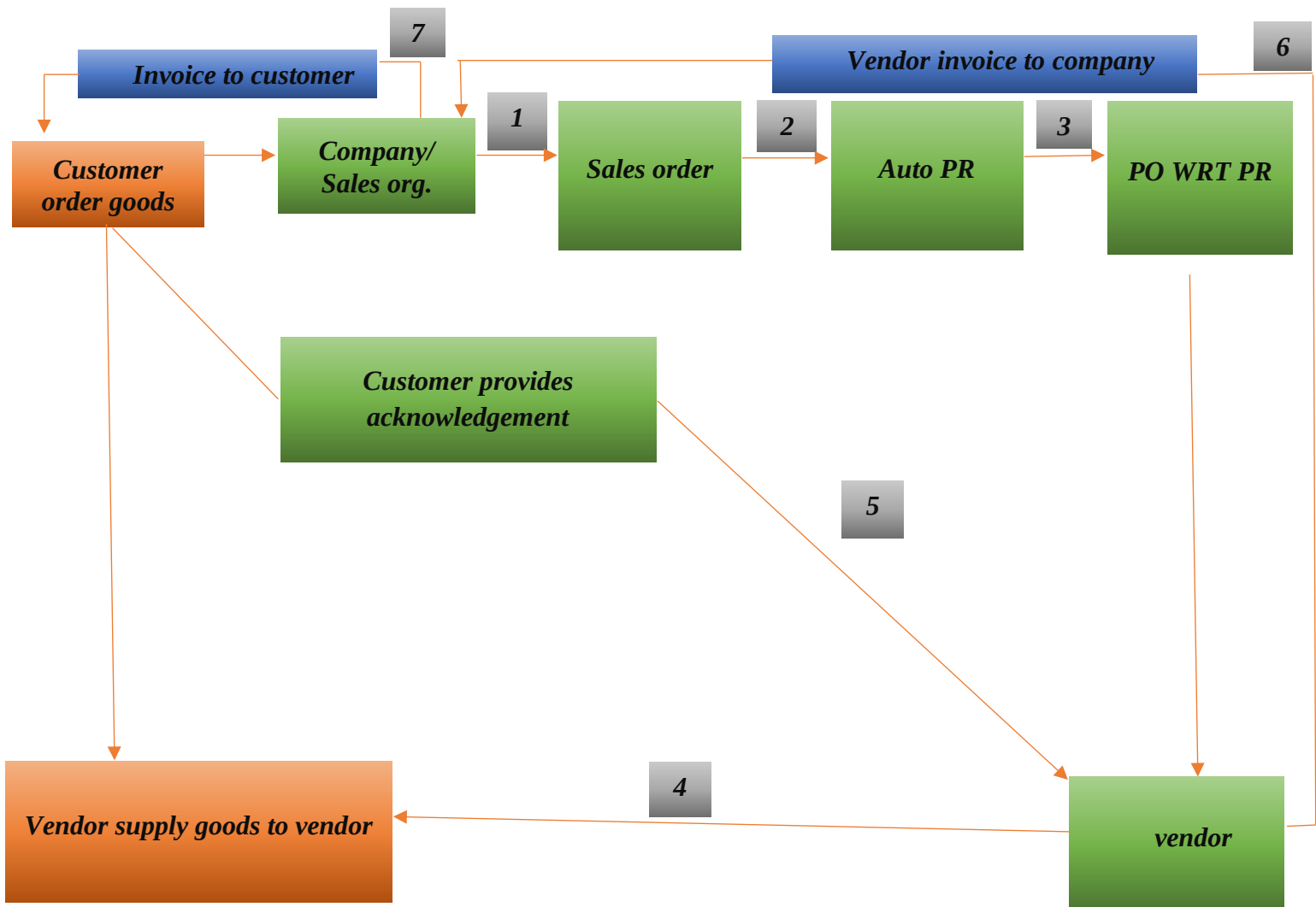
We MM consultant are learn sales and distribution only because of two special procurements.

- Third party process.
- Stock transfer order (STO).

Third party process:

In third party process delivery of goods required by customer is not delivery by company. Instead, the order will transfer to vendor who send goods to customer directly on behalf of company.

Third party process flow



What happens in third party process:

- ❖ Customers order goods to company and sales order will be created (SD – VA01).
- ❖ Purchase requestion will be created automatically once sales order is saved (SD/MM – VA02/VA03).
- ❖ Purchase order is created to the vendor (MM – ME21N).
- ❖ Invoice receipt from vendor (MM/FI – MIRO).
- ❖ Invoice to customer (SD – VF01).

Create customer: (XD01)

Customer is an individual person or individual company or group of companies to whom our client company will sell the goods and services.

Customer will be created by SD team.

We are creating a customer here just to learn third party process and stock transfer order.

- Go to t. code- XD01.
- Select customer account group -0001.
- Give company code, sales organization, distribution channel, division – enter.
- Customer general data:
- In this tab we have to maintain customer name, address along with telephone details.
- Click on company code data icon. Maintain recon. Account – 140000.

Click on sales area data icon:

Sales tab:

- ✓ Maintain currency.
- ✓ Maintain customer pricing procedure – 1.

Shipping tab:

- ✓ Maintain delivery priority -1.
- ✓ Maintain shipping condition – 01.

Billing document:

- ✓ Maintain taxes -1.

-----save the customer-----

Customer master t. codes:

- XD01 – Create customer.
- XD02 – Change customer data.
- XD03 – Display customer data.
- XD04 – Changes.
- XD05 – Block/ Unblock customer.
- XD06 – Flag for deletion.
- XD07 – Change customer account group.
- XD99 – Mass maintenance.
- VCUST – Customer list.

Tables – SE16N:

- KNA1 – Customer general data.
- KNB1 – Customer company code data.
- KNVV – Customer sales data.

Sales order (SO):

Sales order is a document generated by the sales department specifying the details of products or service ordered by the customer.

Third party sales order:

- Order type – OR (standard type).
- Item category for third party – TAS.
- Schedule line – CS.

Note:

As we are aware in third party sales order automatically PR will generate to purchase materials for customer. SD consultant will do following configuration to generate PR automatically.

In schedule line CS, SD consultants will assign PR document type, based on this system will generate PR once sales order is saved.

T. code – VOV6.

Go to t. code- VOV6→ Click on position→ Give your schedule line CS→ Enter→ Double clicks on CS line→ Once PR document type in order type field→ Enter and save.

Note:

We can assign PR standard document type- NB or we can customize new document for third party PR's.

Creation of third-party sales order:

- Go to t. code- VA01.
- Select order type – OR.
- Maintain sales organization, distribution channel, division- Enter.
- Maintain your customer number in sold to party- Enter.

Error:

- ✚ No pricing procedure could be determined.

Solution:

- ✚ Go to t. code – OVKK.
- ✚ Select 0001 sales org. line.
- ✚ Click on copy as icon.
- ✚ Give your own sales org. distribution channel, division – Enter.
- ✚ Save it.

✚ Go back to sales order again press enter.

✚ Error will be resolved.

-----**Error resolved**-----

- Maintain material code, order qty.
- Maintain item category – TAS.
- Maintain plant and price – press enter.
- Click on go to option menu bar.
- Select item – click on schedule line – maintain schedule line CS – Press enters.
- Go back again click edit option in menu bar click on incomplection log.
- Complete missing filed.
- If document is complete then save the sales.

T. codes of sales order:

- VA01 – create sales order.
- VA02 – Change sales order.
- VA03 – Display sales order.
- VA05 – list of sales order.

Tables – SE16N:

- VBAK – Sales header data.
- VBAP – Sales item data.

How to check PR number in third party sales order:

- Go to t. code – VA03 – sales order display mode.
- Give sales order no – enter.
- Double click on line item.
- Go to schedule line tab.
- Check and copy the PR no.
- Closed the sales order.
- Go to t. code – ME53N.
- Click on other purchase requisition icon.
- Give sales order PR no.
- Check PR in display mode.

Creation third party purchase order:

- Go to t. code ME21N.
- Give document type.

- Give vendor code.
- Give purchase organization, purchasing group and company code.

-----press enter-----

Item overview:

- Drag the screen to right and give third party PR number in PR filed – Press enters.
- System will copy data from PR to PO screen.
- Maintain material price, delivery date.
- Deactivate GR based inv - invoice tab.
- Deactivate goods receipt check box – delivery tab.
- Click on check icon and save the PO.
- System will generate PO number at left bottom of the screen.

Post vendor invoice:

Once vendor delivery goods to customer in third party process then customer give acknowledgement. Vendor once received the acknowledgement then will raise invoice to company plant.

- Go to t. code – MIRO.
- Give company code – enter.
- Give invoice date and posting date.
- Invoice number in reference field – enter.

Go to PO reference tab:

- Give the third-party PO no – enter.
- System will copy PO to invoice screen.
- Select tax code and activate calculate check box.
- Copy balance and paste n amount field.
- If balance amount becomes zero in indicator green button.
- Then simulate and post it.

***company create invoice to customer by using t. code – VF01. (sales order).

-----Sales and distribution topic completed-----

STO – stock transfer order

It is one of the special procurement available in SAP MM. STO will be used to transfer material from one plant to another plant within company.

Types of STO:

In STO we have following two types.

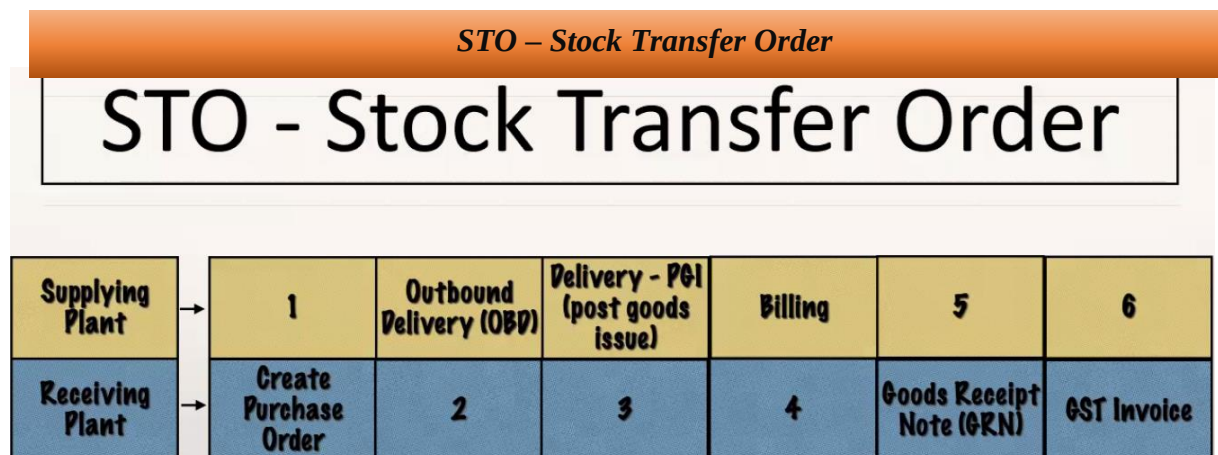
- Intra STO.
- Inter STO.

If material is transferring from one plant to another within same company code, then it is called intra STO.

If material is transferring from one plant to another plant between 2 different company codes, then it is called inter STO.

We can configure STO with MM and SD modules integration.

Process flow:



Consultant role in STO:

1. Setup stock transfer order:
 - ❖ Assign shipping data to plant.
2. Checking rule – 01.

3. Delivery type:

- ❖ NL – Intra STO.
 - ❖ NLCC – Inter STO.
4. Define STO document type by coping standard UB. (Same like NB)

Realtime example:

- ❖ UB – Intra STO – NL.

❖ ZSTO – Inter STO – NLCC.

5. Define and assign number range to STO document type.
6. Define manage and assign field selection to STO document type by copying standard UBF. (Same like NBF)
7. Item category – U.
8. Assign delivery type, checking rule to document type.
9. Assign document type, one step procedure, underdeliver. (Optional, only incase if we want one step process or else leave this step).
10. Define shipping point determination.

Setup stock transfer order:

Assign shipping data to plant:

Under this heading we consultants will assign customer number to receiving plant. Customer will be created with plant information. (Receiving plant is a customer).

Note:

There is no concept of vendor in STO process. We will have only suppling plant and receiving plant.

SD consultant will create a customer with receiving plant information and they will assign to receiving plant.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchase order→ Setup stock transfer order→ Define shipping data for plants→ Enter→ Click on position→ Give Shipping plant or receiving plant and maintain below information:

Fields	Supping plant	Receiving plant
Customer no	Leave blank	Maintain plant customer number
Sales organization	Sales organization	Sales organization
Distribution channel	Distribution channel	Distribution channel
Division	Division	Division

Press enter and save it.

3. Checking rule – 01.

3. Delivery type:

- ❖ NL – Intra STO.
- ❖ NLCC – Inter STO.

4. Define STO document type by coping standard UB. (Same like NB)

Realtime example:

- ❖ UB – Intra STO – NL.
- ❖ ZSTO – Inter STO – NLCC.

5. Define and assign number range to STO document type.

6. Define manage and assign field selection to STO document type by copying standard UBF. (Same like NBF)

7. Item category – U.

8. Assign delivery type, checking rule to document type:

Under this heading we will assign to delivery type to STO document type.

Path:

SPRO→ IMG→ Material management→ Purchasing order→ Setup stock transfer order→ Assign delivery type and checking role→ Enter→ Click on new entries→ Give STO document type, maintain suppling plant, delivery type→ Checking rule-01→ Enter→ Save it.

9. Assign document type, one step procedure, underdeliver. (Optional, only incase if we want one step process or else leave this step):

This step will be performed only if want one step process.

What is 1 step procedure:

In one step procedure when suppling plant issue the goods then automatically GR is done in receiving plant.

What is 2 steps procedure:

In two step procedure suppling plant issue to goods and receiving plant will do GR manually by using T. code MIGO.

***in real time project maximum companies will use 2 step process.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchase order→ Setup transfer order→ Assign document type, one - step procedure, underdeliver tolerance→ Enter→ Click on new entries→ Give your shipping plant, receiving plant, STO documents type (UB) → Activate one step check box→ Press enter and save it.

10. Define shipping point determination:

This step will be performed by SD consultant by using OVL2 t. code.

- Go to T. code – OVL2.
- Click on new entries.
- Give shipping condition (01), Loading GRP- (0001), Suppling plant, maintain shipping point→ Press enter and save it.

STO client role:

STO end user steps:

- ❖ Extend material to both suppling and receiving plant (MM01).
- ❖ Assign suppling plant to vendor for inter STO.
- ❖ Create STO PR (optional) – ME51N.
- ❖ Create STO PO – ME21N.
- ❖ Create outbound delivery (OBD)- VL10B – (SD).
- ❖ Edit delivery (post goods issue) – VL02N – (SD).
- ❖ Create billing – VF01 – (SD).
- ❖ Goods receipt by receiving plant (MIGO).

2. Assign suppling plant to vendor:

Sometime instead of using STO standard document type UB we will use NB document type for STO inter purchase order. So, in NB document type system will ask vendor code but in STO there is no vendor concept then in this case we have to create a vendor with suppling plant information and we need to assign to plant in a vendor.

Path:

Go to purchase tab of vendor master→ Click on extras in menu bar→ Click on add purchasing data→ Maintain plant→ Enter and save it.

4. Create STO purchase order:

We can create PO WRT PR to W/O any reference t. code- ME21N – create PO.

Steps:

- ❖ Go to t. code- ME21N.
- ❖ Select document type UB or your document type.
- ❖ Maintain suppling plant.
- ❖ Maintain purchase organization, purchase group and company code.

-----Enter-----

PO item overview:

- Give item category (U).
- Maintain material code.
- Quantity.
- Delivery date.
- Receiving plant and storage location.
- Press enters.
- Check the shipping tab: in item level if it is available then only click on check icon and save the PO.

*** if shipping tab is not available in STO PO then don't save the PO.

5. Create outbound delivery (OBD):

Under this heading we will learn how to create delivery after receiving order. In real time project delivery will be created by sales end user.

- Go to t. code – VL10B.
- Go to PO tab: STO PO number.
- Remove delivery creation date.
- Execute.
- System will display PO line.
- Select your PO line, click on background icon to create delivery, click on show/hide delivery icon (truck icon).
- Copy delivery no.

6. Edit delivery (goods issue to STO):

Under this heading we will learn how to issue goods against delivery.

- Go to t. code- VL02N.
- Give delivery no – enter.
- Go to picking tab:
- Give suppling plant, storage location, maintain picked qty (same as delivery qty).
- Maintain batch (we will get batch no MMBE t. code).
- Press enter and click on post goods issue (PGI).

Note:**STO accounting entries:**

In STO process accounting entries will not generate at the time of goods receipt but accounting entries will generate at time of goods issue (PGI). Accounting entries are.

- ❖ BSX – Inventory A/c – DR.
- ❖ BSX – Inventory A/c – CR.

Create Billing:

In STO process we will create proforma invoice to send physical goods to receiving plant. This billing will be done by SD end user with reference to delivery.

- Go to t. code – VF01.
- Give Billing type – F8 Pro forma inv dlv.
- Billing date, then give Delivery no in document filed – enter- save it.

Note:

Once billing is created then we will take print out physical goods to receiving plant. Once receiving plant receives goods physically then they will perform goods receipt (GR).

Goods receipt by receiving plant:

Under this heading we will learn how to do GR WRT to delivery (OBD).

- Go to t. code – MIGO.
- Select event goods receipt.
- Select reference document out bound delivery.
- Give delivery number – enter.
- Go to quantity tab: maintain receipt quantity.
- Activate item ok check and post it.

Note:

- Check stock in MMBE t. code.
- No accounting entries will generate in STO GR.
- STO movement type list is as below.

STO – Movement types

SD + MM STO	Intra STO (1st step)	Intra STO (2nd step)	Inter STO (1st step)	Inter STO (2nd step)
	PGI – 647 GR - 101	PGI – 641 GR - 101	PGI – 645 GR - 101	PGI – 643 GR - 101

-----stock transfer order topic completed-----

Outline agreement

It is long term agreement with the vendor for procurement of material with agreed terms and conditions. These terms and conditions are valid only for certain period of time.

An outline agreement can be following two types:

- ❖ Contracts.
- ❖ Scheduling agreement.

1. Contracts:

Contracts is a long terms agreement between vendor and customer (company) with predefined terms and conditions to purchase materials over certain period of time.

Contracts are in two different types:

- ✓ Quantity contracts.
- ✓ Value contracts.

Quantity contracts:

In this type of contract, if the agreement is to purchase certain quantity of materials in certain period of time from vendor.

Value contracts:

In this type of contract, if the agreement is to purchase certain value of materials in certain period of time from vendor.

Consultant role:

- ❖ Define document type by copying standard WK or MK. MK – Quantity contracts, WK – value contracts (same like PO – NB).
- ❖ Define and assign number range to document type.
- ❖ Define, manage and assign field selection to document type by copying standard WKK or MKK. MKK – Quantity contracts, WKK – Value contracts.

Contracts path:

SPRO→ IMG→ Material management→ Purchasing→ Contracts.

Clients' role in contracts:

Under this heading we will learn how to create contracts.

T. codes – ME31K.

Create contracts:

Path:

SAP easy access screen→ SAP Menu→ Logistics→ Material management→ Purchasing→
Outline agreement→ Contracts→ ME31K – Create→ Enter.

Contracts initial screen:

- Give vendor.
- Maintain agreement type – (MK – WK).
- Maintain agreement start date.
- Give purchase org, purchase group – Enter.
- Maintain contracts end date – Press enters.
- Give material code, quantity, price, give plant, and storage location – press enter and save it.

Contracts t. codes:

- ME31K – Create contracts.
- ME32K – Change contracts.
- ME33K – Display contracts.
- ME35K – Release contracts.
- MEMASSCONTRACT – Mass maintenance.
- ME9K – Contracts print.

Create of PO WRT to contracts:

- Go to t. code – ME21N.
- Select document type.
- Go to item overview:
- Drags your screen to right.
- Give your contracts number in agreement number field.

-----press enter-----

- System will copy all data from contracts.
- Maintain quantity and delivery date.
- Press enter and click on check icon and save it.

How to check contracts history:

- Go to t. code – ME33K – Display contracts.
- Give contracts number – enter.
- Select contracts line item.

- Click on release documentation.
- Check the contracts history.

2. Scheduling agreement:

It is a long-term agreement with the vendor for procurement of materials with agreed terms and conditions with predefined delivery date.

Consultant role:

- Define document type of copying standard LP (same like PO- NB).
- Define and assign number range to document type.
- Define, manage and assign field selection to document type by copying standard LPL.

Create scheduling agreement:

T. code – ME31L.

Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material management→ Purchasing→ Outline agreement→ Scheduling agreement→ Create→ ME31L Vendor know→ Enter→ Give vendor→ Maintain document - LP→ Maintain agreement start date→ Maintain purchase org, purchase group→ Enter→ Maintain agreement date→ Enter.

Scheduling agreement overview:

- Maintain material code.
- Maintain target Qty, price, plant, storage location – enter and save it.

Scheduling agreement t. codes:

- ❖ ME31L – Create SA.
- ❖ ME32L – Change SA.
- ❖ ME33L – Display SA.
- ❖ ME35L – Release SA.
- ❖ MEMASSSA – Mass maintenance.
- ❖ ME9L – SA print.

Maintain delivery schedule:

Under this heading we will maintain delivery schedule to the scheduling agreement (delivery date).

T. code – ME38.

Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material management→ Purchasing→ Outline agreement→ Scheduling agreement→ Delivery schedule→ ME38 - maintain→ Enter→ Give scheduling agreement number→ Enter→ Click on delivery schedule icon→ Give your delivery date and maintain schedule qty→ Press enter and save it.

T. codes:

- ME38 – Maintain delivery schedule/ edit.
- ME39 – Display delivery schedule.
- ME9E – Print delivery schedule.

Outline agreement t. codes:

- ME3M – List of agreements by materials.
- ME3L – List of agreement by vendor.
- ME3N – List of agreement.
- ME3K – List of account assignment statements.
- ME3J – List of project agreements (PS module).
- ME3C – List of agreement by material group.

-----Outline agreement topic completed-----

Split valuation

Split valuation will be used to maintain multiple prices (values) for one product in one plant (valuation area).

Example:

If we are buying product from external and manufacturing internally will have different prices and if your client wants to capture both the prices the we will go for split valuation concept.

Consultant steps in split valuation:

1. Activate split valuation – OMWO.
2. Configure split valuation – OMWC.
 - 2.1. Maintain valuation type.
 - 2.2. Valuation category.
 - 2.3. Assign category to valuation area.

1. Activate split valuation:

Under this heading we will activate split material valuation.

T. code – OMWO.

Path:

SPRO→ IMG→ Material management→ Valuation and account determination→ Split valuation → Activate split valuation→ Enter→ Click on split material valuation activate radio button→ save it.

2. Configure split valuation:

Under this heading we will create valuation type (we have created no of valuation types based on different prices). We will link it with valuation area by valuation category.

T. code – OMWC.

Path:

SPRO→ IMG→ Material management→ Valuation and account assignment→ Split valuation→ Configure split valuation→ Enter→

- Click cancel.
- Click on global types.
- Click on create.

- Give valuation type (10 CHAR).
- Maintain account category different 0001 raw material, maintain attributes internal purchase order allowed→ Press enter.
- Again, click on create icon→ Save it.
- Go back 2 times.
- Click on global categories.
- Again, click on create icon.
- Give valuation category (1 CHAR).
- Maintain description.
- Press enter and click on create icon.
- Then save it. (Under valuation category we will valuation type it will link to valuation area we will assign valuation category to every material if we required split valuation).

Again, go back→ Click on types - cat→ Search your valuation type by using page up and page down key→ Click on your valuation type→ Then click on activate icon→ Save→ Click on cat – OUS→ Search your plant by using page up and page down→ Click on your plant→ Click on activate option→ save.

Client role in split valuation:

Create material and maintain valuation category in accounting 1→ Save the material.

- Extension material to valuation type:
- Go to t. code – MM01- 2 time enter.
- Select only accounting 1 view.
- Press enters again, give your plant.
- Valuation type – enter.
- Maintain valuation class – press enter.
- Price control, maintain price – save it.

Consultant role:

Create purchase order by using ME21N T. code and maintain all required fields then go to delivery tab and maintain valuation type – save the purchase order.

In goods receipt screen under material tab and batch tab we can check valuation type copied from purchase order.

-----Split valuation topic completed-----

Physical inventory

Physical inventory is a process of determination the delivery quantities are exact, if there is any difference in physical stock and SAP books stock then we will use physical inventory to MAP both.

Once physical inventory is finished then SAP books stock and physical stock should be same.

Reasons:

- ❖ Management will have exact stock information.
- ❖ Legal reasons – Audits.
- ❖ Type of physical inventory:
 - ❖ Periodic inventory (once in a year).
 - ❖ Continuous inventory (every month materials will be counted).
 - ❖ Cycle counting (slow moving. Fast moving – 4 times in a year).
 - ❖ Inventory sampling (random material).

Steps involved in physical inventory – client's role:

- Create physical inventory – MI01.
- Enter count result – MI04.
- Post the difference – MI07.

Create physical inventory:

Under this heading we will create physical inventory for the material along with plant and batch if available.

T. code – MI01.

Path:

SAP easy access → SAP Menu → Logistics → Material management → Physical inventory → Physical inventory document → MI01 - create → Enter → Give document date, planned count date → Maintain plant and storage location → Press enter → Give material number and batch → Save it.

T. codes:

- MI01 – Create physical inventory.
- MI02 – Edit physical inventory.
- MI03 – Display physical inventory.

Enter count result:

Once physically inventory s created then store keeper will count physical stock and informs to system under this heading.

T. code – MI04.

Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material management→ Physical inventory→ Inventory count→ MI04 - Enter→ Give physical inventory number→ Maintain year→ Enter→ Enter the count result quantities→ save.

- MI04 – Enter count result.
- MI05 – Edit count result.
- MI06 – Display count result.

Post the difference:

Under this heading we will post the physical inventory stock different.

T. codes – MI07.

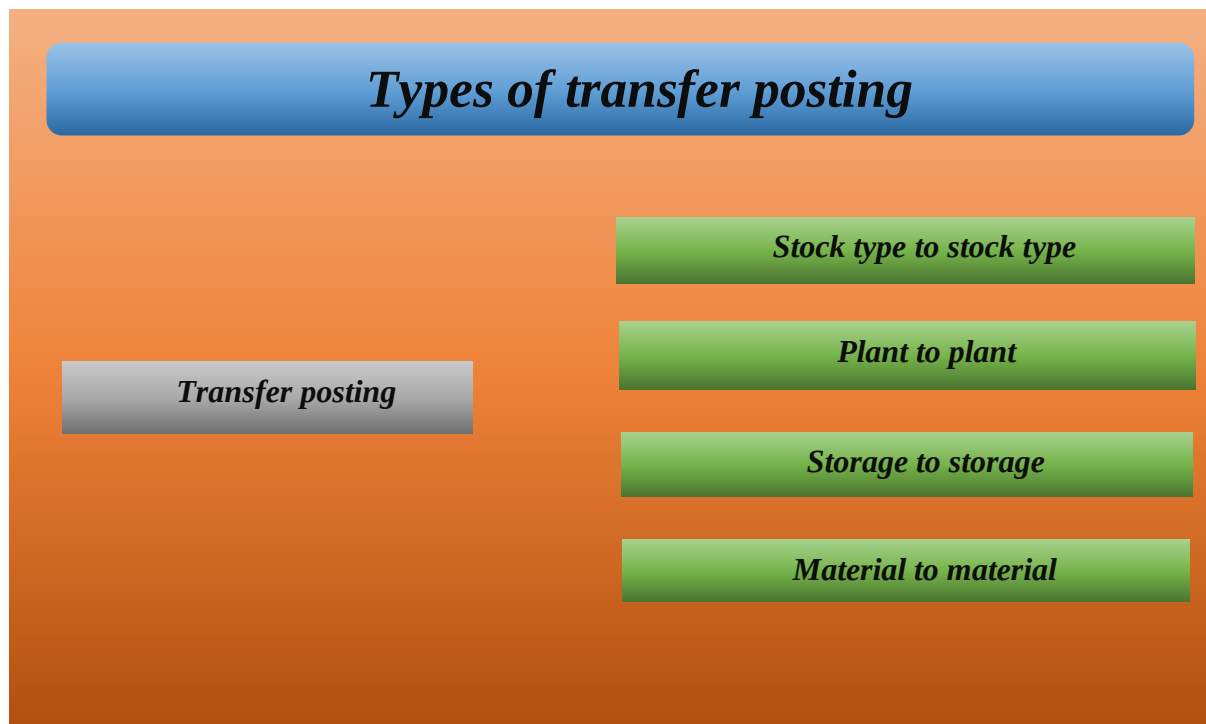
Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material management→ Physical inventory→ Difference→ MI07 - post→ Enter→ Give your physical inventory document number→ Maintain year→ Enter→ Maintain the reason – 0001→ Save it.

-----physical inventory topic completed-----

Transfer posting

Transfer posting will be used transfer goods from one plant to another plant and storage location to another storage location and material to material are change the stock type.



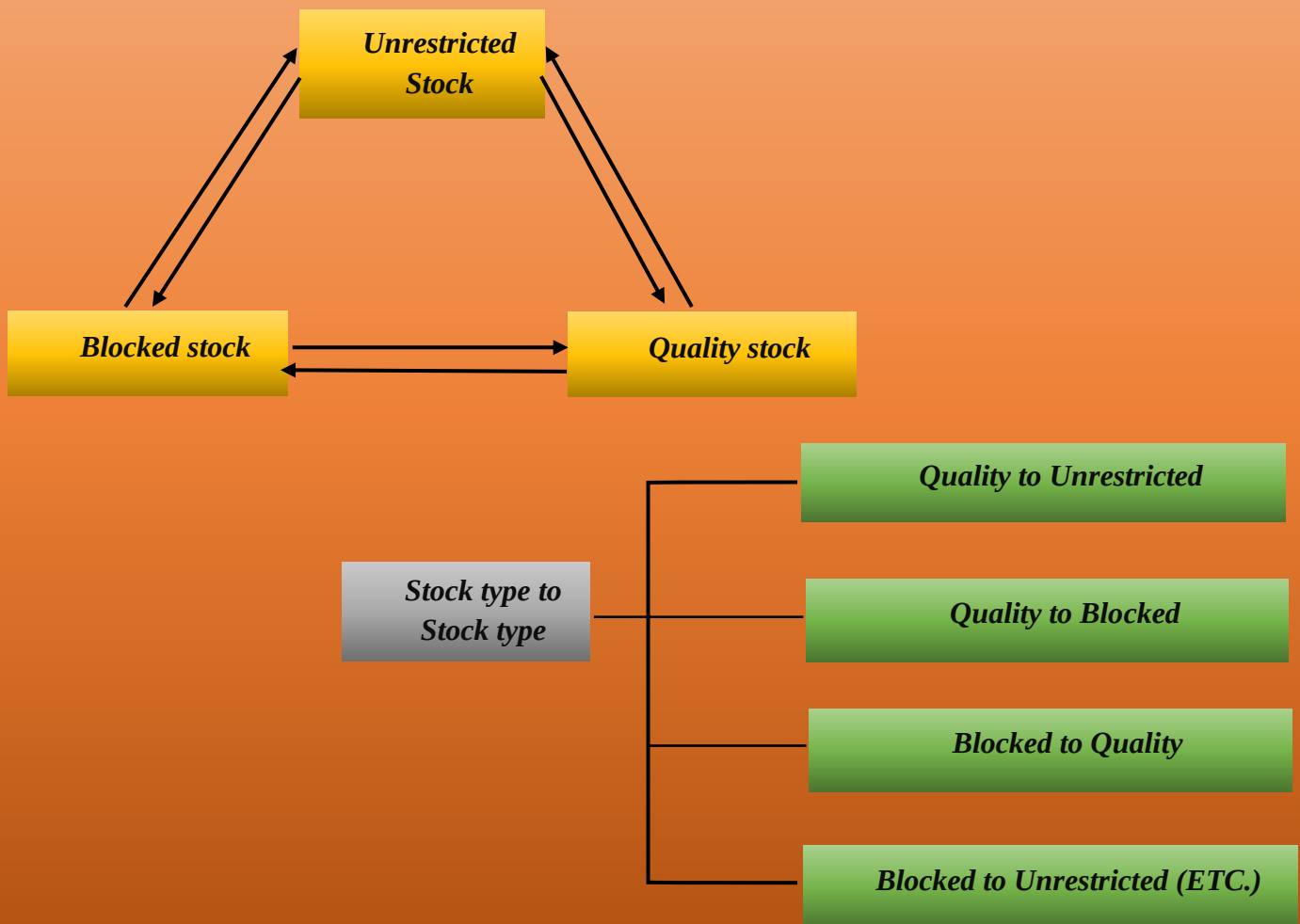
1. Stock type to stock type transfer posting:

In SAP MM – IM will have following types of stocks.

- Unrestricted uses (anyone can use).
- Blocked stock (restricted stock).
- Quality inspection stock (Quality is pending).

Transfer stock from one stock type to another is called stock to stock transfer posting.

Stock type to stock type transfer posting



Quality to unrestricted transfer posting:

We can use MB1B in ECC version and we can use S4 HANA version.

- Go to t. code MB1B.
- Give document date and posting date.
- Maintain movement type – 321.
- Plant and storage location – press enter.
- Give material code, maintain batch – press enter and save it.

Note:

- Check stock in t. code – MMBE.

Plant to plant transfer posting:

- Go to t. code – MIGO.
- Select event transfer posting, reference documents other – press enter.
- Give document date and posting date.

Transfer posting divided two column:

From	Destination
❖ Give material code. ❖ Suppling plant and storage location. ❖ Maintain suppling batch	❖ Material code will copy automatically. ❖ Give receiving plant and storage location. ❖ Receiving batch (same suppling batch and receiving batch).

- Maintain quantity.
- Go to where tab: Give movement type – 301.
- Activate item ok – click on check icon – Then save it.

Note:

- Check stock in t. code – MMBE.
- Plant to plant transfer posting.

Accounting entries are:

- BSX – Inventory A/c – DR (+).
- BSX – Inventory A/c – CR (-).

Storage to storage transfer posting:

We can use this transfer posting to transfer goods to one storage location to another storage location within a plant.

- Go to t. code – MIGO.
- Select event transfer posting, reference documents other – press enter.
- Give document date and posting date.

Transfer posting divided two column:

From	Destination
❖ Give material code.	❖ Material code and plant will copy automatically.
❖ Suppling plant and storage location.	❖ Give receiving storage location.
❖ Maintain suppling batch	❖ Receiving batch (same suppling batch and receiving batch).

- Maintain quantity – press enter.
- Go to where tab: Give movement type – 311.
- Activate item ok – click on check icon – Then save it.

Note:

- Check stock in t. code – MMBE.
- No accounting entries generate.

Material to material transfer posting:

Under this heading we can transfer material quantity from one material code to another. If both materials are same but materials code are different.

We can use same for transferring material quantity one batch to other within plant and same materials.

- Go to t. code – MIGO.
- Select event transfer posting, reference documents other – press enter.
- Give document date and posting date.

Transfer posting divided two column:

From	Destination
❖ Give material code.	❖ Give receiving material code.
❖ Suppling plant and storage location.	❖ Give receiving plant and storage location.
❖ Maintain suppling batch	❖ Receiving batch (same suppling batch and receiving batch).

- ❖ Maintain quantity – press enter.
- ❖ Go to where tab: Give movement type – 309.
- ❖ Go to partner tab: Give vendor.
- ❖ Activate item ok – click on check icon – Then save it.

Note:

- ❖ Check stock in t. code – MMBE.
- ❖ No accounting entries generate.

Activate batch status management

T code – OMCS.

SPRO→ IMG→ Logistics general→ Batch management→ Specify batch level and activate status management→ Enter→ Click on batch status management→ Click on activate radio button-save it

Batch level:

T. code – OMCE.

We will maintain batch number unique at following level:

Definition-batch level:

- Batch unique at plant level.
- Batch unique at material level.
- Batch unique at client for a material.

By default, SAP will activate batch unique at material level.

Path:

SPRO→ IMG→ Logistics general→ Batch management→ Specify batch level and activate status management→ Enter→ Click on batch level→ Then activate batch unique at material level→ Save it.

Batch number assignment:

Activate internal batch number assignment.

T. code – OMCZ.

Path:

SPRO→ IMG-Logistics general→ Batch management→ Batch number assignment→ Activate internal batch number assignment→ Enter→ Double clicks on activate batch number assignment line→ Click on activate radio button→ Save it→ Go back→ Double clicks on internal batch number assignment for assigned goods receipt→ Enter→ Click on positive→ Give your plants→ Enter→ Activate batch no automatically for GRW acct→ Enter and Save.

Maintain internal batch number assignment range:

T code – OMAD.

Path:

SPRO→ IMG→ Logistics general→ Batch management→ Batch number assignment→
Maintain internal batch number assignment range→ Enter→

- Give batch object batch – CLT.
- Click on ranges.
- Click on change intervals.
- Then insert line.
- Give serial no 01.
- Give from and to number range (10 CHAR).
- press enter and save it.

Client role:

1. If we require batch for any material then activate batch management field in material master purchasing view while creating material.
2. Batch number will generate automatically while performing GR.

T. code:

- MSC1N - Create batch.
- MSC2N - Change batch.
- MSC3N - Display batch.
- MSC4N – Changes.

Batch tables - SE16N:

- MCHA – Batches.
- MCHB - Batch stock.
- MSPR - Project stock (PS module).

----- **Activate batch status management topic completed**-----

Automatic Account Determination (AAD) – OBYC

In procurement process cost of goods and service will involve which need to make payment to vendor by the company. The cost being paid to vendor must be posted in the organization systems in a correct GL account every time during procurement so it may lead in wrong selection of G/L account, to avoid this these manual selections we will go for automatic account determination concept.

In AAD, system will select GL account automatically based on following setting and in real time this setting will be done by FI consultants.

- Define valuation control – OMWM.
- Group together valuation areas – OMWD.
- Define valuation class – OMSK.
- Configure automatic posting – OBYC.

Define Valuation Control:

T. code - OMWM.

Path:

SPRO→ IMG→ Material management→ Valuation and account assignment→ Account determination on wizard→ Define valuation control→ Enter→ Click on valuation grouping code active radio button→ Save it.

Group together valuation areas:

In this step valuation area and valuation group code are assigned to company code.

T. code – OMWD.

Path:

SPRO→ IMG→ Material management→ Valuation and account assignment→ Account determination→ Account determination without wizard→ Group together valuation area→ Enter→ Click on position→ Give your plant→ Enter→ Assign valuation group pin code (0001) or leave it blank→ Press enter→ And save it.

Define valuation class:

In SAP standard each material type will have one valuation class.

T. code – OMSK.

Path:

SPRO→ IMG→ Material management→ Valuation account determination→ Account determination→ Account determination without wizard→ Define valuation classes→ Enter→ Click on valuation classes→ Click on new entries→ Give valuation classes (4CHAR) → maintain account category reference (0001) → Maintain valuation classes distribution press and save it.

4.Configuration automatically posting-OBYC AAD related error:

Account determination for entry INT GBB/ BSX/ WRX/ FR1... In 3000/ 7920/ 34000...

Not possible in goods movement.

Solution:

If we miss to assign GL account in OBYC T code for our valuation class assigned in material then this error will trigger. In this case we have to assign GL account to the transaction key with valuation class combination as shown below:

Transaction keys:

- ❖ BSX - Inventory posting.
- ❖ WRX - GR/ IR Cleaning.
- ❖ BSV - Off setting entry.
- ❖ FRI – Freight.
- ❖ KON - Consignment payables.
- ❖ PRD - Price difference etc...

Chart of account → INT (only in practice):

- ❖ Go to t code-OSYC.
- ❖ Double click on transaction key.
- ❖ Chart of account INT.
- ❖ Click on new entries.
- ❖ Give valuation modifier (valuation grouping code) + Valuation class.
- ❖ Maintain GL account→ press and save it.

----- Automatic Account Determination (AAD) – OBYC -----

External service management (ESM)

In every company along with material sometimes they require services which include installations, testing painting works, lighting, housekeeping etc... These services are provided by an external vendor hence this entire process of purchasing services is called external services management.

Key points:

1. Services are always direct consumption.
2. Item category(D) represents services.

1. Service management deals with following terms:

- ❖ Service master goals.
- ❖ Service purchase order.
- ❖ Service entry sheet (SES).

1.1. Service master goals:

Same like material master services master contains the details of services that can be procured from an external vendor. It contains details of vendor, unit of measurement etc...

1.2. Business requirements:

In real time project company procures different types of services like AMC, installation and post testing housekeeping works, outsourcing works etc... we consultants have to show the difference between each service type.

Consultant role:

1. Define service category:

Service category different types of services in Sap (some like material type).

SAP has defined standard service category.

ALL - All service:

We can use standard all or we can customize our own service category by copying standards.

Examples:

- ZAMC - Maintenance services.
- ZINT - Installation and testing etc...

Path:

SPRO→ IMG→ Material management→ External service management→ Service master→ Define service category → Enter→ Select all line→ Click on copy as icon→ Give your own services category (4 CHAR) → Maintain description→ Press enter→ Save it.

Define number range for services master:

In is same like material master we can maintain internal & external number ranges.

T. code - ACNR.

Path:

SPRO→ IMG→ Material management→ External service master→ Service master→ Define number ranges→ Enter→ Click on maintain group→ Click on create icon→ Maintain group name→ give from and to number range→ for internal external if required→ press enter and save it→ go back→ click on ctrl+ F→ give service category→ enter→ click on element/ group→ search your number range group→ double click on your number range→ save it.

Define field selection of service master:

T. code – OXA2.

Path:

SPRO→ IMG→ Material management→ External service management→ Services master→ Field display for service master→ Define field selection for service master (individual maintenance) → Enter→ Manage the field (input)→ And save it.

Client role in service master:

Create service master:

T code- AC03

Path:

SAP easy access screen→ SAP Manu→ Logistics→ Material management→ Service master→ Services→ ACO3 - Service master→ Enter→ Click on create new service icon→ Give service description→ Select service category→ Maintain base unit of measurements→ Maintain material group→ Maintain valuation class (3200) press enter→ Save it.

T. code - AC03→ Create, display, edit, delete, service master.

AC06→ List of service master.

Service master tables – SE16N:

ASMD→ Service master.

Create service PO:

- Go to t code- ME21N.
- Select service document type or standard NB.
- Give vendor code.
- Give pouch org, pouch group and company code.

-----**Enter**-----

Expand item overview / as or go to item overview:

- Give item category (D) - Services.
- Material account assignment – K -Cost enter.
- Material description (services description ex- AMC charges).
- Delivery date.
- Material group (service group).
- Maintain plant.

-----**Enter**-----

Go to service tab in item data:

- Give service master number(optional).
- Give service description directly without services master number.
- Maintain quantity (total service value).
- Maintain unit price (always 1).
- Maintain UOM if it is without service number.

-----**Enter**-----

- Maintain GL account - 417000 (service).
- Maintain cost center - click on continue.
- Click on check item and the save the service PO.

Service entry sheet (SES):

SES is also called as goods receipt. When we order materials then we will do good receipt and if we order service then we will maintain service entry sheet.

Whenever we receive services from vendor then we will maintain services entry sheet to capture services received from vendor.

Consultant role:

Define number ranges service entry sheet.

T. code - OMH9.

Path:

SPRO→ IMG→ Material management→ External service management→ Number ranges→ Define number range for services entry sheet→ Enter→ Click on change internals→ Insert line→ Give serial number, maintain from and to number range (10 CHAR) → Save it → Go back 2times→ assign number range→ Enter→ Assign your number serial number→ In number range for entry sheet column→ Save it.

Define field selection for service entry sheet.

Path:

SPRO→ IMG→ Material management→ External service management→ Define screen layout→ Enter→ Click on position→ Give field selection key ML81→ Enter→ Double clicks on ML81→ Double clicks on field selection group→ Manage the fields→ Continue→ Go back and save it.

Client role in service entry sheet:

Maintain service entry sheet (SES):

Once we receive the services from vendor then we will maintain SES to capture receive services in SAP (same like GR).

T. code – ML81N - Maintain SES:

Path:

SAP easy access screen→ SAP Menu→ Logistics → Material management→ Service entry sheet → ML81N - maintain → Enter.

- ❖ Click on other PO icon.
- ❖ Give service PO no→ Enter.
- ❖ Click on create icon.
- ❖ Maintain short text (1st month payment).
- ❖ Click on Accept. data→ Maintain document date and posting date.
- ❖ Click on service selection icon (bottom of the screen) → Continue immediately.
- ❖ System will display service line item from PO.
- ❖ Select service line item→ Click on services icon→ Continue→ Maintain the value (based on received services) → Enter→ Click on accept icon→ If service entry sheet status is yellow→ Then save the service entry sheet (green flag).

T. codes:

- ML81N – Maintain/ delete/ display SES.
- ML84 – List of service entry sheet.
- ML85 – Collective release.

Tables – SE16N:

- ESSR – Service entry sheet header data.
- ESCC – Service item.
- ESHL – Service header.

Steps involved in service purchase:

- Create service master t. codes – AC03.
- Create vendor t. codes – XK01.
- Create service PR t. codes – ME51N.
- Create service PO t. codes – ME21N.
- Maintain service entry sheet (SES) t. codes – ML81N.
- Invoice receipt t. codes – MIRO.

----- **External service management (ESM) Topic completed**-----

Release strategy

Release strategy will be used to approve purchasing document in SAP system. Purchasing document are like PR, RFQ, PO system contracts and scheduling arrangement.

Possibilities of release strategy configuration:

We can configure release procure (RP) with and without classification RP with classification.

1. RP with classification will be used to approve internal and external purchasing document (PR, PO, RFO, Contracts, SA).
2. PR can be released at header and item level as well.
3. All external purchasing document will be released at header level only.

4. Communication structure:

- ❖ PR – CEBAN.
- ❖ PO – CEKKO (all external purchasing document).
- ❖ SES – CESSR.

PR without classification:

1. RP without classification will be used to approve only PR.
2. PR can only be release at item level.
3. No scope for external purchasing documents.
4. No communication structure.

Consultants' steps involved in release strategy:

1. Edit characteristics – CT04.
2. Edit class – CL02.
3. Define release group.
4. Release codes.
5. Release indicators.
6. Release strategy.
 - 6.1. Release pre – requisites.
 - 6.2. Release statutes.
 - 6.3. Release classification.
 - 6.4. Release simulation.

1. Edit characteristics:

Under this heading we will create any purchasing field character just to communicate purchasing tables to triggered release strategy tab in purchasing document.

T. codes – CT04.

Purchase order value character:

T. codes – CT04.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchase order→ Release procedure for PO→ Edit characteristics→ Enter→ Give characteristics (29 CHAR) → Click on create icon.

Go to additional tab:

- Give communication structure tables name filed.
- Maintain field name (GNETW) for PO value.
- Press enters.
- Then continue – again continue.
- Maintain currency.
- Activate multiple value radio button.
- Activate internal vals allowed.

Go to values tab:

- Maintain the condition value - save it.

Document type character:

- Go to t. code – CT04.
- Give characteristics.
- Click on create icon.

Go to addition tab:

- Give communication structure table name field.
- Maintain filed name.
- Press enters – continue.
- Activate multiple value radio button.

Go to value tab:

- Maintain document type condition.
- Press enters – save it.

Edit class:

Under this heading we will create a class to assign all characteristics together.

Release class type – 032.

T. code – CL02.

Path:

SPRO→ IMG→ Material management→ Purchasing→ purchase order→ Release procedure for PO→ Edit class→ Enter.

- Give purchase order class.
- Maintain class type – 032.
- Click on create icon.

Go to basic data:

- Give class description.

Go to char tab:

- Assign your characteristics – press enter and save the class.

Define release procedure for PO:

Path:

SPRO→ IMG→ Material management→ Purchasing→ Purchase order→ Release procedure for Purchase orders→ Enter→ Define Release procedure for Purchase order→ Enter.

Release group:

- Double click on release group.
- Click on new entries.
- Give release group (2 CHAR).
- Assign your PO class (step).
- Maintain release group description.
- Press enter and save it.
- Go back 2 time.

Release codes:

- Double click on release code.
- Click on new entries.
- Give release group (step -3).

- Maintain release code (2 CHAR, release code representative the people who release purchasing document, up to *8 levels can be used in SAP standard for release purchasing document).
- Maintain description – press enter and save it.

Ex:

- Level – 1 – PE – Purchase executive.
- Level – 2 – PM – Purchase manager.
- Level – 3 – PH – Purchase head.
- Level – 4 – VP – Vice president.
- Go back 2 time.

Release indicator:

- Double click on release indicator line.
- Check the release ID's B (blocked) and R (released) check box should be activated.
- Maintain release indicator's (1 – 6) to release ID's – press enter and save.
- Go back 2 times.

Release strategies:

- Double click on release strategies.
- Click on new entries.
- Give release group (PO).
- Give release strategy.
- Maintain description.
- Maintain release code in a sequence.
- Press enters.
- Click on release prerequisites.
- Draw this box and continue.
- Click on release statistics.
- Check the release indicator assign to release code – continue.
- Click on classification.
- Select condition value – continue.
- Click on release simulation – continue.
- Save the release strategy.

Clients testing in release strategy:

Create PO by using t. code – ME21N.

Maintain all required field.

Click on check icon if PO condition are satisfying with release strategy conditions, then system will add new tab called release strategy in PO header level.

Save the PO.

How the release PO:

In standard SAP we/ client can approve purchase order in following 2 ways.

- Individual release – ME29N.
- Collective release – ME28.

PO individual release:

- Go to t. code – ME29N.
- Click on other purchase order icon.
- Give your release PO number.
- Press enters.
- Click on edit icon if PO is in display mode.

Go to release tab:

- Click on release tick mark.
- Save the PO to effect release (approval).

Individual release means we will approve single purchase order.

Purchase order collective release:

Collective release will be used mainly by manager and heads to release multiple PO's at a time.

- Go to t. code – ME28.
- Give release code.
- Execute and system will display all the approval pending PO's.
- Single click on every PO then clicks on release icon and save.

-----Release strategy topic completed-----

Condition master data (pricing)

Condition master data (pricing) will be used to maintain prices in purchasing document like gross price, discounts, freight charges and other charges etc...

consultant step involved in pricing procedure:

1. Maintain condition tables – M/03.
2. Access sequence – M/07.
3. Define condition types – M/06.
4. Define calculation schema – M/08.
5. Schema group.
 - 5.1. Schema group for vendor.
 - 5.2. Schema group for purchase org.
 - 5.3. Assign schema group purchase org. to purchase organization.
6. Schema determination (pricing procedure).

Maintain condition table:

- Condition table will be maintained if we want condition type value automatically.
- Condition table are key combination of field from purchasing document.
- SAP has already defined -1 – 500 condition tables and if we consultants want to configure new condition tables then we have to use number from 501- 999.

Ex:

- Material + Vendor + Plant – Ex- 688.
- Plant + Purchase organization – Ex- 689.

Create condition tables:

T. codes – M/03.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Condition's→ Define price determination process→ Maintain condition table→ Enter.

- Double click on create condition table line.
- Maintain your condition table (501 to 999).
- Press enters.
- Search key combination (or) search field by using page up and page down keys.

- Then double click on field to select into key combination. (Combination material + vendor + plant).
- Click on general icon.
- **T. codes:**
- M/03 – Create condition table.
- M/04 – Change condition table.
- M/05 – Display condition table.

Maintain access sequence:

Access sequence is a search strategy if will search key combination from most specific to most generic.

In this access sequence we will assign condition table. This access sequence will be assigned in condition type such as gross price, discounts, freight etc... to get automatic price (value).

T. code – M/07 – Maintain access sequence.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Conditions→ Define Price determination process→ Define access sequence→ Enter→ Continue.

- Click on new entries.
- Give access sequence (4 CHAR alpha numeric).
- Maintain description- Enter- Enter.
- Select access sequence line.
- Double click on access (left side).
- Click on new entries give serial number, maintain condition tables- press enter till you get description.
- Select condition table line 1.
- Double click on field (left side).
- Press no of enters based on condition tables available from selection.

Go back and save it.

Define condition type - T. code – M/06:

Condition type is also called pricing element. each and every condition type presents one pricing type such as gross price, discount, freight charges, etc...

SAP has defined below standard condition type which we can use to fulfil client business requirements.

we will maintain price against each condition type in purchasing document.

- PB00 – Gross price automatic.
- PBXX – Gross price manual.
- RA01 – Discount % on gross.
- RA00 – Discount % on net price.
- FRA1 – Freight %.
- FRB1 – Freight value.
- FRC1 – Freight quantity.
- SKTO – Cash discount.
- JCDB – Basic custom only.

Use below t. code or path for creation or setting of condition types M/06.

path:

SPRO→ IMG→ Material management→ Conditions→ Define price determination process→ Define condition→ Enter→ Double click on define condition line→ Enter→ Click on position→ Give standard price condition type (PB00) for gross price→ Enter→ Select PB00 line→ Click on copy as icon→ Give your own condition type (4 CHAR) → Maintain description→ Press enter and save it.

Note:

Activate accruals check box for all delivery cost condition type to capture the delivery cost in separate GL accounts.

Examples:

Freight charges, Insurance cost etc...

Define calculation schemas – M/08:

Under this heading we will maintain condition type in a sequence to calculate price of the product. SAP define below standard calculate schemas.

Which we will not using real time project.

- RM0000 – Domestic purchase.
- JIMPOR – Import purchase.

In real time project we consultants we customize new pricing procedures to fulfil clients’

Requirements by using below t. codes or path.

T. codes – M/08.

Path:

SPRO→ IMG→ Material management→ Purchasing→ Conditions→ Define price determination process→ Define calculate schema→ Enter→ Click on new entries→ Give

pricing procedure (6 CHAR) → Maintain description→ Press enter→ Select pricing procedure line→ Double click on control data→ Click on new entries→ Maintain below control data – 16 steps.

1. Give step – serial no. (10,20,30 etc...).
 2. Maintain counter – Sub step (10.1, 10.2, etc...).
 3. Maintain condition types in sequence (check in below table).
 4. Description – It will copy from condition types.
 5. Give FROM No – from where we have to calculate.
 6. Give TO No – till where we have to calculate.
 7. Activate manual – If we want to enter condition type manually in PO then activate this filed.
 8. Required (Mandatory) – If we activate this field for any condition type then that will become mandatory in the pricing procedure.
 9. Statistical – If we activate this field in pricing procedure for any condition type then it will not allow the value of condition type to be taken into net value calculations.
 10. Activate this field for all delivery cost condition types.
 11. Print – If we want to print any condition type in PO print out then activate this field by maintaining (X).
 12. Subtotal – This field will determine values to be calculated in subtotals. Give subtotal (9) for gross price.
 13. Requirement – It is a routine that is written by ABAP according to business requirement.
 14. Calculation type – It is again a routine that is written by ABAP.
 15. Base type – It will determine that base for calculating the value of condition types alternative to the standard.
 16. 15/16. Account key/ Accruals – If we want to capture the condition values in separate values in separate GL account then maintain transaction key in these fields in pricing procedure.
- (15 – Account key will search GL account and accruals will post the value in that GL account).
- Save it.

----- **Condition master data (pricing) topic competed** -----

Schema group

under this heading we will create schema group for vendor and purchasing organization. Schema group will help to trigger pricing (calculate schema – step -4) in condition tab to purchasing document.

5.1 Schema group for vendor:

In real time project we will create 2 schema group for vendor.

Examples:

DM – Domestic vendors.

IM – Import vendors.

Path:

SPRO→ IMG→ Material management→ Conditions→ Define price determination process→ Define schema group→ Enter→ double Schema: vendor→ Click on new entries→ Give schema group vendor (2 CHAR), maintain description→ Press enter and save it. (This schema group we will assign in our vendor master).

5.2 Schema groups for purchasing organization:

Path:

SPRO→ IMG→ Material management→ Purchasing→ Conditions→ Define price determination process→ Define schema group→ Enter→ Double click schema group for purchasing organizations→ Click on new entries→ Give your schema group purchase organization (4 CHAR) → Maintain description→ Press enter and save it→ Go back 2 times→ Double click on assignment of schema group to purchase organization line→ Click on position→ Give your Purchase organization→ Enter→ And assign you're in schema group Purchase organization→ Press enter and save it.

6. Define schema determination: (pricing procedure)

Under this heading we will assign calculation schema (pricing procedure) and schema group. (Step -4 and step -5).

Path:

SPRO→ IMG→ Material management→ Purchasing→ Conditions→ Define price determination process→ Define schema determination→ Enter→ Click on determine calculation schema for standard purchasing orders→ Click on new entries→ Give schema group vendor and assign pricing procedure→ Press enter and save it.

Client role in condition master data:

Assign schema group vendor in vendor master. It will help to trigger pricing procedure in purchasing order condition tab.

- ✓ Use t. code – XK01 for new vendor.
- ✓ XK01 for already created vendor.

Maintain condition record condition record (Condition master data):

As we ware if any condition type value (gross price or discount etc...) required automatically in purchasing document then, we will assign access sequence in those condition type. Condition types which are maintained access sequence for those condition type we have to maintain condition record. (Nothing but value which will trigger in purchase organization automatically). These condition record will be maintained by purchasing end users.

T. code – MEK1 – Maintain condition records.

path:

SAP easy access screen→ SAP Manu→ Logistics → Material management→ Master data→ Conditions→ Other→ MEK1 – create→ Enter→ Give your condition type→ Which access sequence is assigned→ Click on key combination→ Select key combination→ Enter→ Maintain key combination values→ Maintain amount maintain valid from and to dates→ Press enter and save it.

T. codes:

- MEK1 – Maintain condition record.
- MEK2 – Edit condition record.
- MEK3 – Display condition record.

Tables SE16N:

- KONH – Condition header.
- KONP – Condition item.

-----Schema group topic completed-----

WORK SHOP CONTENT

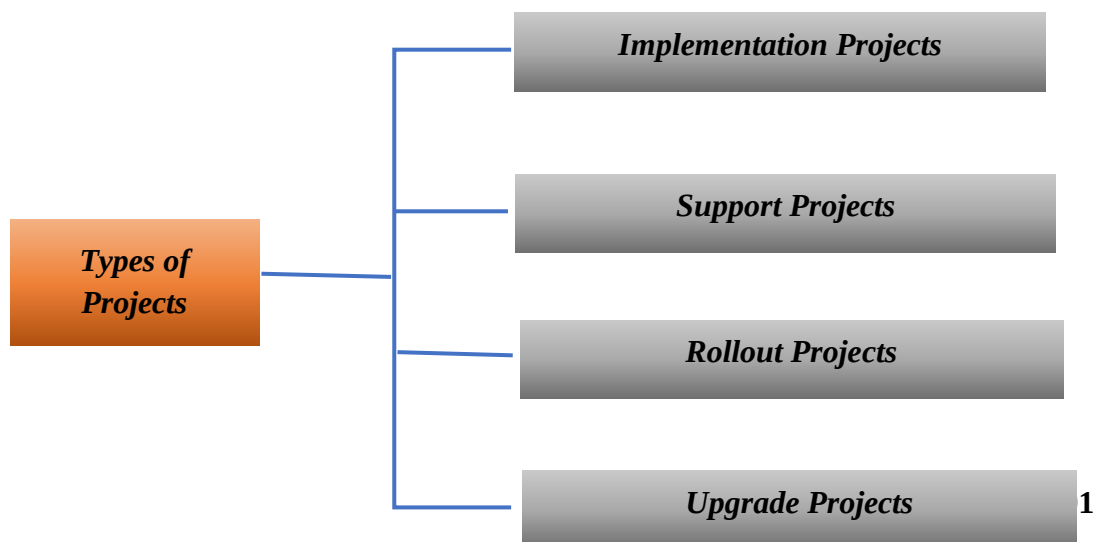
1. Introduction to project/content.
2. Types of projects.
3. Roles & responsibilities in projects.
4. TR – Transport request.
5. Documentation in projects.
6. Meeting in projects.
7. Cutover activities.
8. Cutover templates.
9. Ticketing tool.
10. RICEFW – Report, Interface, Conversions, Enhancements, From and workflow.
11. Data uploads via LSMW, LTMC, BADI and BAPI.
12. Challenges/Issue in projects/ CR.

WHATS IS PROJECTS IN SAP

A project is defined as a task to create a product or to provide a service. Projects will have cost, duration and resources.

TYPES OF PROJECTS IN SAP

In SAP we have below different types of projects:



IMPLEMENTATION PARTNER

1. IBM
2. TCS
3. Accenture
4. Deloitte
5. Mind tree
6. Yash
7. Wipro
8. NTT
9. Infosys
10. Some small IT companies.

WHATS IS IMPLEMENTATION PROJECTS

If any client/customer wants to turn up their business operation system into SAP then they will raise request to implementation partner (any IT company) to convert operating system into SAP.

Converting client's business operations into SAP system for first time is called implementation projects.

Implementation projects takes 8 months to 12 months duration based on project size with team size 15+ members and completely it depends on project size & budget.

EXAMPLES OF CLIENTS

1. MRF	1. LG Polymers
2. HERO	2. DR. Reddy's
3. Asian paints	3. Adani Groups
4. Jindal steels	4. Zandu
5. Sony	5. Cipla
6. Titan	6. HAL
7. Nestle	7. Mankind Pharma
8. Honda	8. BHEL
9. Bajaj	9. GAIL
10. TVS motors	10. BSNL
11. Ford	11. BOSCH
12. Hindustan Motors	12. BEML
13. Maruti	13. CEAT
14. Big Bazaar	14. DMART
15. Coca cola	
16. Tata Chemicals	
17. Ambuja Cements	
18. Ultra Tech	

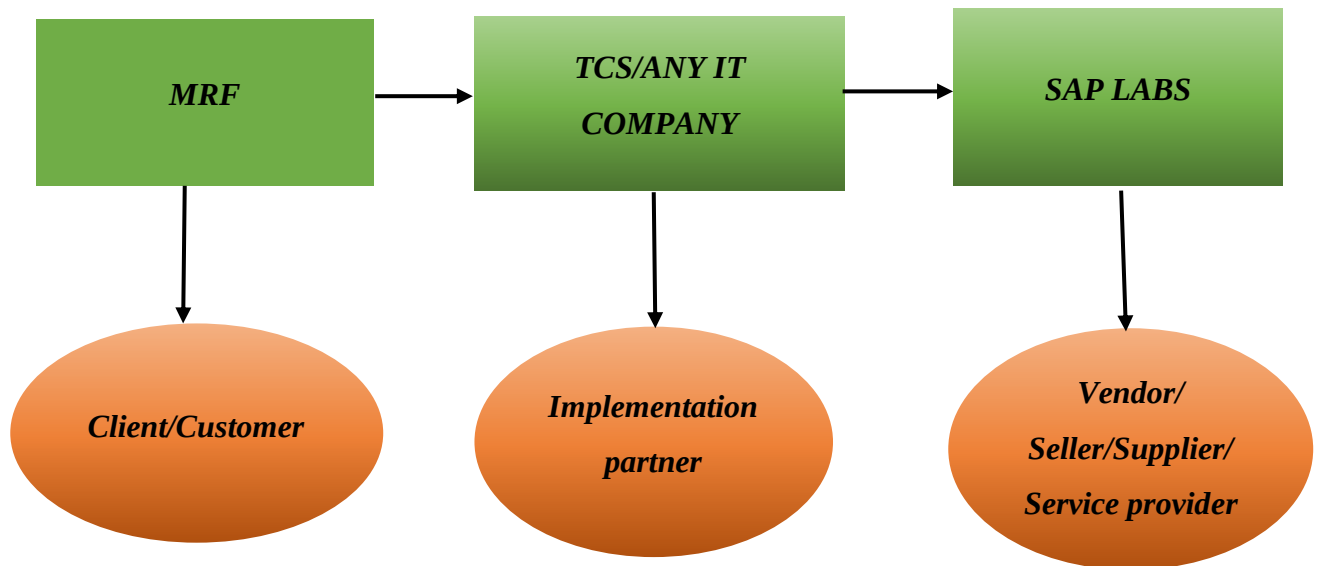
Methodologies in SAP

Implementation partner required proper methodology (route map) to implement SAP to companies (Any industry). In SAP we have following route map to implement.

- ❖ ASAP – Accelerated SAP.
- ❖ AGILE
- ❖ SCRUM
- ❖ S – IMPL
- ❖ RUN SIMPLE
- ❖ ACTIVATION (S/4 HANA) – 2016.

* In ECC version all companies used ASAP methodology but after S4 HANA started using ACTIVATE.

Examples



What is ASAP Methodology?

ASAP is a route map or a methodology used to implement SAP to companies in different stages.

Stages is also called as phases.

In ASAP we have 5 different route map/ phases. Clients' business process will be implemented into SAP in 5 different stages.

Phases in ASAP Methodology?

- ❖ Project preparation phase.
- ❖ Business blueprint phase (BBP).
- ❖ Realization phase.
- ❖ Final preparation phase.
- ❖ Go live phase.

* In between final preparation and go live we will get cutover phase.

Project preparation phase

In project preparation phase both the management team will involve to finalise infrastructure, license, consultants, projects timeline etc...

Once consultants got finalised then both clients end user team and implementation consultants' team will approach to have introduction and project overview discussion which is called **KICK- OFF** meeting.

In the **KICK- OFF** meeting both the teams will introduce each other than clients project manager will have their company's business overview.

Once project overview is done then we consultants have to prepare questionnaire (question list to know client business process in details). Every module consultant will prepare their own questionnaire.

Business Blue Print Phase

Business blue print is a phase as well a document.

BBP document consists of AS-IS and **TO-BE** documents. AS-IS document consists of clients

Current non- SAP business and **TO-BE** document consists of clients SAP business operations.

Once questions list is ready then we consultants will drop an email to clients and end user to know the answers for the questions and once client started to explain business process, we consultants will capture in a document called **AS-IS**. This will be done by individual module consultants.

While mapping **AS-IS and TO-BE** process we will get **GAP** in client's requirement. We will prepare a separate document for **GAP** topics.

Once BBP document is ready then we consultants will explain to client users and project manager if they satisfied with the **TO-BE** document then they will get approval (sign off) to next phase. This entire explanation process happens in the form of meeting called sign-off meeting.

What is Transport Request (TR)?

Each and every configuration in development server – golden client will be stored under TR – transport request.

TR will be used to move the configuration from one server to another server (DEV – QAS – PRD).

The data maintained, edited or deleted or copy in IMG screen of development server will store in TR form.

We have following ways to select TR to store configuration:

Last used TR.

We can create new TR by selecting creation option.

We can select existing TR's by using own request options.

Types of TR's:

We have parent TR (main TR) and child TR (sub-TR). Every configuration will store under child TR of main TR.

If we want to move TR from development server to quality server, we need to release TR but if we want to move from quality server to production TR release is not required.

What is transport request (TR)

How to release TR's:

If we want to move TR from development server to quality server, we need to release TR but if we want to move from quality server to production TR release is not required.

Use T. code SE01/ SE09/ SE10 to check, delete and release TR's. Go to SE01→ Give user→ Then activate modifiable status check box then press enter→ System will display all TR's list related to the user→ Select main TR then expand (+) now select sub-TR→ Click on release icon to release icon to release sub-TR now select main TR again and release it.

Note:

Once TR is released then we can't use to store the data again. It will be used only to move data.

What is transport request (TR)

How to move TR's:

We can move TR's from client to client and server to server.

Client to client TR movement:

If we are moving the configuration from golden client to testing client of development then we will use this client-to-client concept.

T. code – SCC1:

Go to T. code – SCC1 in testing client→ Maintain source list (golden client code), give main TR, activate including subtask check box→ Client on start immediately option.

Server to server TR movement:

If we are moving data for DEV – QAS – PRD servers then we will use the concept. We have to release TR's move data from server to server.

T. code – STMS_IMPORT:

Login into destination server then go to T. code – STMS_IMPORT, select the TR then click import option.

In real time project TR's will be moved by BASIS consultants but we individual module consultants should share TR's list in excel in proper sequence.

Final Preparation Phase

Once all the TRs are transferred to quality server then we will start 4 phases i.e., final preparation phase.

Before starting this phase, we consultants have to provide training to end users, once training is done, we will give 15-day time for testing.

If this phase client end users of all the department will perform testing in quality server.

If they found everything okay then they will give acceptance note called UAT.

Client will do testing in quality server.

UAT – User Acceptance Test. Please find below UAT format but company to company format may get change.

Once UAT is done then we consultants will start cutover activities and prepare for go live.

<i>UAT FORMAT</i>						
<i>S.no</i>	<i>Topic</i>	<i>T. code</i>	<i>Input</i>	<i>Expected results</i>	<i>Actual result</i>	<i>Status</i>
1	Creation of Material master	MM01	IS, MT, Field etc...	Material 0001 created	Material 0001 created	Accepted
2	Change material	MM02	Material code, plant, & SL	Material 0001 changed	Material 0001 changed	Accepted
3	Mass maintenance	MM17	Material codes	Vendor 123 created	Vendor 123 created	Accepted
4	Vendor creation	XX01				
5	Change of vendor	XX02				

Cutover activities

1. Decide cutover date.
2. Decide **go live** date.
3. Move all the TRs to production server in a sequence, this will be done by BASIC team.
4. Once TRs are moved then we consultants have to create again number ranges, characteristics, class, condition tables etc in production server manually.
5. Data upload – We consultants have to upload clients non-SAP (legacy system) data into SAP is called data uploads, from MM we will upload below data:
 - ❖ Material Master data.
 - ❖ Vendor Master data.
 - ❖ Initial stock upload (opening stock).
6. Open orders and GR creation capturing.
7. Once everything uploaded and moved in production then we will hand over the project to client called **go live**.

What is cutover and Golive dates?

The date on which the clients legacy data extracted and uploaded in sap is called cutover date. Once cutover is decided then clients will not enter any data in their legacy system, they will stop all their payments and purchases between cutover and golive dates. If their is any data left even after golive then it will be entered manually.

The date on which first entry is posted in sap, it is called golive date.

Golive and Post golive support

After all cutover activities and gaps fulfilment we will handover the project to clients called golive phase. Once golive is done then clients will enter all business transaction entires in sap.

After golive we consultants have to give support to clients for safe entry in sap. This will be for 4 weeks to 6 months depends on agreement between clients and implementation partner. Providing support immediately for golive is call post golive support.

Once post golive support is done then actual support project will start.

Meetings in Implementation project

1. Kick off meeting - Phase 1
2. Sign - off meeting - phase 2
3. Golive meeting. (Project handover meeting)

Documentations in Implementation project

1. Questionnaire (list of questions)
2. AS - IS and TO - BE
3. BBP document
4. User manuals
5. Test cases
6. UAT - user acceptance test
7. Configuration document
8. T. Codes list
9. Reports list

Templates in Implementation project

1. **Material Master data upload**
2. **Vendor master data upload**
3. **Initial stock upload template**

**** Templates will be used to upload the data from Non SAP to SAP. Templates will be in excel format and will be prepared by individual module consultants.**

What is GAP in Implementation project?

As we are aware after receiving clients requirement we consultants will search solution in standard SAP. If there is no solution in standard sap then it is called GAP.

To fulfil this GAP we will do following activities:

1. We will check for alternative (work around)
2. We will take help of ABAPers by preparing FSD document.

List of GAP's in realtime implementation project:

1. PO printout
2. In Collective Number (10 char) field of RFQ, The number in series should be automatically generated by the system while the document is saved.
3. PR print out
4. Delivery challan printout for subcontracting
5. Purchase register (MM+FI) - Report

What is RICEFW?

In realtime projects RICEFW team is also called ABAP team.

RICEFW team helps to fill GAP in the project.

RICEFW stands for

R - Reports

I - Interface

C - Conversions

E - Enhancements

F - Forms

W - Workflows

This RICEFW concept is applicable to all the modules. Development of these concepts required separate time and team. Separate team will work on development of programs, scripts.

What is RICEFW?

R - Reports -

Developing new reports or database output (reports) by executing program. If clients requires any new reports for their business then we consultants will take the help of ABAPers to develop.

Standards reports - Ex. ME2N,MM60,MB51,MB5B,MB52,ME20, etc..

Customised report - developed by functional and ABAP consultants (new reports) Ex. Purchase register, subcontracting report etc. .

I - Interface - in some projects /clients some of the business process like storing of material, planning, invoice uploads are executed in external system (non sap system). Linking non sap and sap system to transfer data is called interface. IDocs concept used to transfer data.

Functional consultants will provide what data needs to be transferred or what data needs to be received , ex. What fields, what numbers etc to the ABAP team.

What is RICEFW?

C - Conversions - conversion means converting data from one format to another format or from one system to another system. Ex. Transferring data from clients legacy system to SAP system. During implementation project clients will change their business operations from non sap to sap then we consultants have to upload whatever data is available in non sap into SAP. Ex. Material, vendors, stock upload, open PO's etc. . .

Uploading tool is LSMW, LTMC, BDC etc . .

E - Enhancements - Enhancements means we are adding our functionality to standards sap screens .

F - Forms - forms are nothing but printouts ex. PO print, Pr print, DC print etc. .

W - Workflows - workflow means flow of work. Workflow suggest the name who will work next. It will flow the work from one person to another person.

Sequence of work. To configure this workflow concept functional module will coordinate with abap consultant and provides the details.

Note - functional module consultants will provide requirements to ABAP consultants in the form of FSD - Functional specification document.

Support Project

After go-live clients will get support of consultants called go-live support. Once go-live support is done then actual support will start which is called AMS - Annual maintenance support or AMC - Annual Maintenance contract.

In this AMS companies (service provider/it company and clients) will go for an agreement called SLA - Service Level Agreement. Consultants will work as per SLA.

In support project consultants will help to resolve the clients issues in SAP system.

Issues are called tickets in SAP. Tickets will be exchanged by using ticket tool.

SAP Support Project which includes handling tickets is called Issue Tracking system. The errors or bugs forwarded by the end user to the support team are prioritised under three severity. They are High, Medium and Low. Each and every severity has got its time limits before that we have to fix the error.

SAP also provides **SAP Solution Manager (SOLMAN)**, which is also used as Ticketing Tool. General concept of Tickets: Handling tickets is called Issue Tracking system. The errors or bugs forwarded by the end user to the support team are prioritised under three severity High, Medium and Low.

Work Process in Support project

1. The customer or the end user logs a call through any ticketing tool or by mail (RADIX) (service Now).
2. Each one of the support team is a part of support group.
3. Each consultant and client will have access to ticketing tool.
4. Whenever a client logs a call or creates a ticket in tool he /she has to mention to which work group ex. PTP, OTC, production, FI, CO etc..
5. Once the calls came to the work group the support consultant or the team needs to send an IR (Initial Response) to the user depending upon the priority of the calls. (Top, High, Med, Low, None) then the error is fixed, debugged by the support consultant or the team.
6. Then after testing properly by generating TR (Transport Request through the basis and transfer to further servers)
7. Then it is informed to the end user/customer/super user about the changes, which have moved to the production server, by CTS - change and transport system process.

Note:

In summary, if any configuration or customization is required to solve the issue, then the consultant have to work on DEV Client, then the end user will test it in the QAS client and after approval the BASIS consultant will move changes to PRODUCTION system.

Realtime Example of support project

End user not able to Create STO purchase order for a supplying plant from a New plant, since they got shipping point error in PO, hence the document becomes INCOMPLETE and he will not be able to proceed further like PO, DELIVERY, BILLING) ETC..

Steps to be followed.

1. Set priority like low, medium and high.
2. Now you need to solve this ticket. You would analyze the problem and identify that the SAP configuration has to be done for the new plant.
3. You would request a transport for DEV CLIENT to BASIS. You do the change and Request one more Transport to BASIS for QA client. The End user will test the same by creating a purchase order for the new plant and approve it.
4. Finally, you request a transport to move the changes to PRODUCTION. Once the change is deployed in production the TICKET is closed. This is just an small example. In real time sometimes we will get issues with severity HIGH in your day-day support.

Ticketing Tool format

This screenshot shows a 'User New record' form. The form is divided into two main columns. The left column contains fields for 'User ID', 'First name', 'Last name', 'Title', 'Department', and 'Password'. Below these are checkboxes for 'Password needs reset', 'Locked out', 'Active' (checked), 'Web service access only', and 'Internal Integration User'. The right column contains fields for 'Email', 'Language' (set to '-- None --'), 'Calendar integration' (set to 'Outlook'), 'Time zone' (set to 'System (America/Los_Angeles)'), 'Date format' (set to 'System (yyyy-MM-dd)'), 'Business phone', 'Mobile phone', and a 'Photo' field with a 'Click to add...' link. A 'Submit' button is located at the bottom left of the form.

User ID		Email	
First name		Language	-- None --
Last name		Calendar integration	Outlook
Title		Time zone	System (America/Los_Angeles)
Department		Date format	System (yyyy-MM-dd)
Password		Business phone	
Password needs reset	☐	Mobile phone	
Locked out	☐	Photo	Click to add...
Active	☒		
Web service access only	☐		
Internal Integration User	☐		

Submit

Ticketing Tool format

This screenshot shows a 'User New record' form, identical to the one above. The form is divided into two main columns. The left column contains fields for 'User ID', 'First name', 'Last name', 'Title', 'Department', and 'Password'. Below these are checkboxes for 'Password needs reset', 'Locked out', 'Active' (checked), 'Web service access only', and 'Internal Integration User'. The right column contains fields for 'Email', 'Language' (set to '-- None --'), 'Calendar integration' (set to 'Outlook'), 'Time zone' (set to 'System (America/Los_Angeles)'), 'Date format' (set to 'System (yyyy-MM-dd)'), 'Business phone', 'Mobile phone', and a 'Photo' field with a 'Click to add...' link. A 'Submit' button is located at the bottom left of the form.

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Password		Business phone	
Password needs reset	☐	Mobile phone	
Locked out	☐	Photo	Click to add...
Active	☒		
Web service access only	☐		
Internal Integration User	☐		

Submit

Ticketing Tool format

The screenshot displays a web-based incident management interface. At the top, there's a header with a back arrow, a menu icon, the text 'Incident INC0010112', and action buttons: 'Follow', 'Update', 'Close Incident', and 'Delete'. Below the header, the form is organized into two columns. The left column contains fields for 'Number' (INC0010112), 'Caller' (survey user), 'Category' (Inquiry / Help), 'Subcategory' (None), 'Service', and 'Configuration Item'. The right column contains 'Contact type' (None), 'State' (Resolved), 'Impact' (3 - Low), 'Urgency' (3 - Low), 'Priority' (5 - Planning), 'Assignment group', and 'Assigned to'. At the bottom, there's a 'Short description' field with the text 'Assessment : ATF Assessor' and a larger 'Description' text area. A 'Related Search Results' link is at the bottom right.

Rollout Project

If client is extending their business such as adding new business or new branch and if they want to use SAP then we consultant have to go for rollout project to extend client business operations to new business or branch.

We have 2 different types of roll outs:

1. Company code rollout
2. Plant rollout

In this rollout project our consultants role is to create new plant & assignment along with other plant configurations.

We have to extend master data's to new plants and company code.

Upgrade Project

If clients wants to upgrade to latest version of SAP then we have to go upgrade project.

In this upgrade project BASIS consultants will take clients data backup then servers will be upgraded in DEV-QAS-PRD sequence. One upgrade is done we functional module consultants will check the latest changes and informs to client.

What is CR?

CR stands for change request.

Change request concept will come in support project. after implementation, if clients wants to add or modify to the existing configuration in sap.

Points to be noted

1. Before CR, consultants should do analysis, effects on existing systems settings.
2. Consultants should get approval from authorities to perform CR.
3. CR may include ABAP changes r only functional changes.
4. CR request can be maintained in solution manager(SOLMAN) if it is configured in project.
5. CHaRM (change request management) is a tool delivered with SAP solution manager and it is a proper approval process and it tracks change requests, transport requests.

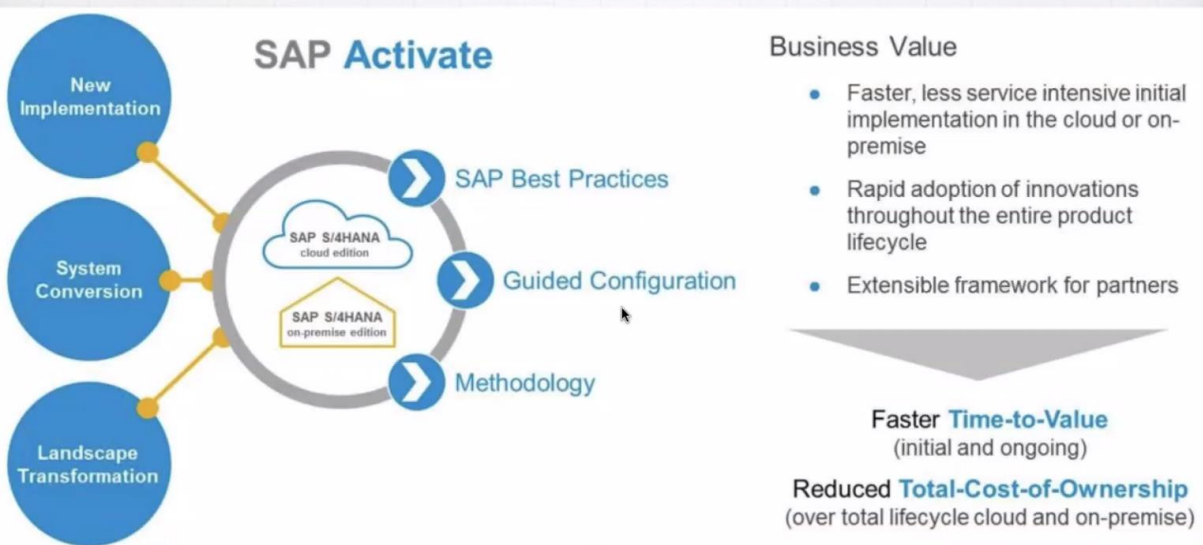
What is S4 HANA?

SAP S4 HANA stands for SAP business suite 4 SAP HANA. It brings the next big wave of innovation to SAP customers, similar to R2 and R3

SAP S4 HANA abbreviations:

S stands for SIMPLE and it is fourth version of SAP business suite after R1, R2, R3 and ERP architecture.

What is S4 HANA?



Benefits of S4 HANA?

S4 HANA allows you to :

1. Reinvent business models
2. Drive new revenues and profits
3. Connect with customers through any channel to deliver value
4. Access the Internet of Things(IOT) and Big Data
5. Simplify your processes, drive them in real-time and adapt instantly
6. Gain insight to any data, in real-time – supporting instant decision making

SAP S/4HANA will simplify your IT landscape and reduce cost of ownership (TCO), through:

1. Reducing your data footprint
2. Working with larger data sets in one system saving hardware costs, operational costs, and time
3. Choice of deployment: cloud, on premise, or hybrid to drive quick time-to-value
4. Innovation is also made simple with an open platform (SAP HANA Cloud Platform) to drive advanced applications – for example, predicting, recommending, and simulating – while protecting existing investments.
5. Finally business users can leverage a simple, role-based user experience based on modern design principles that minimise training and increases productivity.

S4 HANA next generation business suite

SAP S/4HANA, The Next Generation Business Suite

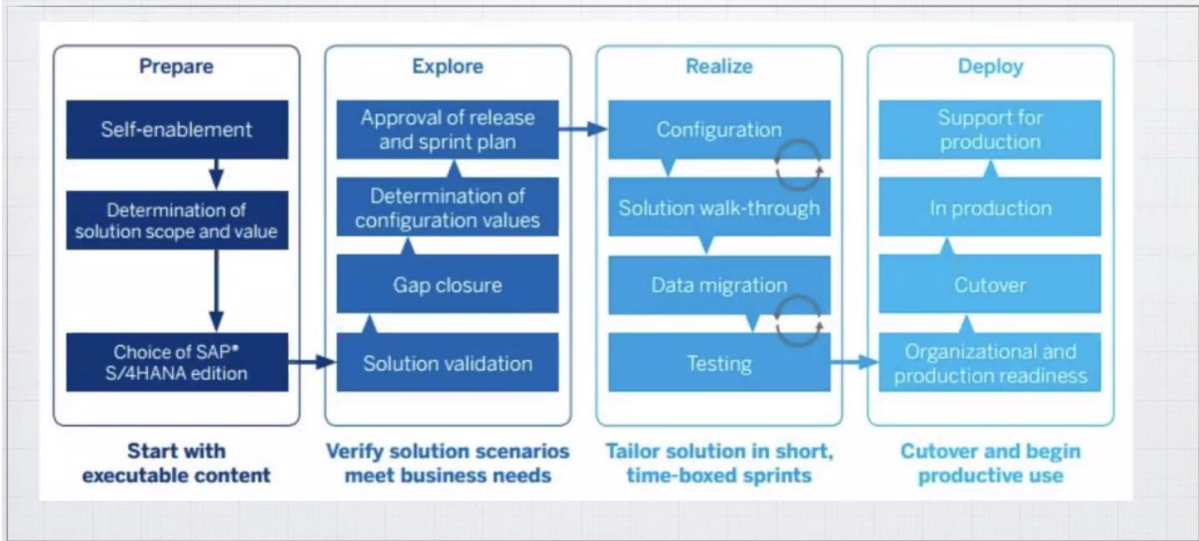


Activate Methodology

Previously we used ASAP methodology in traditional system but S4 HANA in we started using ACTIVATE methodology. SAP Activate provides ready to use Guided configuration, key deliverable and Accelerators (documents, templates) which is supported for SAP S/4 HANA Cloud and On-Premise solution

SAP Activate is the innovation adoption framework that expedites SAP S/4HANA implementations throughout the customer lifecycle. It offers ready-to-run digitized business and technology processes, guided configuration, and next-generation methodology.

Phases in Activate Methodology



MM simplification in S4 HANA

1. New material type SERV is introduced in place of DIEN - services.
2. Material number range length is extended to 40 CHAR earlier it was 18 CHAR.
3. Material ledger made mandatory in material master accounting 1 view (FI change)
4. Vendor master is removed completely in S4 HANA in that place business partner concept is introduced. Same for customer master also (SD)
5. T code changes in SAP MM module in S4 HANA

XK01,XK02,XK03,XK04,XK05,XK06,XK07,MK01,MK02,MK03,MK04,MK05,MK06,FK01,FK02,FK03,FK04,FK05,FK06 are replaced with T. Code BP.

MB1A,MB1B, MB1C,MB02,MB03,MBRL,MBST are replaced with MI00.

Batch management codes.

MSC1,MSC2,MSC3 replaced with MSC1N,MSC2N,MSC3N.

MM simplification in S4 HANA?

1. SAP introduced new output management in S4 Hana. BRF+ - business rule framework (T. Code - OPD). It is unified solution across SD,MM FI.
2. It is fully integrated with FIORI.
3. User can send multiple messages to multiple recipients using multiple channels at same time.
4. User can configure without ABAP program.
5. SAP has introduced new table MATDOC in S4 HANA. It is a central table contains all material documents and history.
6. In data uploads SAP introduced LTMC tool. In this tool template will be available readymade but in LSMW templates to be prepared manually.

Note:

1. S4 HANA can run only on HANA data base but ECC with can be run ORACLE DB2 etc...

S4 HANA is in memory capability data base.

** In memory data base means the data much forward compare to traditional data base.

2. S4 HANA combined the data structure of different components. Ex. FICO, COPA, ML at in to single line table called for S4 HANA.

Some notes Pending

What is Business Partner?

Business partner is a add I SAP S4 HANA. It is the replacement of vendor and customer. As we are aware SAP as removed vendor and customer from ECC along with all their T. codes.

Consultant steps involved in business partner:

- ❖ Define BP roles.
- ❖ Define BP role group.
- ❖ Define number ranges for business partner.
- ❖ Assign number ranges.
- ❖ Define filed selection of business partner.
- ❖ Define BP role direction vendor to BP.
- ❖ Define vendor account group (T. codes – OMSG).
- ❖ Define and assign number ranges to VAG (T. codes- OMSJ).
- ❖ Master data synchronization/ assign BP role to vendor account group.

1. Define BP roles:

Business partner role define the functionality of vendor SAP has defined below standard BP roles.

- FLVN00 – Company code BP role.
- FLVN01 – Purchasing BP role.

We can use above standard or we can customize new BP roles by copying standard only.

In real time projects we will use standard only.

In case if want customize BP role below path.

Path:

SPRO→ IMG→ Cross Application components→ SAP Business partner→ Business partner → Basic setting→ Business partner role→ Define BP role→ Click on position→ Give standard BP role (FLVN00, FLVN01) → Enter→ Select BP role line→ Click on copy as icon→ Give your own BP role (6 CHAR), maintain description→ Enter→ Save it.

2. Define BP role grouping:

Under this heading we are going to group together BP roles. BP roles (if is like vendor account group).

Path:

SPRO→ IMG→ Cross Application components→ SAP Business partner→ Business partner → Basic setting→ Business partner role→ Define BP role grouping→ Enter→ Click on new entries→ Give BP role grouping (6 CHAR) (same like vendor account group) → Maintain title→ Maintain description→ Press enter→ Double click BP grouping →→ BP role→ Again click on new entries→ Assign BP role→ Press enter→ And save it.

3. Define number ranges for Business partner:

Under this heading we are going to define number ranges for business partner.

Path:

SPRO→ IMG→ Cross Application components→ SAP Business partner→ Business partner → Basic setting→ Number ranges and grouping→ Define number ranges→ Enter→ Click on change intervals→ Insert line→ Give serial number→ Give from and TO number range→ Press enter→ Save it→ Go back 2 times→ Select define grouping and assign number ranges→ Enter→ Click on new entries→ Give grouping (4 CHAR) → Give grouping name, grouping description→ Assign number ranges serial number→ Press enter→ Save it.

5. Define field selection of business partner:

As we know field selection used to manage field of business partner.

Path:

SPRO→ IMG→ Cross Application components→ SAP Business partner→ Business partner → Basic setting→ Field grouping→ Configure field attributes per BP role→ Enter→ Double clicks on configure field attributes of BP roles→ Click on position→ Give BP role→ Press enter→ Double clicks on BP role→ Manage the field→ Save it.

8. Define BP role direction vendor BP:

In this we will assign BP role to vendor account group. If we do this then when we create vendor in S4 HANA. (T. code – BP) then automatically vendor code will generate in ECC also (T code – XK01).

- a. Define vendor account group. (OMSG)
- b. Define number ranges for vendor (these number should same as S4 HANA number ranges) and activate external number ranges (OMSJ).
- c. Assign number ranges to vendor account group (OMSJ).
- d. Define and assign partner functions.

9. Master data synchronization/ assign BP role to vendor account group:

Vendor this heading we will assign BP role group and vendor account group. If we do this setting then when we create BP then automatically vendor will be created in XK01.

Path:

SPRO→ IMG→ Cross application component's→ Master data synchronization→ Customer/ Vendor integration→ Business partner setting→ Setting for vendor integration→ Field assignment for vendor integration→ Assign keys→ Define number assignment for direction BP to vendor→ Enter→ Click on new entries→ Assign number ranges group→ Assign vendor account group→ Activate same NOS check box→ Press enter and save it.

Clients' role in Business partner:

Business partner generally used to manage master data for BP centrally like customer, vendor/ suppliers. BP is a single point of entry to create, edit, display master data for Business partner. Business partner can person, Organization, group in which a business partner interested.

In SAP ECC R/3 system vendor and customer will be created with different transaction code like MK01, FK01, XK01 for vendor and VD01, FD01, XD01 for customer but in S4 HANA. We can create vendor and customer by executing single transaction code "BP".

Create Business partner (supplier):

- Go to T. code BP.
- Select vendor category organization.
- Select BP role to create BP.
- Immediately click create.
- Select grouping for number ranges.

General data:

In general data address tab:

In this tab will maintain vendor name, address along with communication details.

Identification tab: (This used for India projects)

Give Tax category – IN3, maintain vendor GST number under tax number heading.

Click on company code, then give your finance information→ Now maintain company code→ Enter→ Maintain RECON AC→ 160000→ Click on purchasing tab, give

purchasing organization→ Enter→ Maintain order, currency, INR, terms of payment, incoterms and other required filed→ Press enter and save the BP.

Note: No changes in tables. All area same.

T. codes:

BP – Create/ Edit/ Display/ Block/ Unblock/ Flag for deletion/ Undelete Business partner.

BP tables: SE16N.

- BP000 – Business partner, General data.
- BUT000 – General data 1.
- BUTOBANK – Bank details.
- BP030 – Business partner – Address.

-----**Business partner topic completed**-----

-

S4 HANA Migration approach:

We have following approaches.

- ❖ Green field.
- ❖ Brown field.
- ❖ Hybrid approach.

1. Green field:

There are few scenarios for depending on few key factors, such as SAP usage of company, development choice, size and state of the data and future business needs.

The green field approach means starting from new. This can be done on – premises or in this could.

Any customization previously done on SAP ECC will be lost.

What is Brown field approach:

Brown field means upgrading and existing system. This approach allows to migrate existing SAP workflow and system over the newest version of S4 HANA.

What is Hybrid approach:

This is most widely used deployment model. In Hybrid approach we can select the best part of green field and rest part of brown implementation. This is best for large enterprises who deals with huge data.

The down side to this Hybrid approach is we used third party tool to do this job.

Mass maintenance:

Mass maintenance are existing transaction code available in SAP which is used to change the information to multiple record at once.

It is used change all the field which are storing field.

Mass maintenance t. codes:

- MM17 – Material mass maintenance.
- XK99 – Vendor mass maintenance.
- MEMASSIN – PIR mass maintenance.
- MEMASSRQ – Purchase Requisition mass maintenance.
- MEMASSPO – Purchase Order mass maintenance.

----- Mass maintenance topic completed-----

Sub sequent debit/ sub sequent credit:

Sub sequent debit/ credit will be performed for only those PO's which are done with invoice. These are an additional invoice or credit memo is received for a transaction that has already been posted.

T. code: MIRO

If vendor is given money back to us then we post S. credit if we have to make extra payment to supplier other than invoice then we have post S. debit.

-----Sub sequent debit/sub sequent credit topic completed-----

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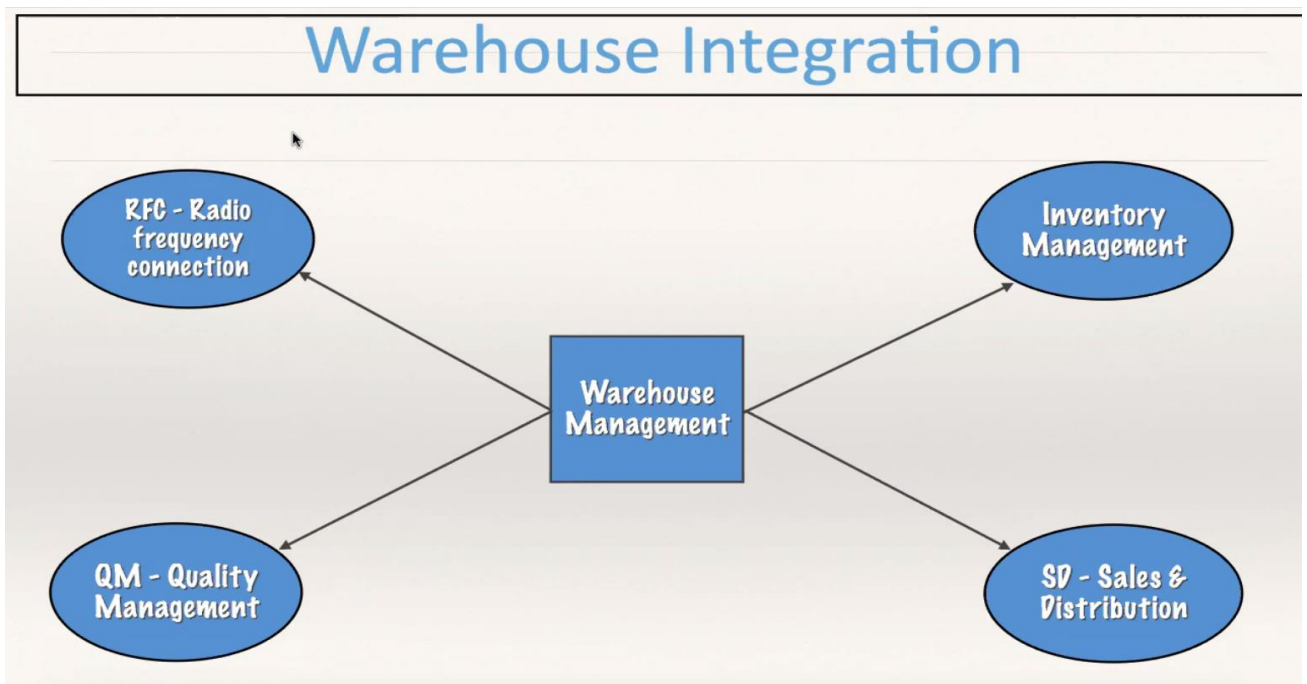
Warehouse Management:

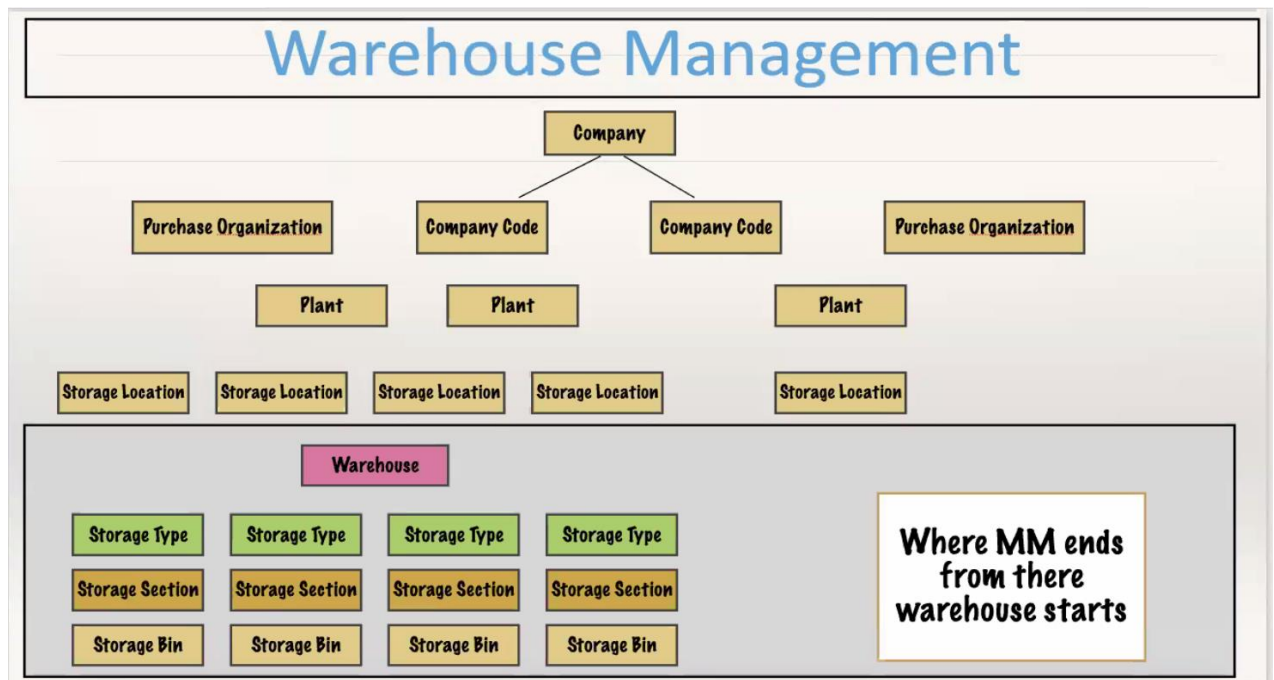
Warehouse Management is one of the modules in SAP which will be used to store companies' material in racks and bins.

Warehouse Management streamlines and control all warehousing process to create an efficient operation. It allows user to manage all structures present in a warehousing operation, regardless of their complexity, while monitoring stock difference and controlling hazardous material.

Benefits:

1. Faster inventory process.
2. Accurate goods tracking.
3. Warehouse automation.
4. Better organization.
5. Customize to fit your operation.
6. Reduce warehouse costs.
7. Increase visibility.
8. Improve compliance.





Define WM Organization structure:

Define Warehouse number:

Warehouse is a complex which is used to store all plant material in single location.

Warehouse no. will be created 3 CHAR code 25 CHAR description. Use below path to copy warehouse from 001 standard.

Path:

SPRO → IMG → Enterprises structure → Definition → Logistics execution → Define copy/delete/check warehouse number → Enter → Double click copy/ delete/ check warehouse number → Click on copy or object icon → Give from standard warehouse number 001 → Give to warehouse number your own (3 CHAR) code → Press enter → Continue → Click on yes to continue → Go back double click on define Warehouse line → Click on position → Give your own Warehouse number → Enter → Change your' s Warehouse name → Press enter and save it.

Assign warehouse number to plant/ storage location:

In SAP we can store multiple plant and storage location material in one warehouse.

Path:

SPRO → IMG → Enterprises structure → Assignment → Logistics execution → Assign warehouse number plant/ storage location → Enter → Click on new entries → Give your plant and give your storage location and assign Warehouse number → and save it.

Storage type:

Number of areas can be defined within the warehouse to store the products is called storage type.

We can have multiple storage location types under one warehouse to store fast moving material, bulk materials, slow moving material etc...

In real time projects or here in practice when we copy warehouse then automatically storage types also will copy. We can check storage types in below path.

Path:

SPRO→ IMG→ Logistics execution→ Warehouse management→ master data→ Define storage type→ Enter→ Click on position→ Give your warehouse number→ Enter→ Check storage type copied from standard type warehouse.

Define storage section:

The storage type can be divided into multiple area called storage section. Storage section contains storage bins where materials are stored physically like pallet goods, fast moving materials slow moving materials etc...

In real time projects or here in practice when we copy warehouse then automatically storage types and storage section will copy. We can check storage section in below path.

Path:

SPRO→ IMG→ Logistics execution→ Warehouse management→ master data→ Define storage section→ Enter→ Click on position→ Give your storage type number→ Enter→ Check storage section.

Define number ranges in warehouse:

Under this heading we will maintain warehouse number like such as TR, TO, PCN etc...

Path:

SPRO→ IMG→ Logistics execution→ Warehouse management→ Mater data→ Define number range→ Enter→ Click on for Transfer requirement→ Give your warehouse number→ Click on change intervals→ Insert line→ Give serial number→ Give from and to number range (10 CHAR) Internals→ Save it→ Go back 2 times→ Click on for transfer order→ Give your warehouse number→ Click on change intervals→ Insert line→ Give serial number→ Give from and to number range (10 CHAR) internals→ Save it→ Go back 2 times→ Click on assignment number range→ Click on position→ Give your warehouse number→ Assign number range serial number→ Save it.

Define storage Bin:

Storage bin is part of storage section which is used to store products physically.

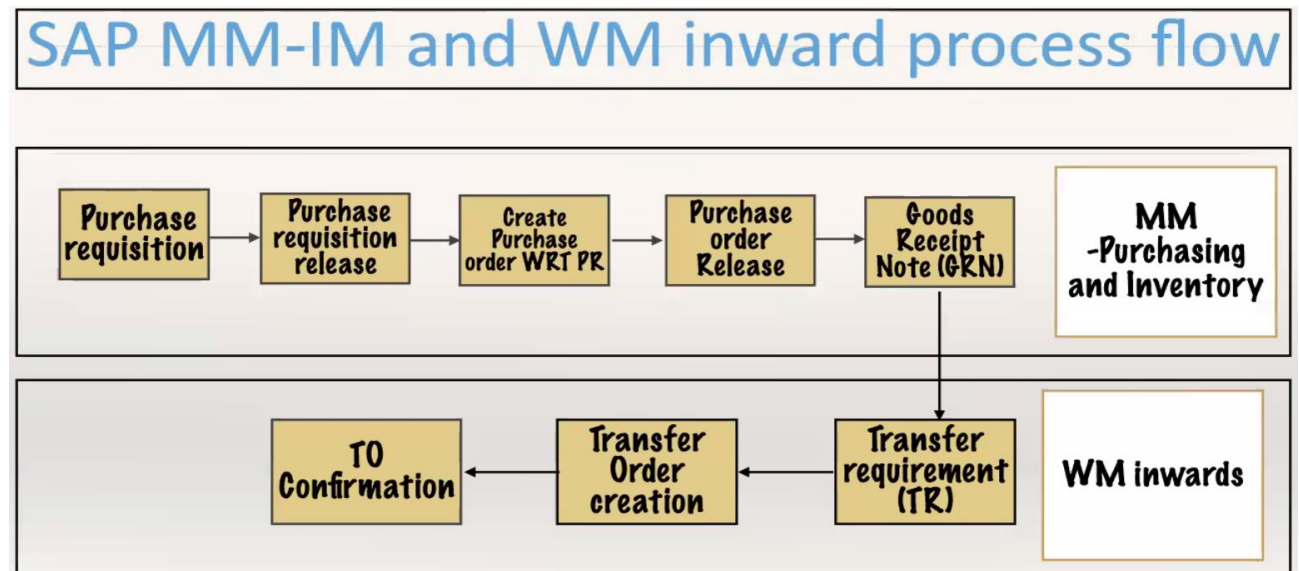
T. codes – LS01N.

Path:

SAP easy access screen→ SAP Menu→ Logistics→ Logistics execution→ Master data→ Warehouse→ Storage Bin→ Create- LS01N- manually→ Enter→ Give warehouse number→ Give storage type→ Give storage Bin (10 CHAR) → Give storage section→ Maintain stor. Bin type (E1) → Maintain total capacity bin→ Press enter→ Save.

T. codes:

- LS01N – Create Bin.
- LS02N- Change Bn.
- LS03N- Display Bin.



T. codes:

- ❖ LB01 – Create TR.
- ❖ LB02 – Change TR.
- ❖ LB03 – Display TR.
- ❖ LB10 – TRs for storage type.
- ❖ LB11 – TR for material.
- ❖ LT01 – Create TO.
- ❖ LT12 – TO Confirmation.
- ❖ LX02 – Warehouse stock.
- ❖ LT21 – Display TO.

LSMW- Legacy System Migration Workshop:

LSMW is a tool in the SAP software which is used to transfer data from non-SAP to SAP through spreadsheets etc...

T. code – LSMW

Recording (one time activity for every transaction MM01, MM02, XK01, XK02 etc...)

Uploading- no of time we can data (create data).

LSMW- Recording:

Under this heading we consultants we record T. code and field required for uploading the data in to SAP system thru LSMW. Recording is an always a one-time activity.

To record and upload data we will use same T. code- LSMW.

Go to T. code – LSMW.

Give project, sub project, object. (We are given own code to identify purpose the recording)

Click on create icon.

Give project name, sub project name, object name.

Continue→ And execute.

Step-1:

System will select step-1 Maintain object attributes:

- Execute.
- Click on display<—> change icon.
- Click on Batch input recording radio button.
- Click on overview icon. (mountain)
- Click on create recording icon.
- Give recording (10 CHAR).
- Maintain recording description→ Press enters.
- Give transaction code to record (Ex- MM01, XK01 etc...)
- Press enters.
- Record the required field save the master data.
- Click on default all icon.
- Check the recording field and save it.
- Go back 2 time.

- Click on recording field.
- Then click on space bar on your keyboard.
- Then system will display our recording names.
- Selected press enters and save the step-1.
- Go back→ Click on object overview icon.
- Then click on tables.
- Copy all the recorded fields→ From column 3 to 6.
- Open your Excel sheet copy to paste the excel.
- Copy the excel data.
- Open new spread sheet re name with Template.
- Paste the transport mode.
- Save the excel sheet on desktop.
- Save the excel template end-user to get the legacy data.
- Go back to SAP.
- Go back again in SAP.

Step-2:

System will select step-2 Maintain source structure:

- Execute.
- Click on display<—> change icon.
- Click on create icon.
- Give source structure.
- Maintain description→ Continue.
- And save the step-2.
- Go back.

Step-3:

System will select step-3 Maintain source field:

- Execute.
- Click on display<—> create icon.
- Click on source structure.
- Click on table maintenance icon.
- Go to Excel.
- Copy the recorded fields.
- Go back to SAP and pasting in required columns.

- Press enter and save it.
- Go back again save the step-3.
- Go back.

Step-4:

System will select step-4 Maintain structure relations:

- Execute.
- Click on display<—> change icon.
- Double click on structure relation line→ Read the information→ Press enters.
- Save the step-4.
- Go back.

Step-5:

System will select step-5 Maintain field mapping and conversion rules:

- Execute.
- Click on display<—> change icon.
- Go to Extra in Manu bar.
- Click on auto field mapping→ Continue.
- Click on accept proposal icon.
- For the all field→ Once auto field mapping completed→ Continue.
- Save the step-5. (Recorded completed)

Step-6 ignore it.

Maintain Fixed Values, Translations, User-Defined Routines.

Note:

1. Prepare the template in excel in share with client to get data to be upload.
2. Go back to excel after receiving from client and check the format length and other field save the desktop.
3. Reopening the excel copy the uploading data, open a note pad and paste it then click back space save the note pad on desktop close the note pad.
4. Now everything is ready upload the data.

LSMW upload:

- We will below steps to upload note pad data into SAP through LSMW in t. codes.
- Go to T. codes- LSMW.
- Give project, sub project, object.

- Execute.

Step-7.

System will select step-7 Specify files:

- Execute.
- Then click on display<—> create icon.
- Click on legacy data on pc line.
- Click on create icon.
- Select the file from desktop.
- Maintain file name.
- Activate tabular radio button.
- Continue and save it step-7.
- Go back.

Step-8.

System will select step-8 Assign files:

- Execute.
- Click on display<—> create icon.
- Double click on source structure.
- Read the information→ Continue.
- Save the step-8.
- Go back.

Step-9.

System will select step-9 Read data:

- Execute→ Again execute.
- Click allow check the file note pad count. (Not written 0)
- Go back 2 times.

Step-10.

System will select step-10 Display read data:

- Execute→ Continue.
- Double click on any line company the data with note pad.
- Go back 2 times.

Step-11.

System will select step-11 Convert data:

- Execute→ Again execute.

- Check the convert data count is should be note pad count.
- Go back 2 time.

Step-12.

System will select step-12 Display converted mode:

- Execute→ Continue.
- Double click on any line check the data your note pad.
- Go back 2 times.

Step-13.

System will select step-13 Create batch input session:

- Execute→ Again execute.
- System will create a batch→ Continue.

Step-14.

System will select step-14 Run batch input session:

- Execute.
- Select session line→ Click on process icon.
- Select display error only radio button.
- Activate extended log, expert mode.
- Click on process again.
- Once data uploaded is done system will display a message batch input session competed.
- Then exit batch session.

-----LSMW topic completed-----

What is LTMC?

LTMC stand for legacy transfer migration cockpit. It is a tool introduced S4 HANA which is transfer data from non-SAP to SAP same like LSMW.

We can use T. code – LTMC.

In SAP LTMC migration templates are readymade available just we need to download those templates from SAP (Excel) then add the data and upload it.

We can download LTMC templates by using LTMC web page by using option create for starting new migration project. Provide project title & retention time and click on create. So finally, templates download in XML format.

What is LTMOM?

LTMOM- Legacy transfer Migration Object Modular:

T. code – LTMOM.

LTMOM allow us to create additional field or customization fields in structure.

What is BAPI and BADI?

BAPI stands for business application programming interface. It works as an interface to transfer data from SAP-to-SAP systems or SAP to non- SAP system.

BADI stands for Business AD- in. it is source code of SAP which helps to improve existing ABAP programs. These improvement helps business.

Ex: if there is no standard code for customer requirements then in BADI it is possible to develop solutions according to requirements without disturbing existing codes.

-----LTMC topic completed-----

Production Planning module (PP)

PP is one of the important modules in SAP that helps businesses to plan the manufacturing, sales and distribution of goods... SAP PP activities include material requirements, planning, bill of material, routing and capacity planning.

SAP PP is an important module in SAP. It tracks and makes a record of the manufacturing process flows, for example, the planned and actual costs. Also, goods movements from the conversion of raw material to semi- finished goods.

It is fully integrated with the other SAP modules like SD, MM, QM, FICO and PM.

SAP MM has a deep integration with SAP PP for manufacturing organizations.

Key points of SAP MM and SAP PP integration are:

- ❖ Bill of materials (BOM).
- ❖ Material requirement planning for external procurement (MRP).
- ❖ Stock determination.

Organization structure in SAP PP:

In any production manufacturing location will be plants and storage location within the plants.

Importance of plant and storage location in production planning module:

- All production master data is created at plant level.
- Planning activities are also performed at plant level.
- Production confirmation process and related goods movement occur at plant and storage location level.

Master data in SAP PP:

- ❖ Material master.
- ❖ BOM- Bill of material.
- ❖ Work center.
- ❖ Routing.
- ❖ Production version.

Bom of materials (BOM):

BOM is used to maintain components list used to manufacture final products.

Stock determination:

Stock determination enables you to implement various strategies to withdrawn materials for goods issues and stock transfers based on the material requirements entered. The system determines how and when the material should be withdrawn, and from which storage location and stocks.

Based on the stock determination strategy, the system makes decision on material withdrawal depending on the material, plant and the business process.

T. codes- MD04.

Agenda:

- ❖ Overview of MRP.
- ❖ Basic of MRP.
- ❖ Different MRP procedures.
- ❖ What is CBP and how it is different from MRP.
- ❖ Execution of MRP.
- ❖ MRP data in material master.

-----**Production Planning topic completed**-----

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Material Requirement Planning

(MRP):

Once we maintain the MRP views (MRP I, MRP II, MRP III) in material master, we can generate the automatic reservation and purchase requisition after running the MRP. MRP depends upon the lead time, safety stock, reorder point, procurement type, special procurement, lot sizing and delivery time for materials. We can run the MRP as a background job for automatic creation of PR and reservations. MRP is applicable for both in-house as well as external material procurement.

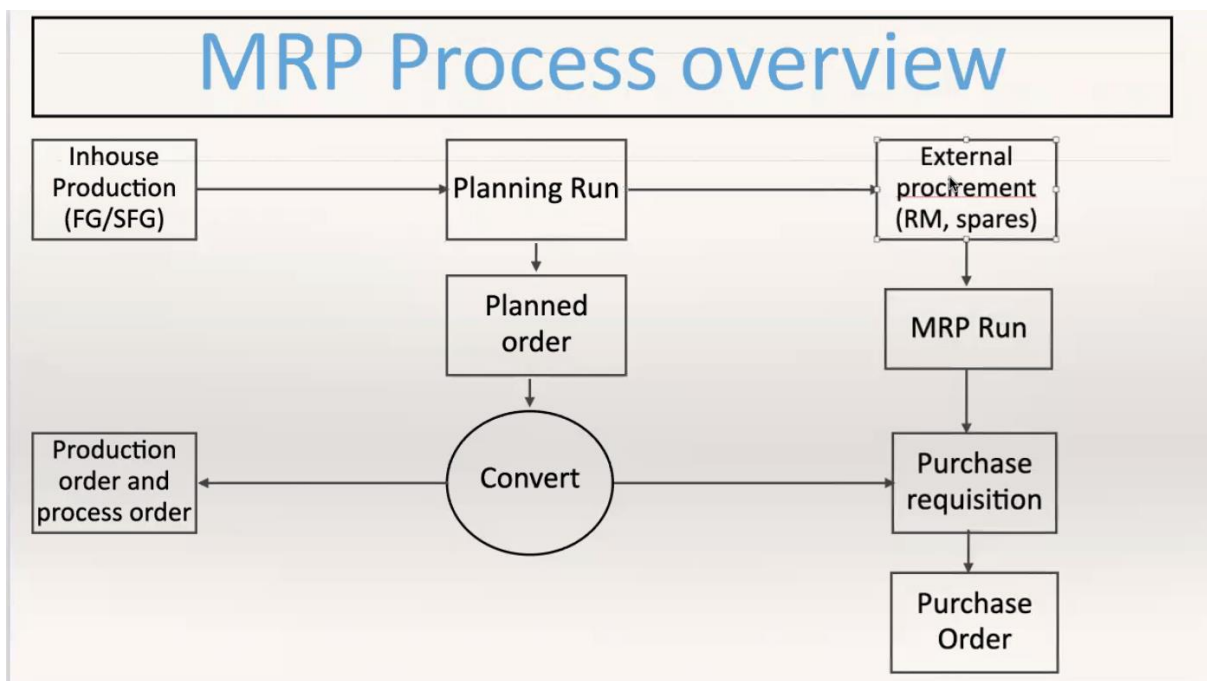
The main purpose of MRP is to guarantee material availability.

MRP determines:

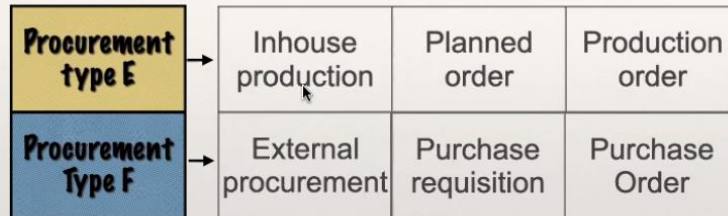
What material needed?

How much quantity needed?

When the material is required?



MRP Process overview



****We will add procurement type in MRP view of material master**

E - In-house production

F - External procurement

MRP Run

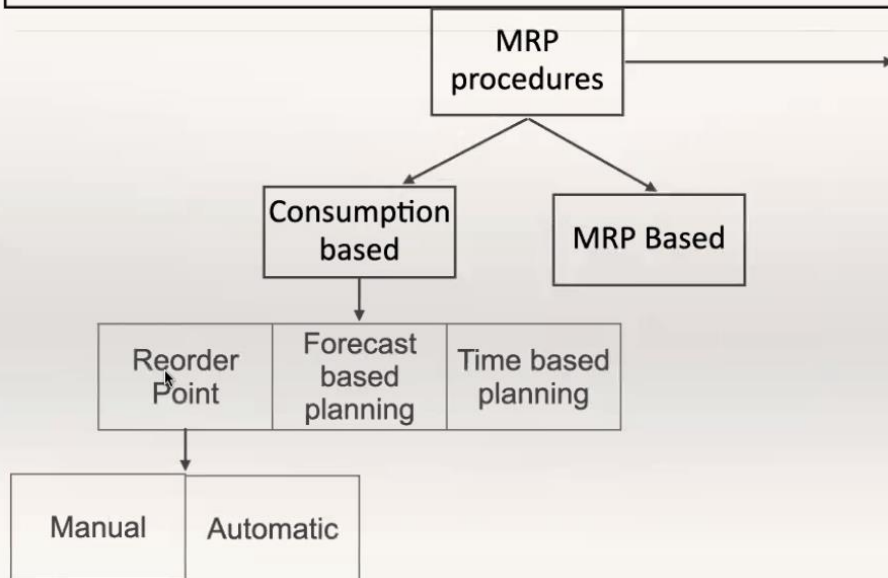
Total planning MRP run for a Plant -
Background- MDBT

Total planning MRP run for a Plant -
Foreground - MD01

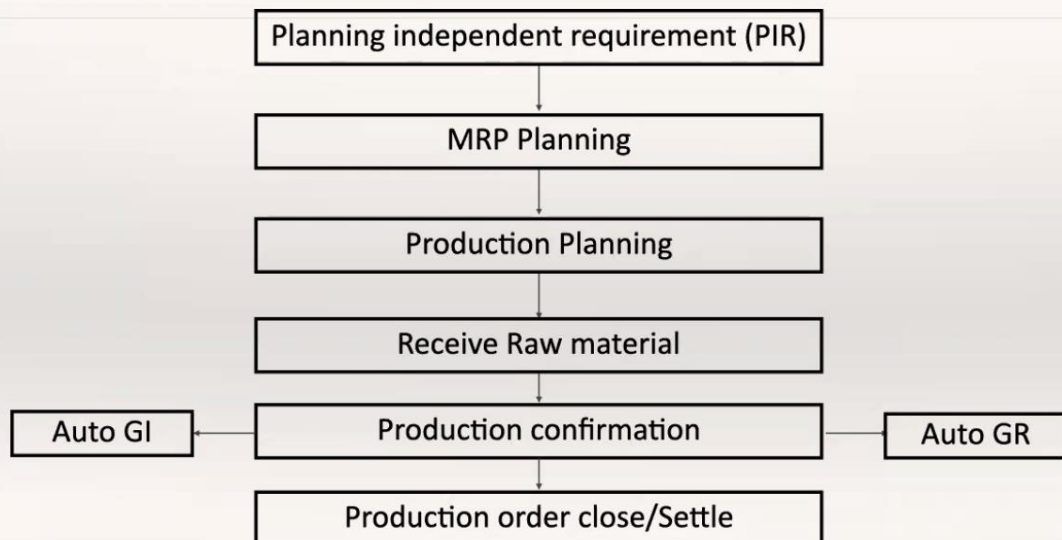
MRP for single material - MD02

****all the transaction codes will be under scope of PP module**

MRP Procedures



Production Planning Process



-----Production planning topic completed-----

GST- Goods and Service Tax

What is GST?

The GST bill was passed in parliament on 29th of march 2017 and came into effected on 1st July 2017.

GST- (Goods and Service Tax):

Goods and Service Tax is a direct tax throughout India to replace existing multiple taxes levied by the central and state government.

Applicable GST rates are determined by the combination Nature of purchase and Harmonized System of Nomenclature (HSN) code assigned to material.

Types of GST's:

- ❖ SGST- State Goods and Service Tax.
 - ❖ CGST- Central Goods and Service Tax.
 - ❖ IGST- Integrated Goods and Service Tax.
 - ❖ UGST- Union Territory Goods and Service Tax.
1. **SGST**- It is used by the state government on intra state (same state) transaction. The revenue collected by state government through GST is earned by the state government.
 2. **CGST**- This is for central government on intra – state goods and service transactions. The central government collects the revenue generated through central goods and service tax. This revenue is shared between central and state government.
Ex: If same state dealer is selling the goods to same state customer, then SGST and CGST will get calculate. SGST amount will be to state and CGST amount will be too central.
 3. **IGST**- Integrated goods and service tax is calculated on interstate (Between 2 states) goods and service transaction. It is applicable imports and export also.

How to create HSN code in SAP:

HSN code stands for Harmonised system of Nomenclature. This is introduced to classify goods and services all over the world. HSN is a 6digit code which identifies the products nature, based on HSC code tax percentage will be decided.

How to create HSN code:

Path:

- Go to T. code- J1ID.

- Select chapter – ID radio button.
- Click on maintain icon.
- Click on new.
- Give chapter ID (HSN code).
- Maintain unit of measurement.
- Maintain description.
- Press enter and save it.

*** We will assign this HSN codes in Material Master.

*** When we are creating PO then we can see HSN code in PO India tab. In this tab HSN code will copy from material master.

How to assign HSN code in Material Master:

- Go to T. code - MM01.
- Give your existing material code.
- Press enters.
- Click on select view.
- Select foreign trade: Export data, foreign trade, Import data view.
- Press enters.
- Give your plant→ Enter.
- Assign your HSN code in central code field.
- Press enter and save it.

-----Goods and Service Tax topic completed-----